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(54) **ILLUMINATION DEVICE AND METHOD FOR CALIBRATING AN ILLUMINATION DEVICE OVER CHANGES IN TEMPERATURE, DRIVE CURRENT, AND TIME**

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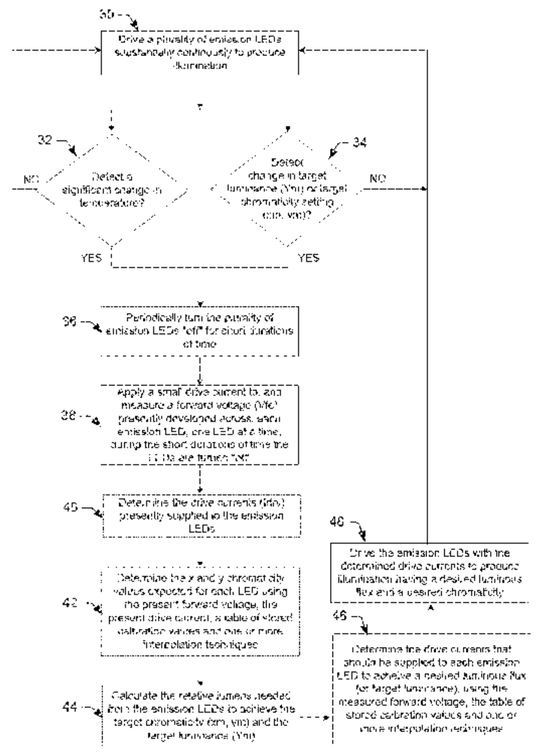
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(57) **ABSTRACT**

An illumination device and method are provided herein for calibrating individual LEDs in the illumination device to obtain a desired luminous flux and a desired chromaticity of the device over changes in drive current, temperature, and over time as the LEDs age. The calibration method may include subjecting the illumination device to a first ambient temperature, successively applying at least three different drive currents to a first LED to produce illumination at three or more different levels of brightness, obtaining a plurality of optical measurements from the illumination produced by the first LED at each of the at least three different drive currents, obtaining a plurality of electrical measurements from the photodetector and storing results of the obtaining steps within the illumination device to calibrate the first LED at the first ambient temperature. The plurality of optical
(Continued)



measurements may generally include luminous flux and chromaticity, the plurality of electrical measurements may generally include induced photocurrents and forward voltages, and the calibration method steps may be repeated for each LED included within the illumination device and upon subjecting the illumination device to a second ambient temperature.

18 Claims, 14 Drawing Sheets

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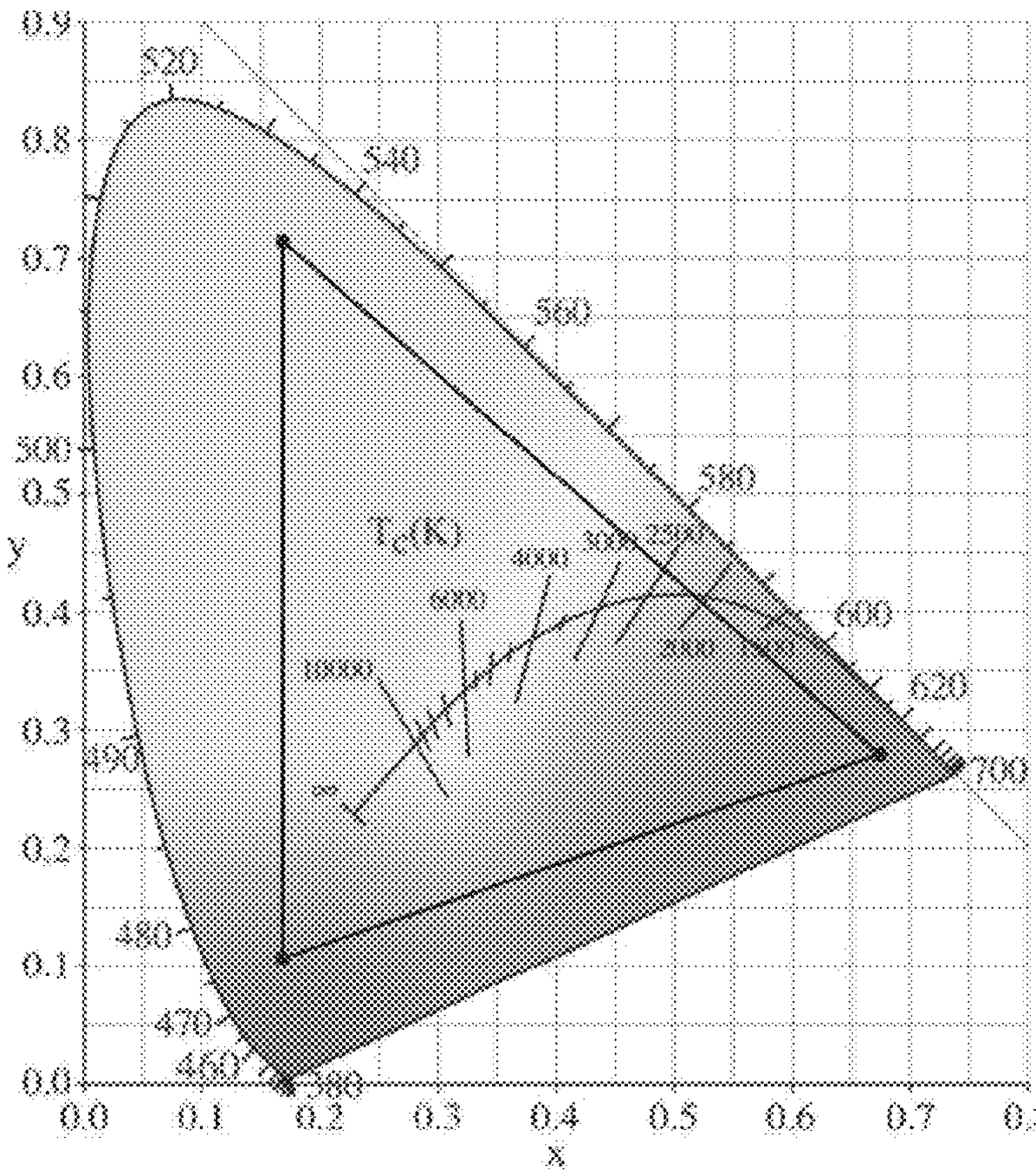

0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
y
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
x
520
540
560
580
600
620
700
500
490
480
470
460
380
T_c(K)

FIG. 1

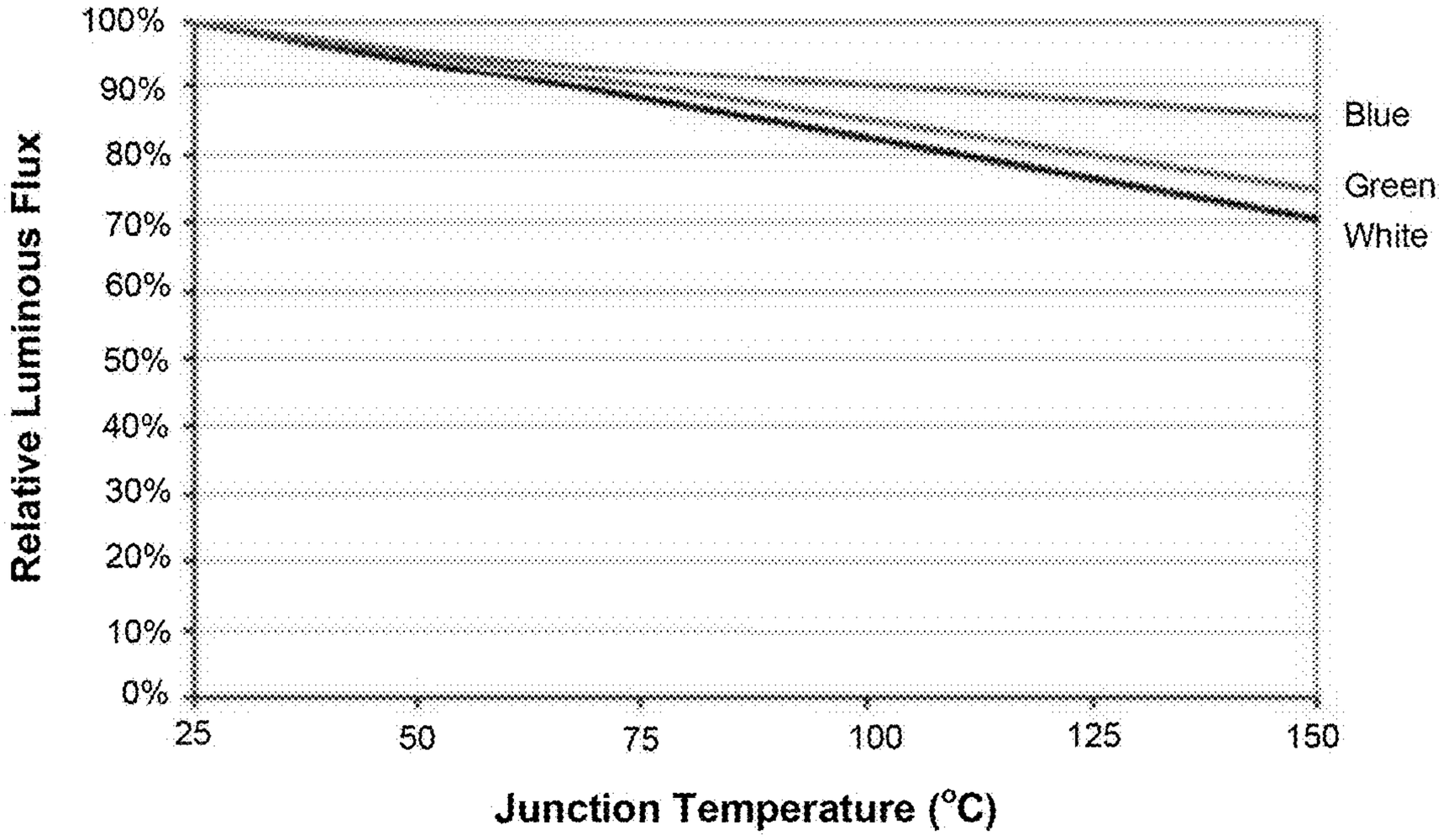

100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
Relative Luminous Flux
25
50
75
100
125
150
Junction Temperature (°C)
Blue
Green
White

FIG. 2

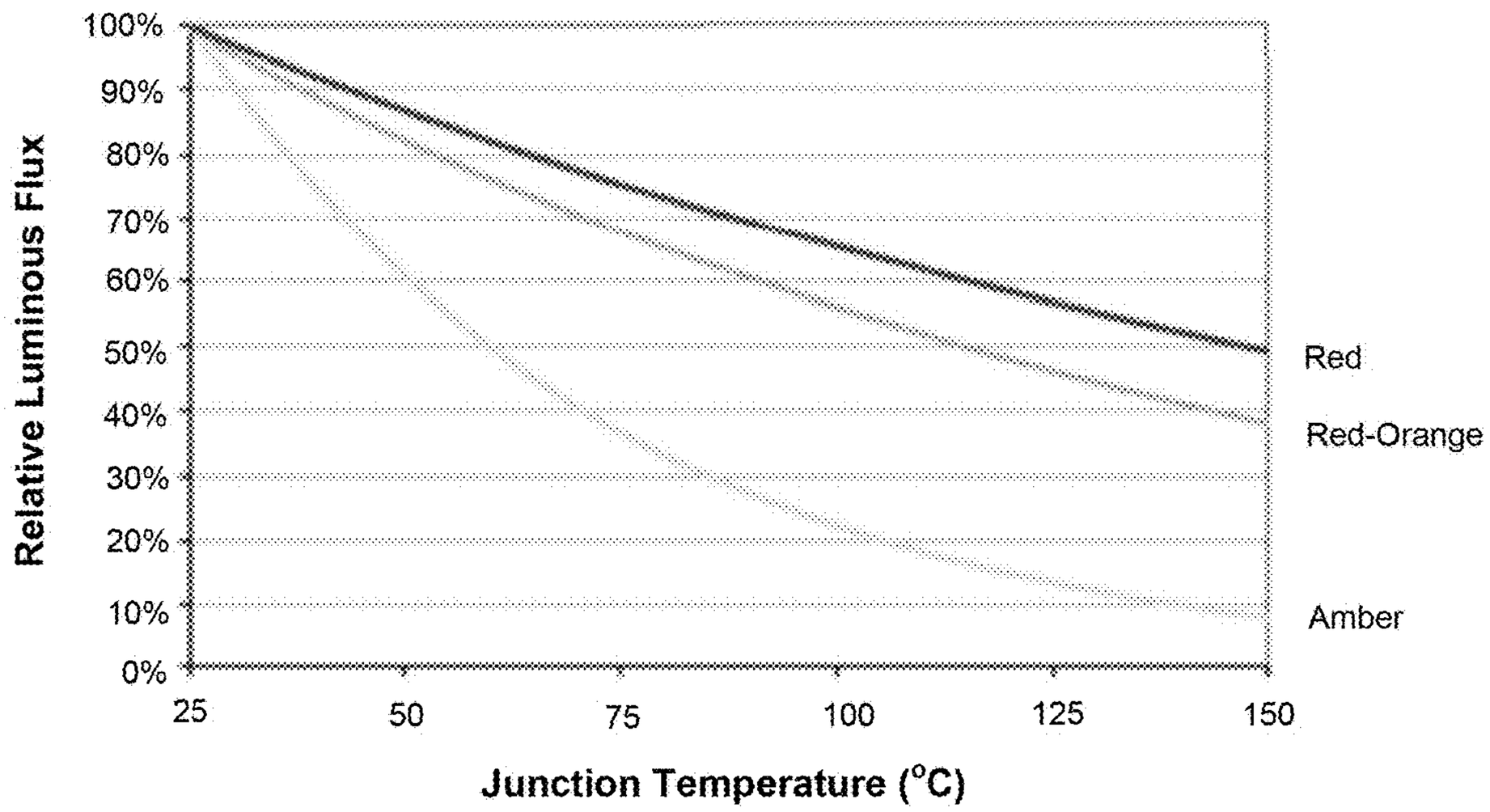

100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
Relative Luminous Flux
25
50
75
100
125
150
Junction Temperature (°C)
Red
Red-Orange
Amber

FIG. 3

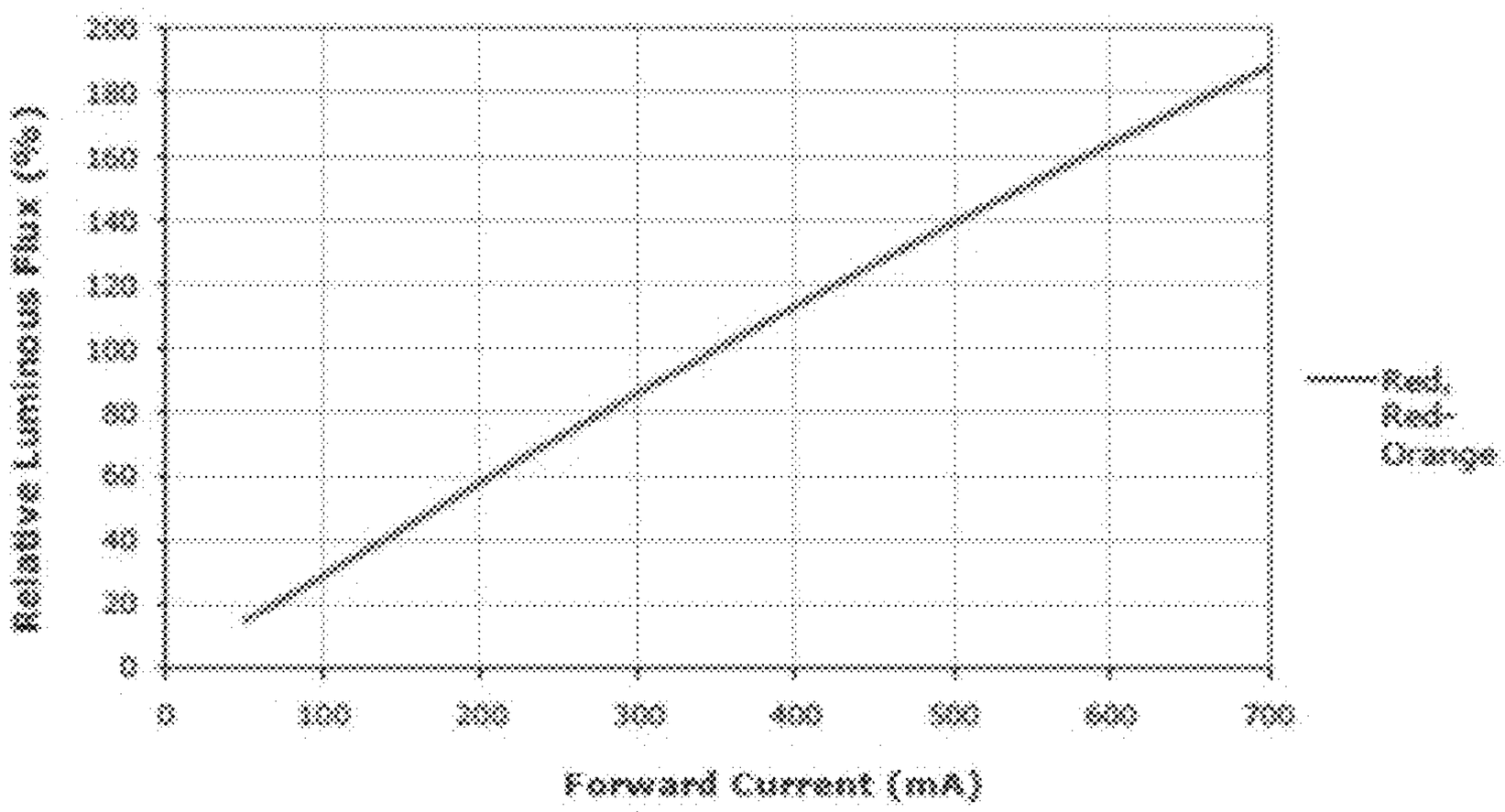

Relative Luminous Flux (%)
0
20
40
60
80
100
120
140
160
180
200
0
100
200
300
400
500
600
700
Forward Current (mA)
Red,
Red-
Orange

FIG. 4

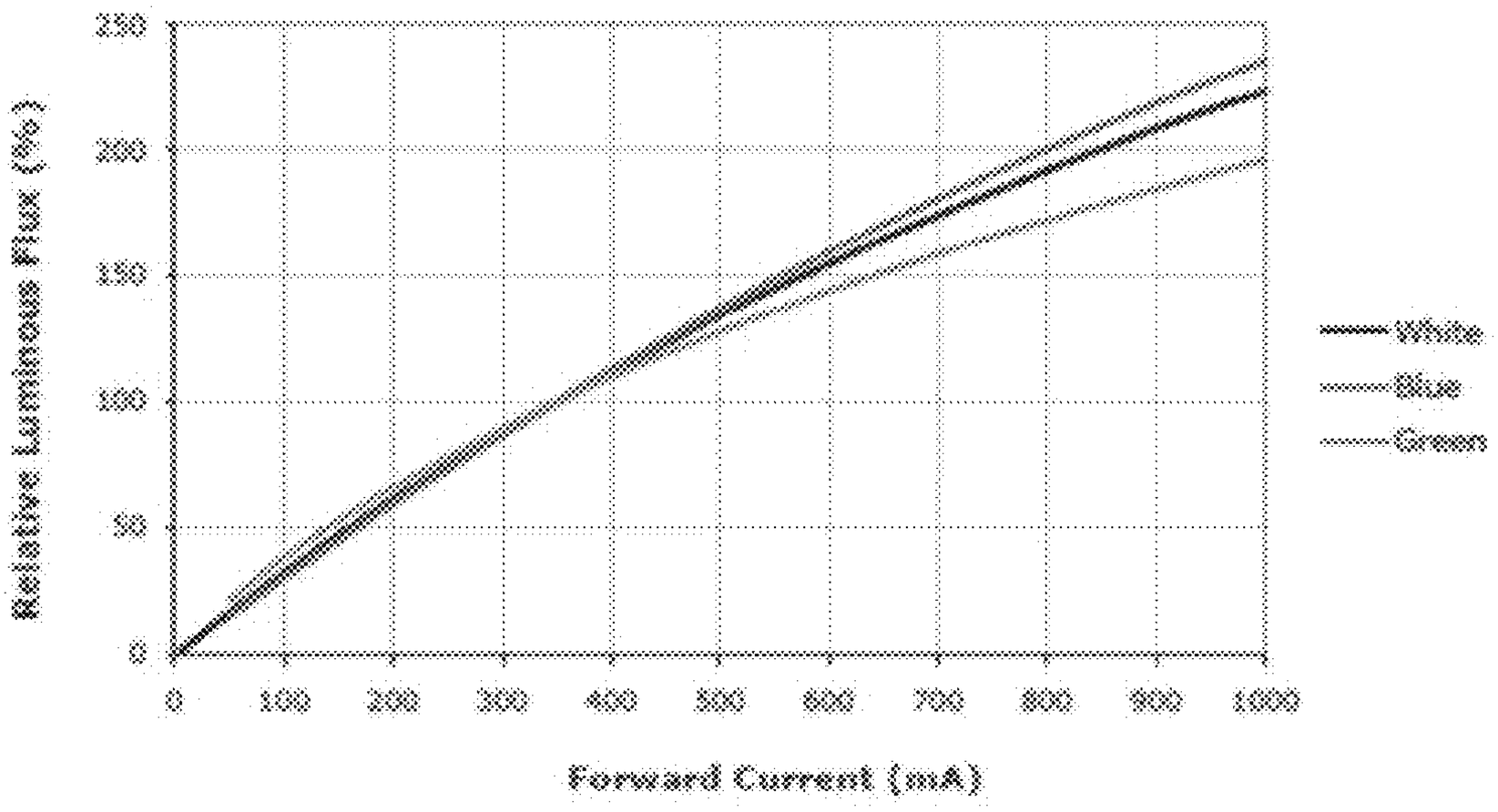

Relative Luminous Flux (%)
0
50
100
150
200
250
0
100
200
300
400
500
600
700
800
900
1000
Forward Current (mA)
White
Blue
Green

FIG. 5

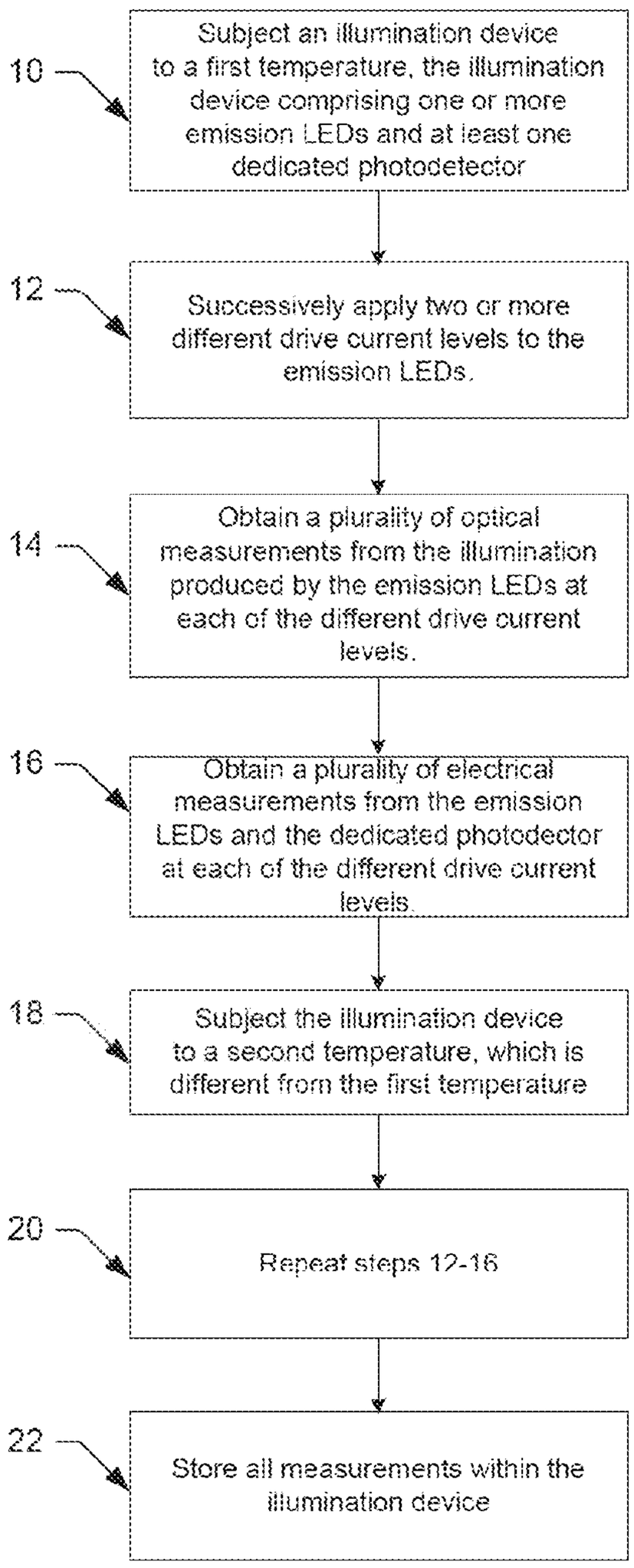

10
Subject an illumination device to a first temperature, the illumination device comprising one or more emission LEDs and at least one dedicated photodetector
12
Successively apply two or more different drive current levels to the emission LEDs.
14
Obtain a plurality of optical measurements from the illumination produced by the emission LEDs at each of the different drive current levels.
16
Obtain a plurality of electrical measurements from the emission LEDs and the dedicated photodector at each of the different drive current levels.
18
Subject the illumination device to a second temperature, which is different from the first temperature
20
Repeat steps 12-16
22
Store all measurements within the illumination device

FIG. 6

Temp	Drive	Param	White	Blue	Green	Red
T0	10%	Luma	x	x	x	x
		X chrom	x	x	x	x
		Y chrom	x	x	x	x
		Iph_d1	x	x	x	x
		Vfd1	x	x	x	x
		Vfe	x	x	x	x
		Iph_d2	x	x		
		Vfd2	x	x		
	30%	Luma	x	x	x	x
		X chrom	x	x	x	x
		Y chrom	x	x	x	x
		Iph_d1	x	x	x	x
		Vfd1	x	x	x	x
		Vfe	x	x	x	x
		Iph_d2	x	x		
		Vfd2	x	x		
	100%	Luma	x	x	x	x
		X chrom	x	x	x	x
		Y chrom	x	x	x	x
		Iph_d1	x	x	x	x
		Vfd1	x	x	x	x
		Vfe	x	x	x	x
		Iph_d2	x	x		
		Vfd2	x	x		
T1	10%	Luma	x	x	x	x
		X chrom	x	x	x	x
		Y chrom	x	x	x	x
		Iph_d1	x	x	x	x
		Vfd1	x	x	x	x
		Vfe	x	x	x	x
		Iph_d2	x	x		
		Vfd2	x	x		
	30%	Luma	x	x	x	x
		X chrom	x	x	x	x
		Y chrom	x	x	x	x
		Iph_d1	x	x	x	x
		Vfd1	x	x	x	x
		Vfe	x	x	x	x
		Iph_d2	x	x		
		Vfd2	x	x		
	100%	Luma	x	x	x	x
		X chrom	x	x	x	x
		Y chrom	x	x	x	x
		Iph_d1	x	x	x	x
		Vfd1	x	x	x	x
		Vfe	x	x	x	x
		Iph_d2	x	x		
		Vfd2	x	x		

FIG. 7

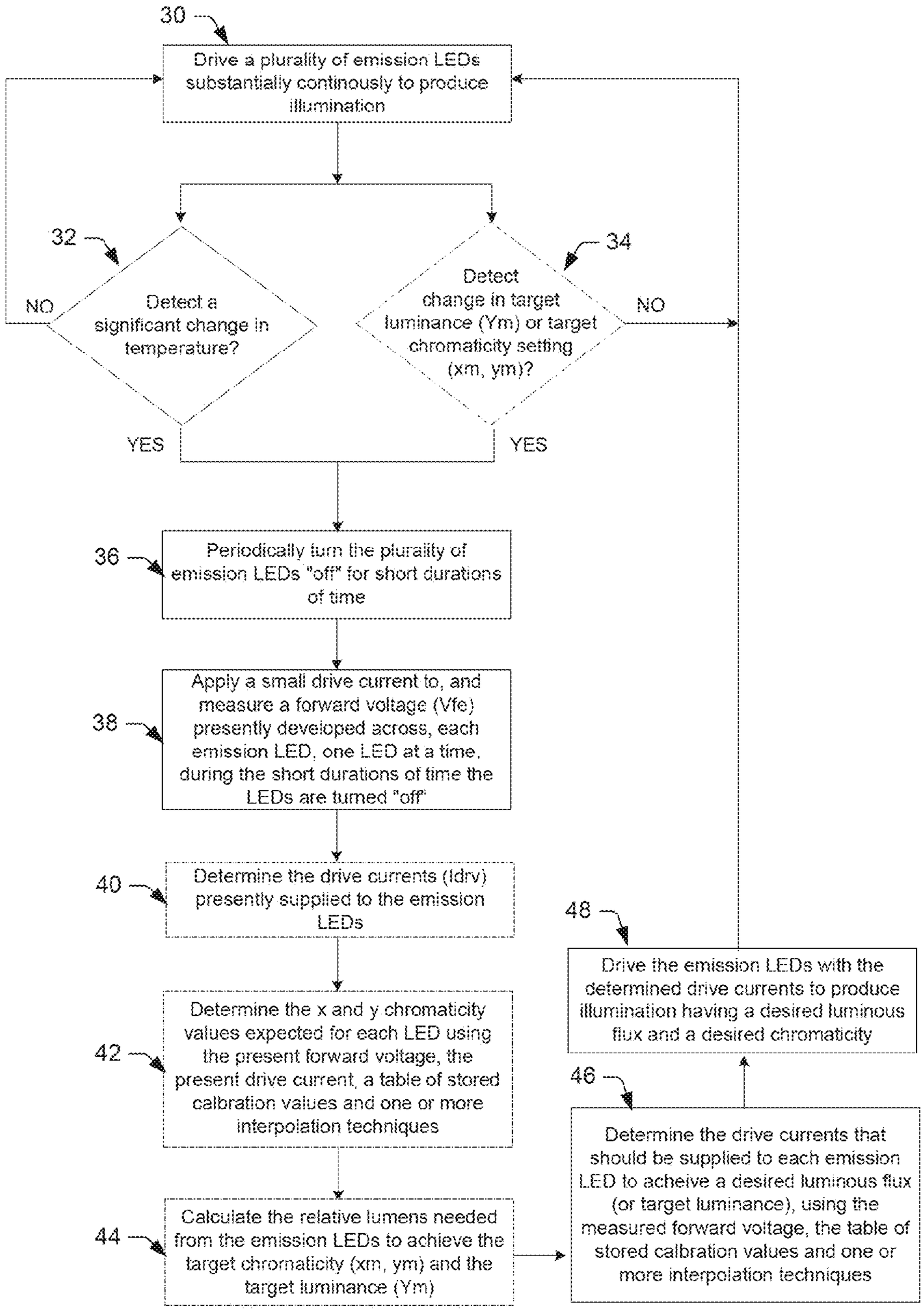

30
Drive a plurality of emission LEDs substantially continously to produce illumination
32
Detect a significant change in temperature?
NO
YES
34
Detect change in target luminance (Ym) or target chromaticity setting (xm, ym)?
NO
YES
36
Periodically turn the plurality of emission LEDs "off" for short durations of time
38
Apply a small drive current to, and measure a forward voltage (Vfe) presently developed across, each emission LED, one LED at a time, during the short durations of time the LEDs are turned "off"
40
Determine the drive currents (Idrv) presently supplied to the emission LEDs
42
Determine the x and y chromaticity values expected for each LED using the present forward voltage, the present drive current, a table of stored calbration values and one or more interpolation techniques
44
Calculate the relative lumens needed from the emission LEDs to achieve the target chromaticity (xm, ym) and the target luminance (Ym)
46
Determine the drive currents that should be supplied to each emission LED to acheive a desired luminous flux (or target luminance), using the measured forward voltage, the table of stored calbration values and one or more interpolation techniques
48
Drive the emission LEDs with the determined drive currents to produce illumination having a desired luminous flux and a desired chromaticity

FIG. 8

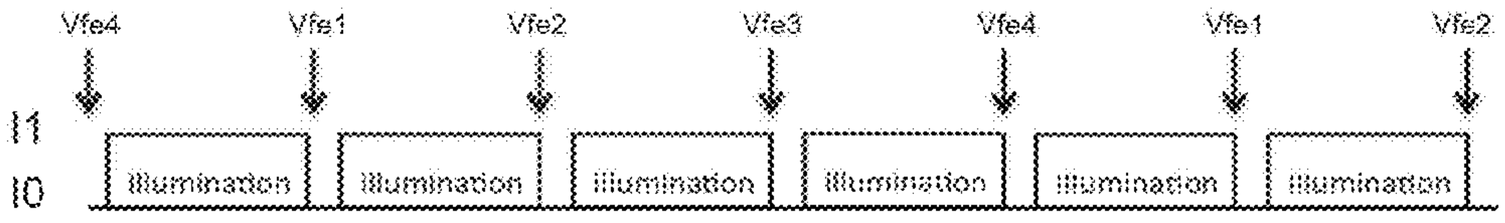

Vfe4
Vfe1
Vfe2
Vfe3
Vfe4
Vfe1
Vfe2
I1
I0
Illumination
Illumination
Illumination
Illumination
Illumination
Illumination

FIG. 9

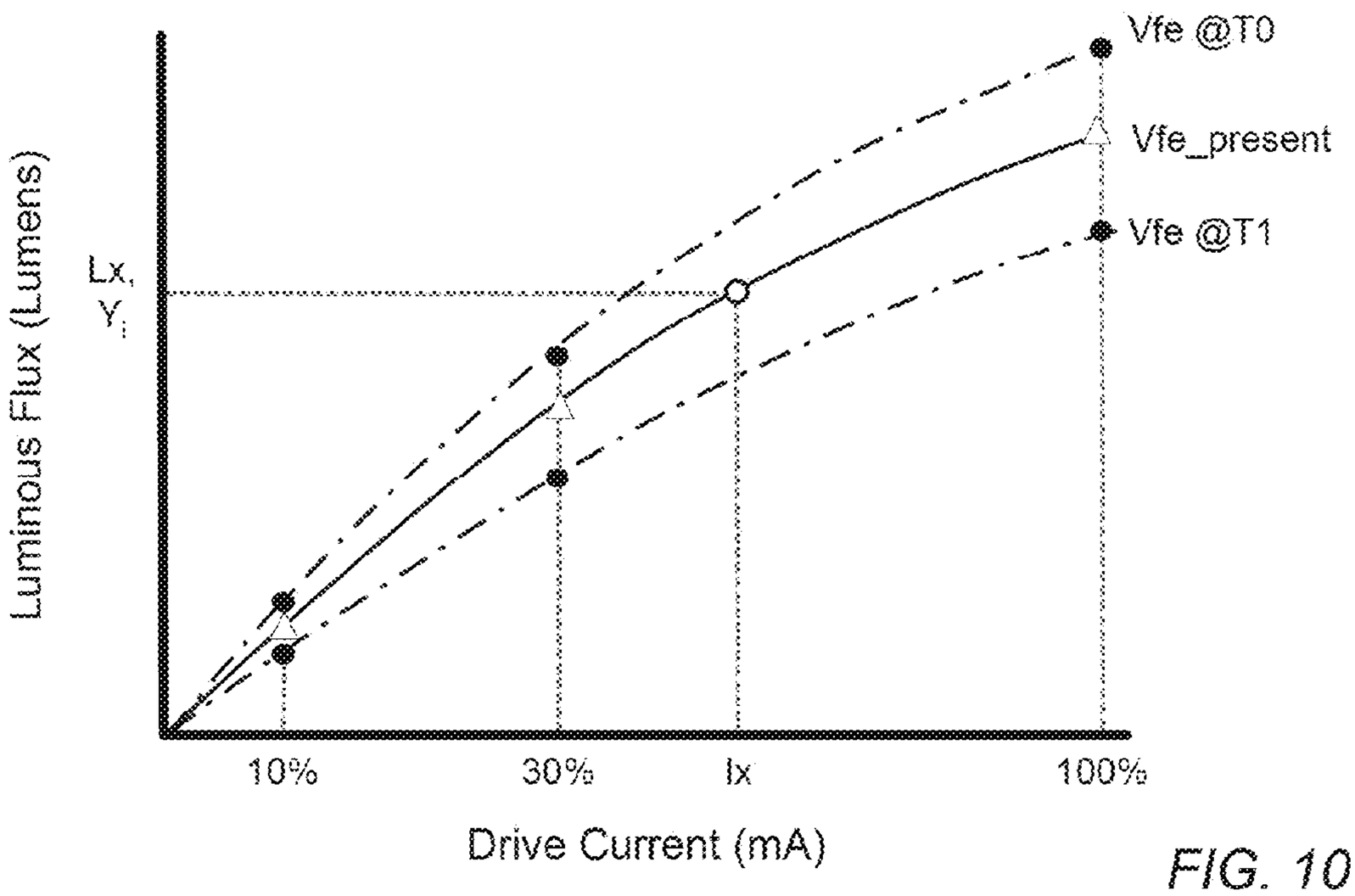

Vfe @T0
Vfe_present
Vfe @T1
Lx,
Yx
Luminous Flux (Lumens)
10%
30%
Ix
100%
Drive Current (mA)

FIG. 10

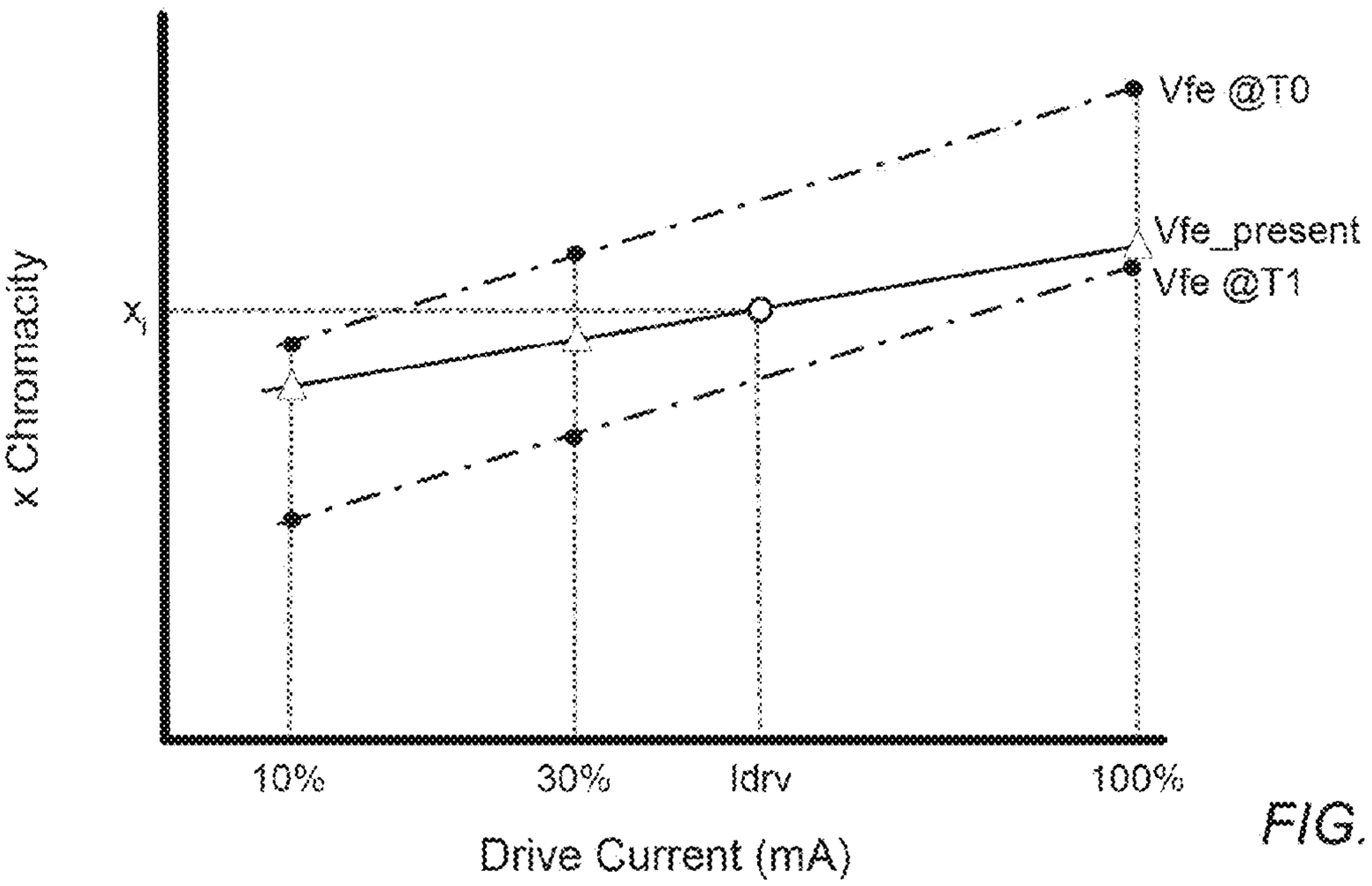

x Chromacity
x_i
Vfe @T0
Vfe_present
Vfe @T1
10%
30%
Idrv
100%
Drive Current (mA)

FIG. 11

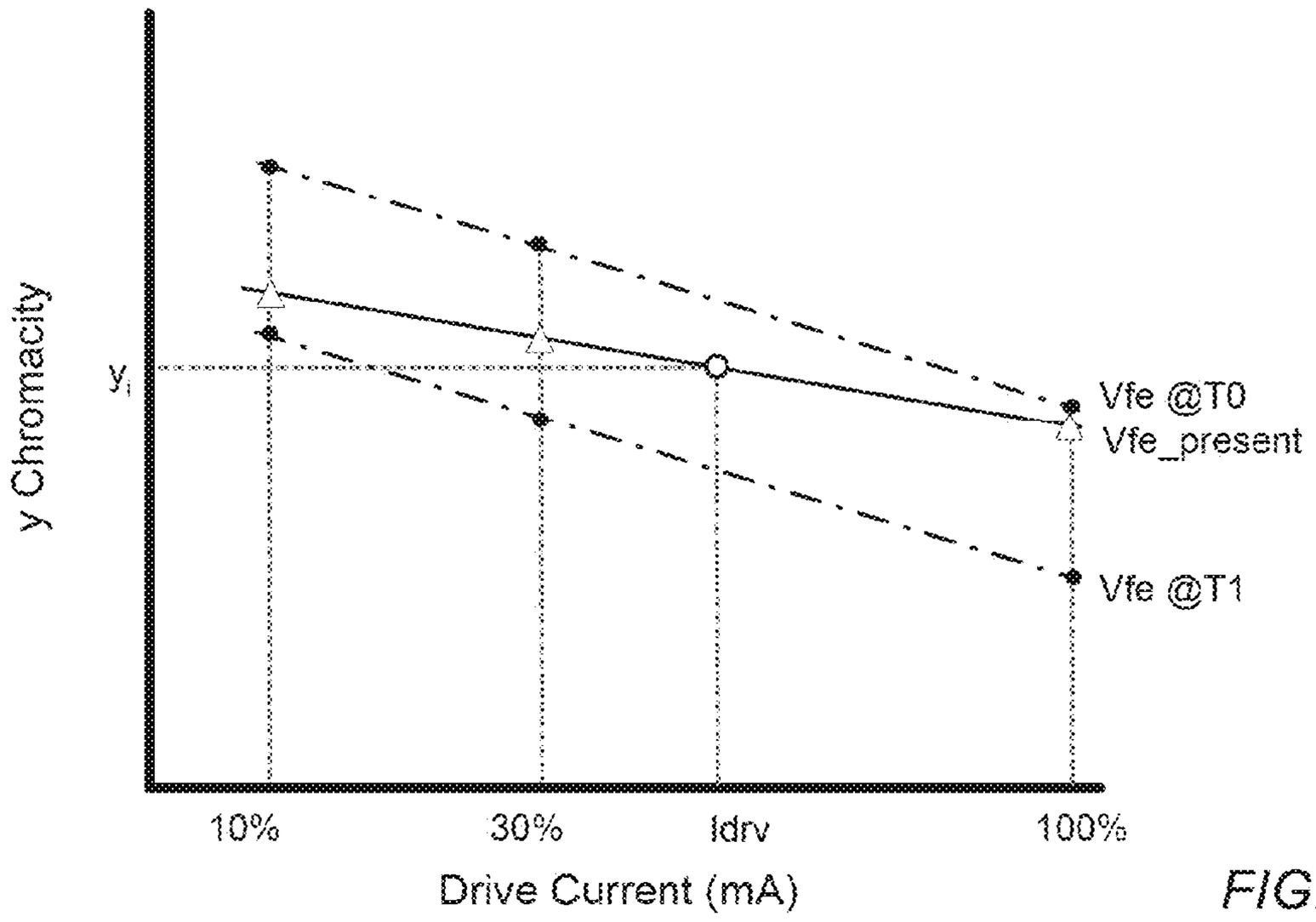

y Chromacity
y_i
Vfe @T0
Vfe_present
Vfe @T1
10%
30%
Idrv
100%
Drive Current (mA)

FIG. 12

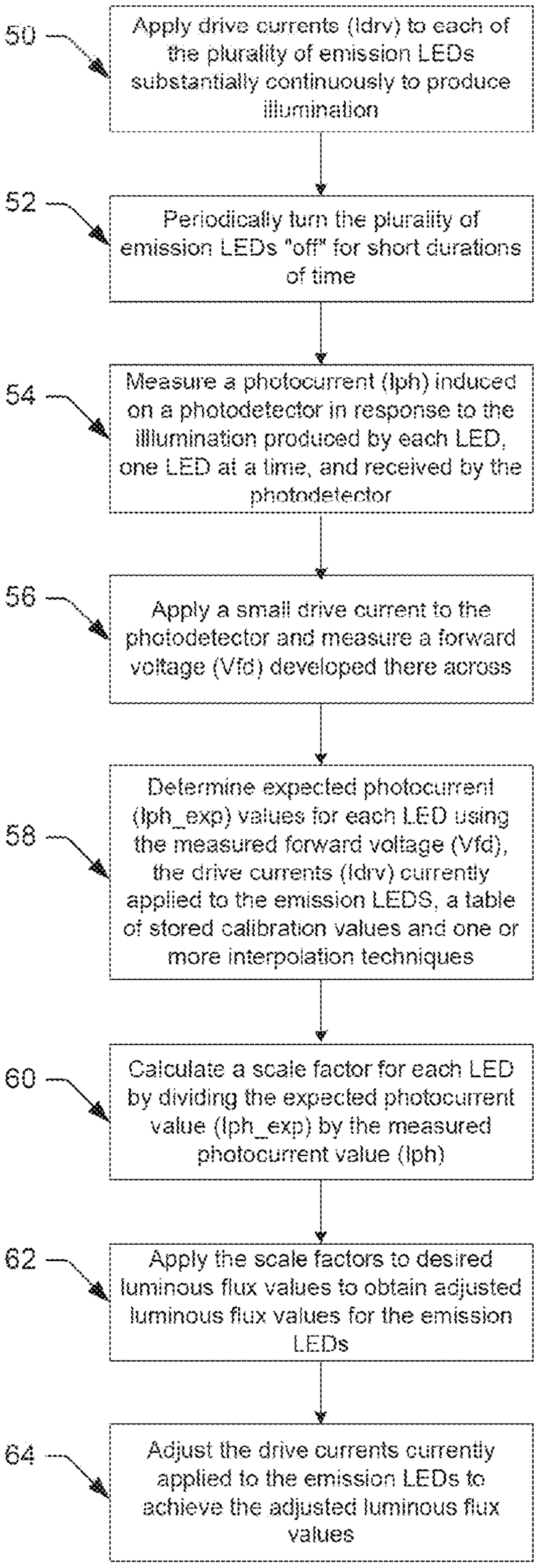

50
Apply drive currents (Idrv) to each of the plurality of emission LEDs substantially continuously to produce illumination
52
Periodically turn the plurality of emission LEDs "off" for short durations of time
54
Measure a photocurrent (Iph) induced on a photodetector in response to the illumination produced by each LED, one LED at a time, and received by the photodetector
56
Apply a small drive current to the photodetector and measure a forward voltage (Vfd) developed there across
58
Determine expected photocurrent (Iph_exp) values for each LED using the measured forward voltage (Vfd), the drive currents (Idrv) currently applied to the emission LEDS, a table of stored calibration values and one or more interpolation techniques
60
Calculate a scale factor for each LED by dividing the expected photocurrent value (Iph_exp) by the measured photocurrent value (Iph)
62
Apply the scale factors to desired luminous flux values to obtain adjusted luminous flux values for the emission LEDs
64
Adjust the drive currents currently applied to the emission LEDs to achieve the adjusted luminous flux values

FIG. 13

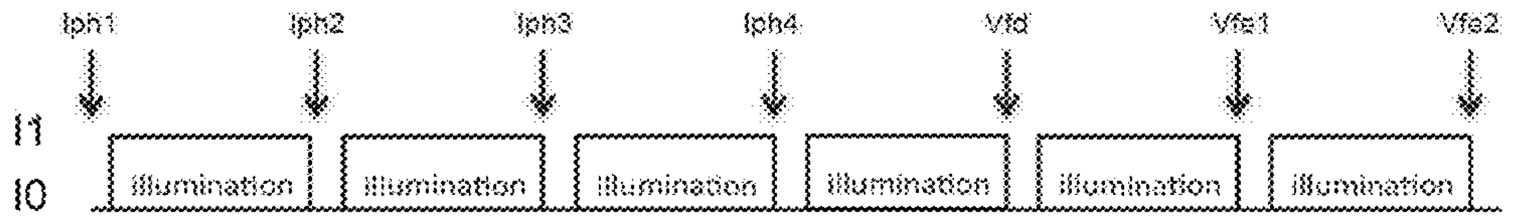

Iph1
Iph2
Iph3
Iph4
Vfd
Vfe1
Vfe2
I1
I0
illumination
illumination
illumination
illumination
illumination
illumination

FIG. 14

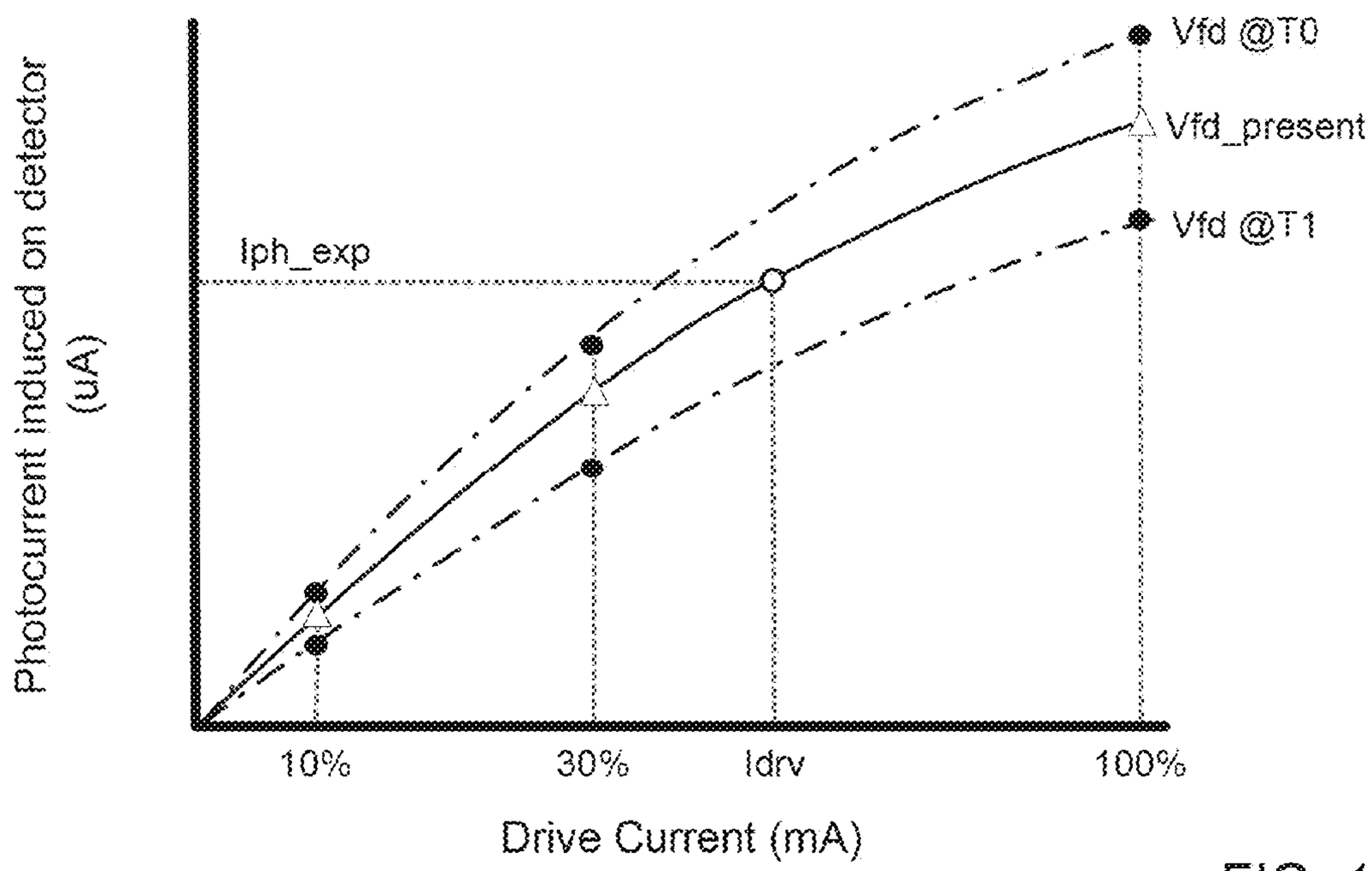

Photocurrent induced on detector (uA)
Vfd @T0
Vfd_present
Vfd @T1
Iph_exp
10%
30%
Idrv
100%
Drive Current (mA)

FIG. 15

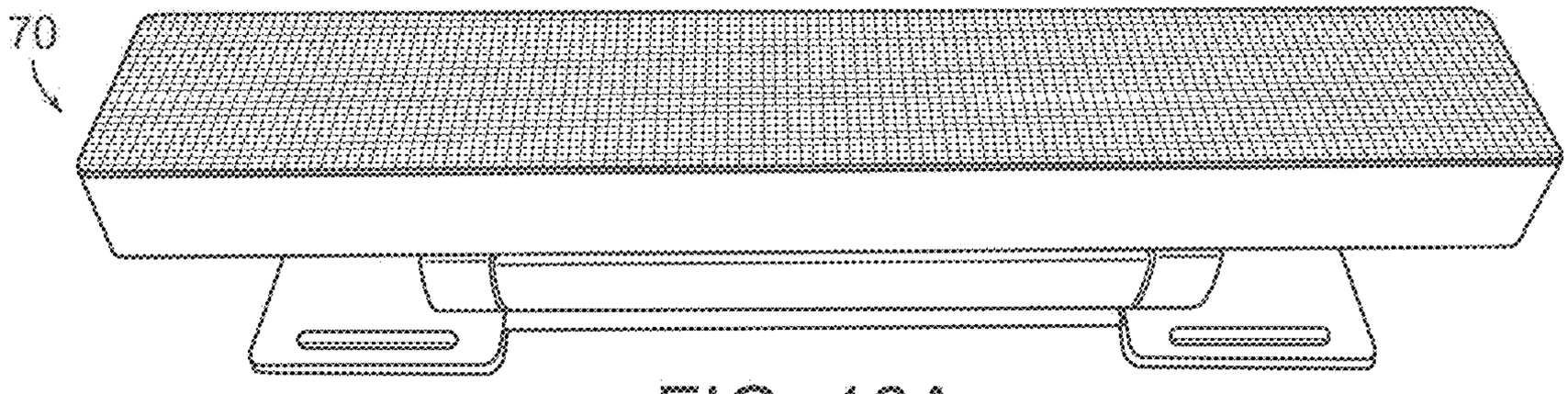


FIG. 16A

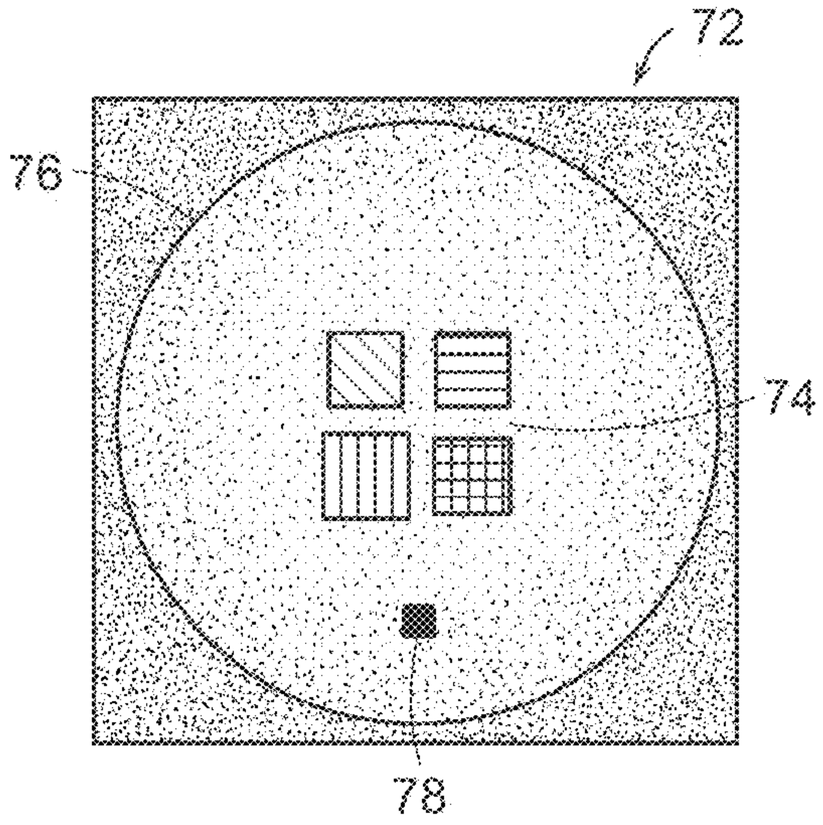


FIG. 16B

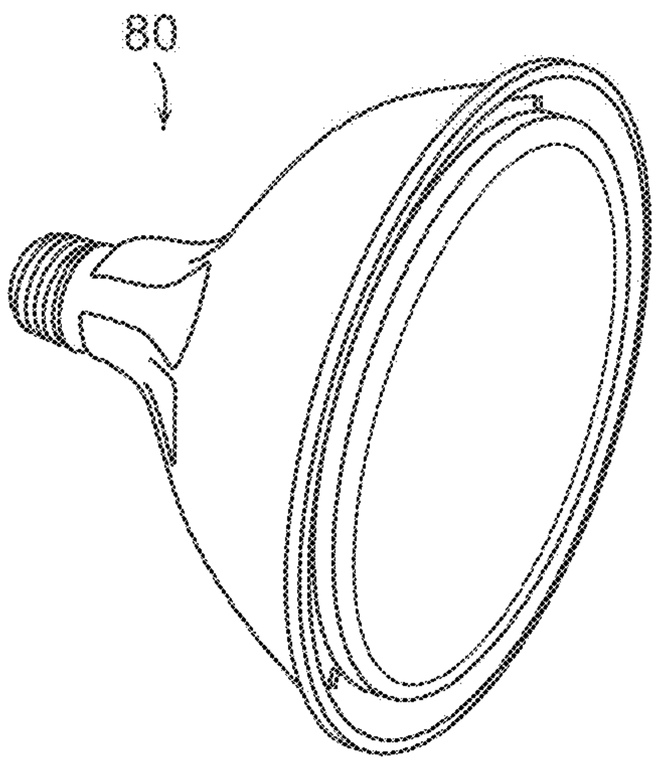


FIG. 17A

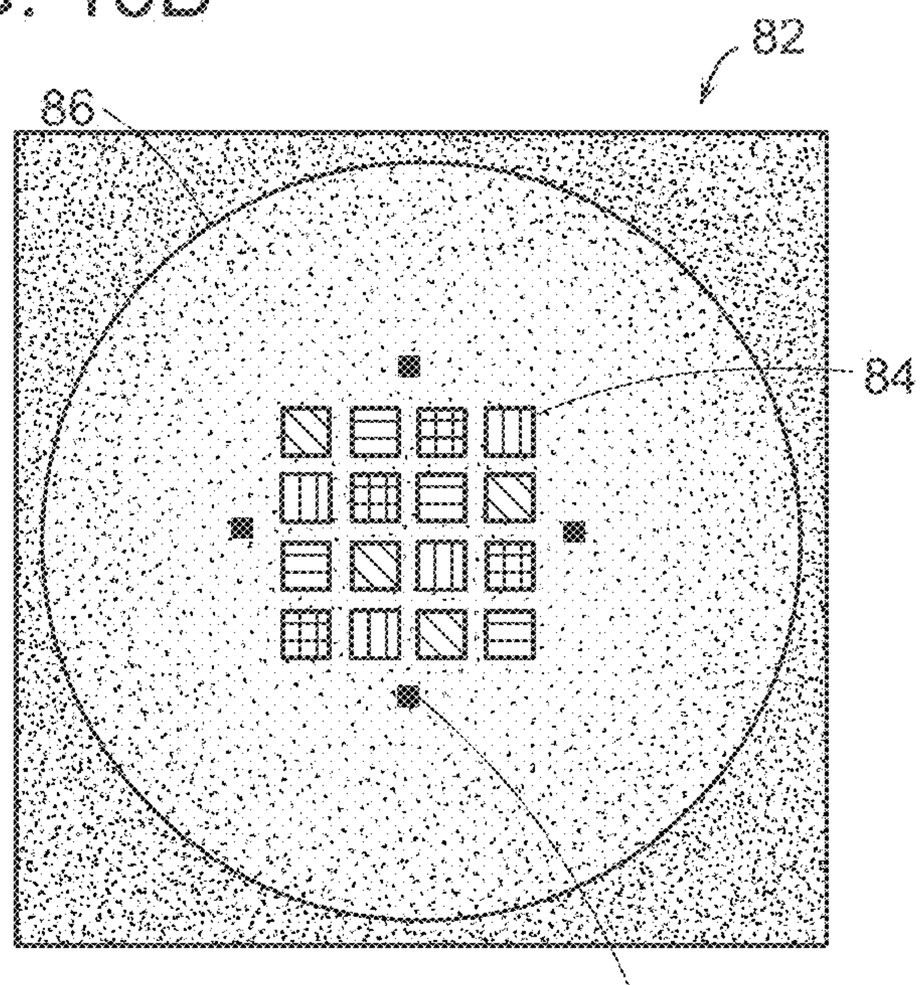


FIG. 17B

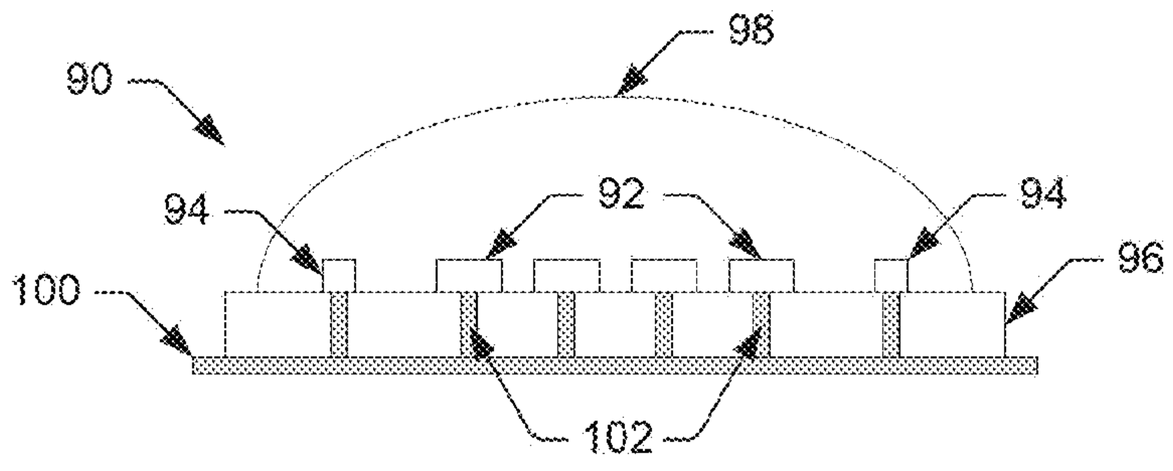


FIG. 18A

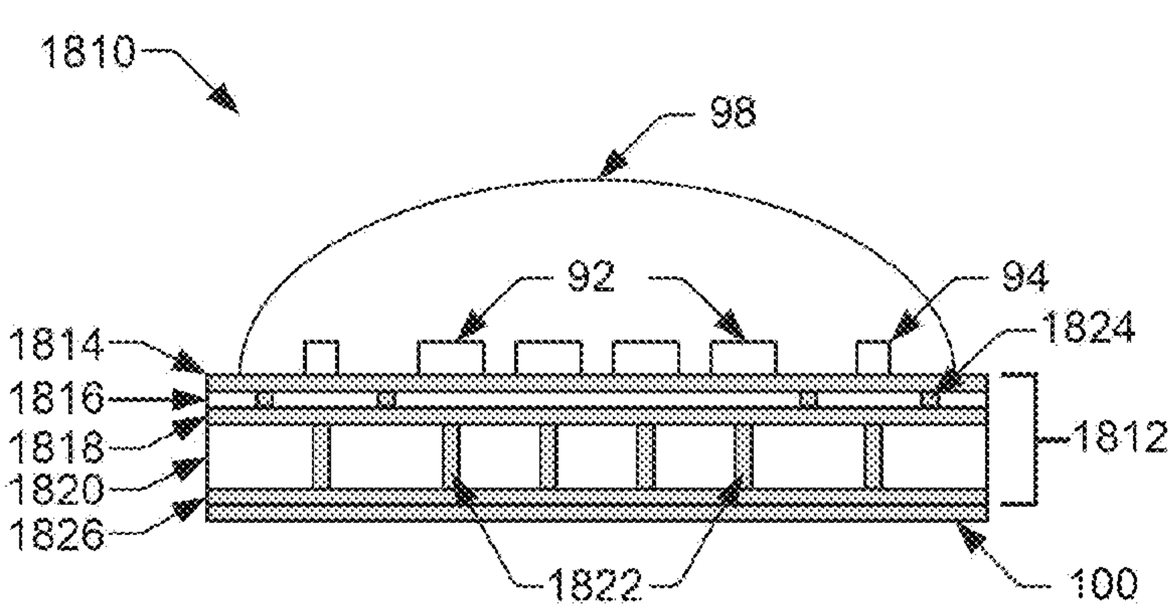


FIG. 18B

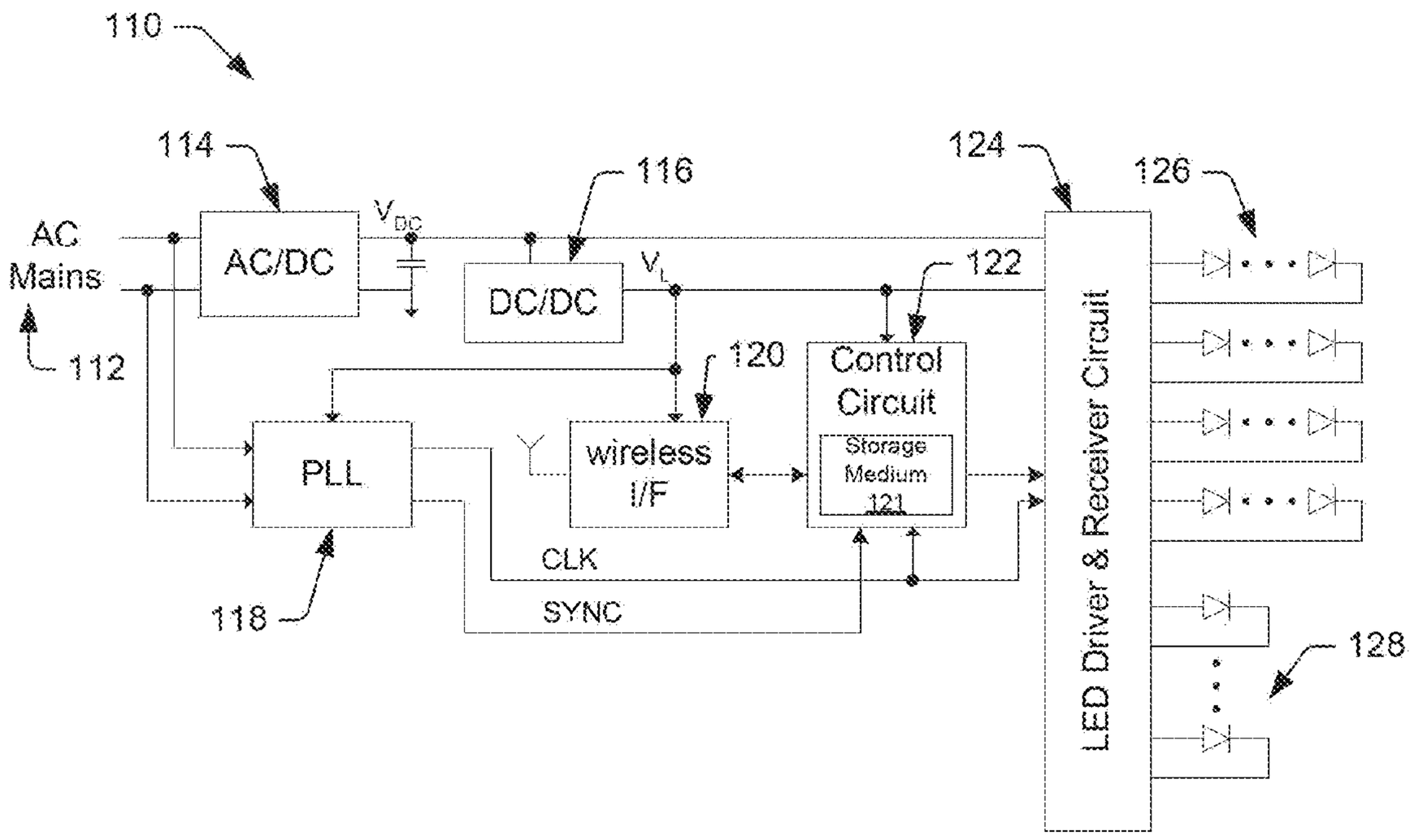


FIG. 19

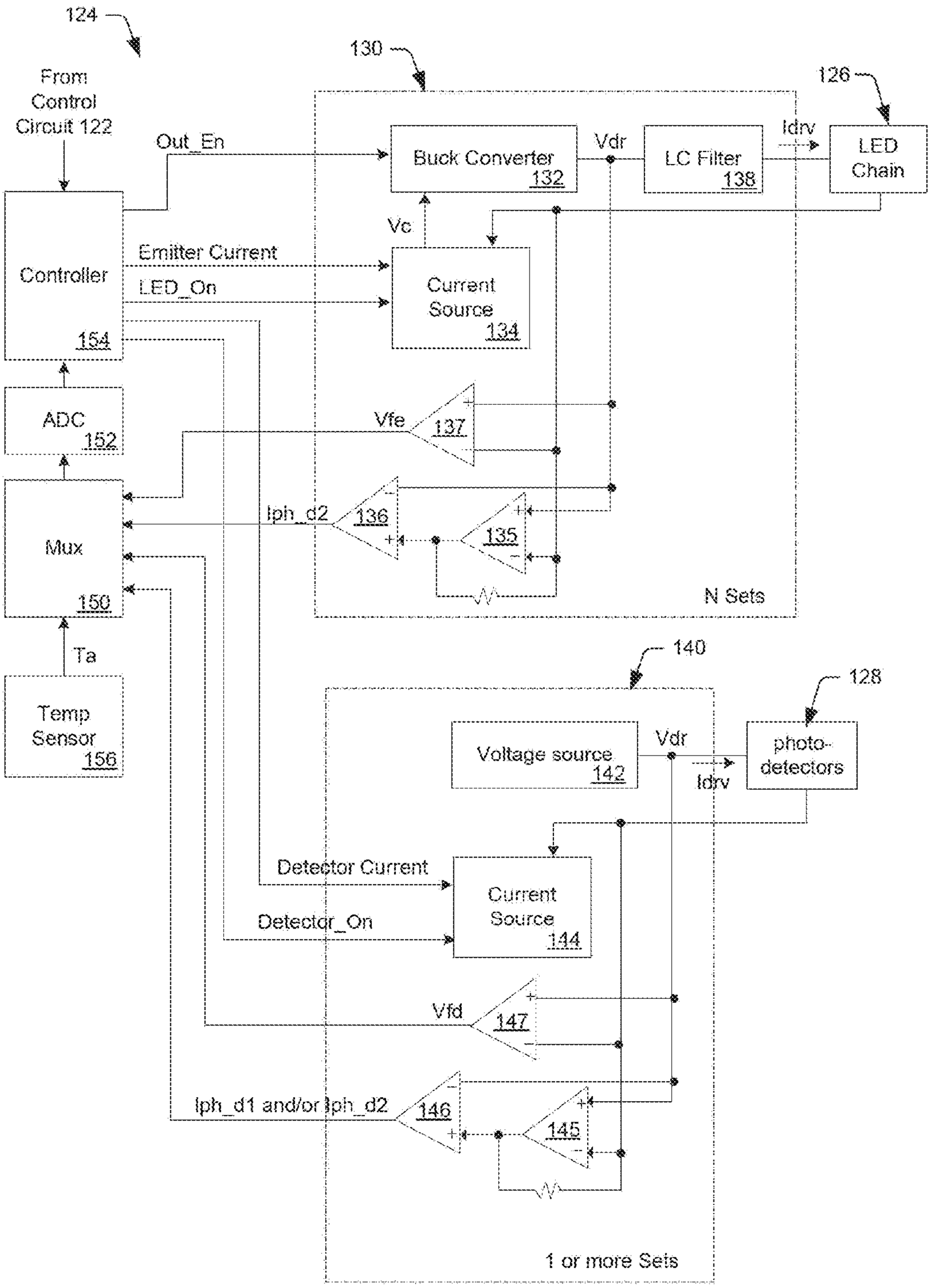

124
130
126
From Control Circuit 122
Out_En
Buck Converter 132
Vdr
LC Filter 138
Idrv
LED Chain
Vc
Emitter Current
LED_On
Controller
154
Current Source 134
ADC 152
Vfe
137
Mux 150
Iph_d2
136
135
N Sets
Ta
Temp Sensor 156
140
128
Voltage source 142
Vdr
photo-detectors
Idrv
Detector Current
Detector_On
Current Source 144
Vfd
147
Iph_d1 and/or Iph_d2
146
145
1 or more Sets

FIG. 20

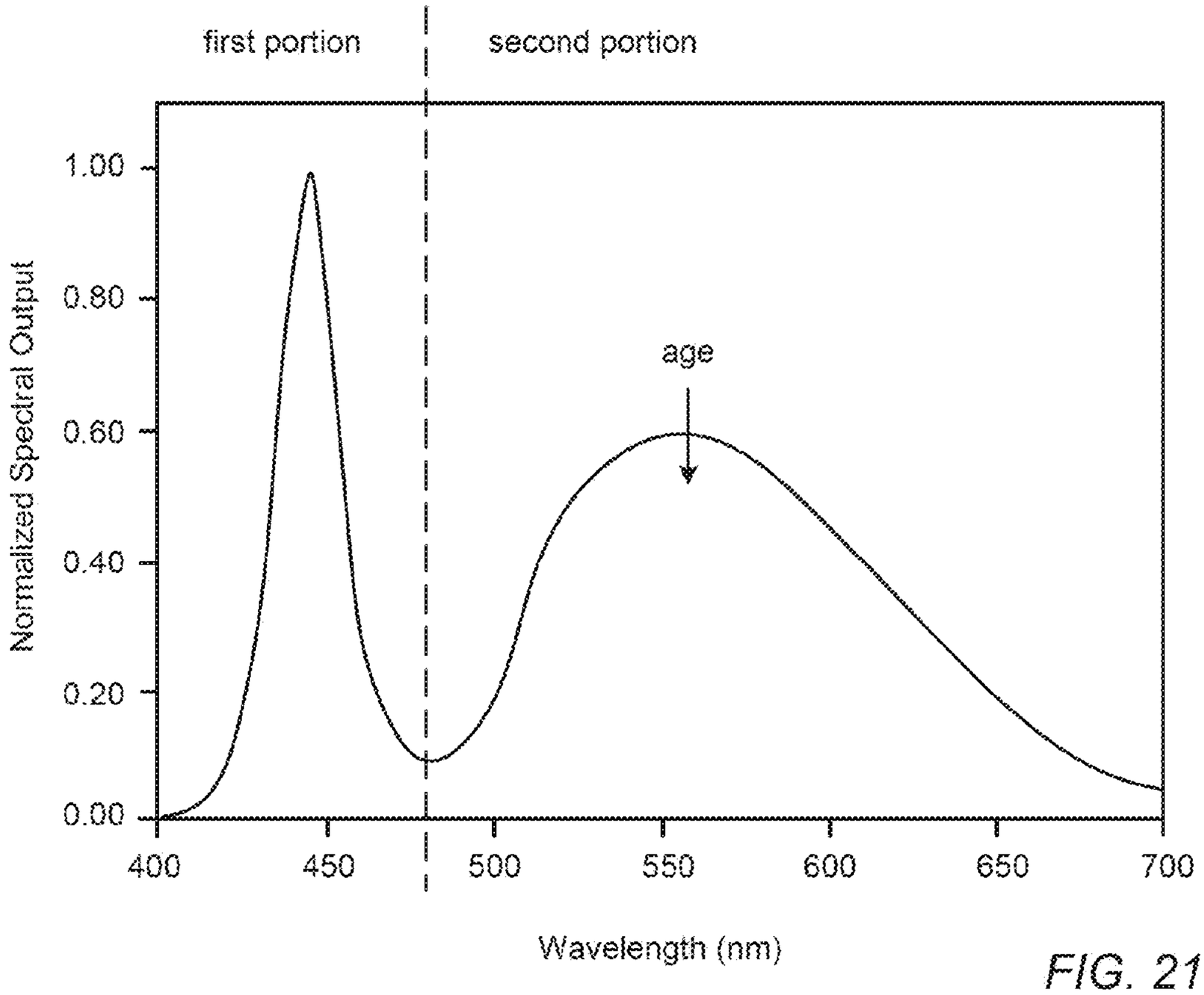

first portion
second portion
Normalized Spectral Output
1.00
0.80
0.60
0.40
0.20
0.00
age
400
450
500
550
600
650
700
Wavelength (nm)

FIG. 21

ILLUMINATION DEVICE AND METHOD FOR CALIBRATING AN ILLUMINATION DEVICE OVER CHANGES IN TEMPERATURE, DRIVE CURRENT, AND TIME

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/669,976 (now U.S. patent Ser. No. 11/723,134), which is a continuation of U.S. application Ser. No. 16/819,497 (now U.S. Pat. No. 11,252,8050), which is a continuation of U.S. application Ser. No. 15/388,607 (now U.S. Pat. No. 10,595,372, issued Mar. 17, 2020), which is a divisional of U.S. application Ser. No. 14/314,451 (now U.S. Pat. No. 9,557,214). U.S. application Ser. No. 14/314,451 is related to commonly assigned U.S. application Ser. No. 13/970,944 (now U.S. Pat. No. 9,237,620), Ser. No. 13/970,964 (now U.S. Pat. No. 9,651,632), and Ser. No. 13/970,990 (now U.S. Pat. No. 9,578,724), U.S. Provisional Application No. 61/886,471, and U.S. application Ser. No. 14/097,339 (now U.S. Pat. No. 9,360,174, issued Jun. 7, 2016). The entirety of each of these applications is incorporated herein by reference.

BACKGROUND

Field

This invention relates to illumination devices comprising a plurality of light-emitting diodes (LEDs) and, more particularly, to illumination devices and methods for calibrating individual LEDs in the illumination device to obtain a desired luminous flux and a desired chromaticity of the device over changes in drive current, temperature, and over time as the LEDs age.

Description of the Relevant Art

The following descriptions and examples are provided as background only and are intended to reveal information that is believed to be of possible relevance to the present invention. No admission is necessarily intended, or should be construed, that any of the following information constitutes prior art impacting the patentable character of the subject matter claimed herein.

Lamps and displays using LEDs (light-emitting diodes) for illumination are becoming increasingly popular in many different markets. LEDs provide a number of advantages over traditional light sources, such as incandescent and fluorescent light bulbs, including low power consumption, long lifetime, no hazardous materials, and additional specific advantages for different applications. When used for general illumination, LEDs provide the opportunity to adjust the color (e.g., from white, to blue, to green, etc.) or the color temperature (e.g., from "warm white" to "cool white") to produce different lighting effects.

Although LEDs have many advantages over conventional light sources, one disadvantage of LEDs is that their output characteristics (e.g., luminous flux and chromaticity) vary over changes in drive current, temperature, and over time as the LEDs age. These effects are particularly evident in multi-colored LED illumination devices, which combine a number of differently-colored emission LEDs into a single package.

An example of a multi-colored LED illumination device is one in which two or more different colors of LEDs are combined within the same package to produce white or near-white light. There are many different types of white light lamps on the market, some of which combine red, green, and blue (RGB) LEDs, red, green, blue, and yellow (RGBY) LEDs, phosphor-converted white and red (WR) LEDs, RGBW LEDs, etc. By combining different colors of LEDs within the same package and driving the differently-colored LEDs with different drive currents, these lamps may be configured to generate white or near-white light within a wide gamut of color points or correlated color temperatures (CCTs) ranging from "warm white" (e.g., roughly 2600 K-3700 K), to "neutral white" (e.g., 3700 K-5000 K) to "cool white" (e.g., 5000 K-8300 K). Some multi-colored LED illumination devices also enable the brightness and/or color of the illumination to be changed to a particular set point. These tunable illumination devices should all produce the same color and color rendering index (CRI) when set to a particular dimming level and chromaticity setting (or color set point) on a standardized chromaticity diagram.

A chromaticity diagram maps the gamut of colors the human eye can perceive in terms of chromaticity coordinates and spectral wavelengths. The spectral wavelengths of all saturated colors are distributed around the edge of an outlined space (called the "gamut" of human vision), which encompasses all the hues perceived by the human eye. The curved edge of the gamut is called the spectral locus and corresponds to monochromatic light, with each point representing a pure hue of a single wavelength. The straight edge on the lower part of the gamut is called the line of purples. These colors, although they are on the border of the gamut, have no counterpart in monochromatic light. Less saturated colors appear in the interior of the figure, with white and near-white colors near the center.

In the 1931 CIE Chromaticity Diagram, colors within the gamut of human vision are mapped in terms of chromaticity coordinates (x, y). For example, a red (R) LED with a peak wavelength of 625 nm may have a chromaticity coordinate of (0.69, 0.31), a green (G) LED with a peak wavelength of 528 nm may have a chromaticity coordinate of (0.18, 0.73), and a blue (B) LED with a peak wavelength of 460 nm may have a chromaticity coordinate of (0.14, 0.04). The chromaticity coordinates (i.e., color points) that lie along the blackbody locus obey Planck's equation, $E(\lambda)=A\lambda^{-5}/(e^{(B/T)}-1)$. Color points that lie on or near the blackbody locus provide a range of white or near-white light with color temperatures ranging between approximately 2500 K and 10,000 K. These color points are typically achieved by mixing light from two or more differently colored LEDs. For example, light emitted from the RGB LEDs shown in FIG. **1** may be mixed to produce a substantially white light with a color temperature in the range of about 2500 K to about 5000 K.

Although an illumination device is typically configured to produce a range of white or near-white color temperatures arranged along the blackbody curve (e.g., about 2500 K to 5000 K), some illumination devices may be configured to produce any color within the color gamut 18 (triangle) formed by the individual LEDs (e.g., RGB). The chromaticity coordinates of the combined light, e.g., (0.437, 0.404) for 3000 K white light, define the target chromaticity or color set point at which the device is intended to operate. In some devices, the target chromaticity or color set point may be changed by altering the ratio of drive currents supplied to the individual LEDs.

In general, the target chromaticity of the illumination device may be changed by adjusting the drive current levels (in current dimming) or duty cycle (in PWM dimming) supplied to one or more of the emission LEDs. For example, an illumination device comprising RGB LEDs may be configured to produce "warmer" white light by increasing the drive current supplied to the red LEDs and decreasing the drive currents supplied to the blue and/or green LEDs. Since adjusting the drive currents also affects the lumen output and temperature of the illumination device, the target chromaticity must be carefully calibrated and controlled to ensure that the actual chromaticity equals the target value. Most prior art illumination devices fail to provide an accurate calibration and compensation method for controlling the color of the illumination device.

Some prior art illumination devices also provide dimming capabilities, i.e., the ability to change the brightness or luminous flux output from the emission LEDs, in addition to (or instead of) color tuning. In most cases, the dimming level is changed by adjusting the drive current levels (in current dimming) or the duty cycle of the drive currents (in PWM dimming) supplied to all emission LEDs to produce a target dimming level. However, adjusting the supplied drive currents changes the chromaticity of the illumination, and this change in chromaticity differs for different LED devices and different dimming methods. For example, the chromaticity of an RGB LED illumination device may change rather significantly with changes drive current level and duty cycle, while the chromaticity of a phosphor-converted white LED illumination device is more consistent. In order to maintain a consistent target chromaticity, a range of target chromaticity values must be carefully calibrated over a range of target dimming levels.

In practice, the lumen output and chromaticity produced by prior art illumination devices often differs from the target dimming level and target chromaticity setting, due to changes in temperature and over time as the LEDs age. In general, changes in temperature affect the lumen output and chromaticity of all phosphor-converted and non-phosphor-converted LEDs. While prior art devices may perform some level of temperature compensation, they fail to provide accurate results by failing to recognize that temperature affects the lumen output and chromaticity of different colors of LEDs differently. Moreover, these prior art devices fail to account for chromaticity shifts in the illumination produced by phosphor-converted LEDs, which result from phosphor aging. As a consequence, these prior art devices cannot maintain a desired luminous flux and a desired chromaticity for an LED illumination device over operating conditions and over the lifetime of the illumination device.

A need remains for improved illumination devices and methods for calibrating and compensating individual LEDs within an LED illumination device, so as to accurately maintain a desired luminous flux and a desired chromaticity for the illumination device over changes in temperature, changes in drive current, and over time, as the LEDs age. This need is particularly warranted in multi-color LED illumination devices, since different colors of LEDs are affected differently by temperature and age, and in tunable illumination devices that enable the target dimming level and/or the target chromaticity setting to be changed by adjusting the drive currents supplied to one or more of the LEDs, since changes in drive current inherently affect the lumen output, color, and temperature of the illumination device.

SUMMARY

The following description of various embodiments of an illumination device and a method for calibrating an illumination device is not to be construed in any way as limiting the subject matter of the appended claims.

According to one embodiment, a method is provided herein for calibrating individual light-emitting diodes (LEDs) in an LED illumination device, so that a desired luminous flux and a desired chromaticity of the device can be maintained over changes in drive current, temperature, and over time as the LEDs age. In general, the illumination device described herein may include a plurality of emission LEDs, or a plurality of chains of emission LEDs, and at least one dedicated photodetector. For the sake of simplicity, the term "LED" will be used herein to refer to a single LED or a chain of serially-connected LEDs supplied with the same drive current. Although the calibration method described below refers only to a first LED, the method may be performed for each emission LED included within the illumination device.

According to one embodiment, the calibration method described herein may include subjecting the illumination device to a first ambient temperature and successively applying at least three different drive currents to a first LED of the illumination device to produce illumination at three or more different levels of brightness. At each of the at least three different drive currents, the calibration method may obtain a plurality of optical measurements from the illumination produced by the first LED and a plurality of electrical measurements from the at least one dedicated photodetector. In some embodiments, the plurality of optical measurements may comprise a plurality of luminous flux and chromaticity measurements, which are obtained from the illumination produced by the first LED at each of the at least three different drive currents.

In some embodiments, the plurality of electrical measurements may comprise a plurality of photocurrents, which are induced on the photodetector by the illumination produced by the first LED at each of the at least three different drive currents, and a plurality of forward voltages that are measured across the photodetector before, during, or after each induced photocurrent is measured. The forward voltages are generally measured across the photodetector by applying a non-operative drive current to the photodetector. The non-operative drive current applied to the photodetector for measuring forward voltages may range between approximately 100 μA and approximately 1 mA, depending on the number of photodetectors included within the illumination device, the size of the photodetectors, and the manner in which they are connected.

In some embodiments, the method may also measure a forward voltage that develops across the anode and cathode of the first LED by applying a non-operative drive current to the first LED sometime before, during, or after each induced photocurrent is measured from the photodetector. The non-operative drive current applied to the first LED for measuring forward voltage may range between approximately 1 mA and approximately 10 mA, depending on the size of the first LED.

In general, the drive currents supplied to the first LED for obtaining luminous flux, chromaticity, and induced photocurrent measurements may be operative drive current levels (e.g., about 20 mA to about 500 mA), and thus, may be substantially greater than the non-operative drive current (e.g., about 0.1 mA to about 1 mA) supplied to the photodetector and the non-operative drive current (e.g., about 0.1 mA to about mA) supplied to the first LED to measure forward voltage. In some cases, increasingly greater drive current levels may be successively applied to the first LED to obtain the luminous flux, chromaticity, and induced

photocurrent measurements. In other cases, the luminous flux, chromaticity, and induced photocurrent measurements may be obtained upon successively applying decreasing levels of drive current to the first LED. The order in which the drive currents are applied during the optical and electrical measurements is largely unimportant, only that the drive currents be different from one another.

In one embodiment, three luminous flux measurements, three x-chromaticity measurements, three y-chromaticity measurements, and three induced photocurrent measurements may be obtained from the first LED upon successively applying a first operative drive current, a second operative drive current that is substantially less than the first operative drive current, and a third operative drive current that is substantially less than the second operative drive current, to the first LED. In one example, the first operative drive current may be substantially equal to a maximum drive current associated with the first LED (e.g., about 500 mA, depending on LED part number and manufacturer), the second operative drive current may be about 30% of the maximum drive current, and the third operative drive current may be about 10% of the maximum drive current. Although examples are provided herein, the calibration method is not limited to any particular value or any particular number of drive current levels and may apply substantially any value and/or any number of drive current levels to the first LED within the operating current level range of that LED. In addition, the calibration method is not limited measuring chromaticity in only the CIE 1931 XYZ color space and may additionally or alternatively measure chromaticity in any color space that can be used to describe the gamut of an LED illumination device comprising combinations of emission LEDs as described herein.

In some embodiments, the first LED may be a phosphor-converted LED and the calibration method may be configured for calibrating separate portions of the phosphor-converted LED spectrum, as if it were two separate LEDs. For example, a spectrum of the illumination emitted from the phosphor-converted LED may include a first portion having a first peak emission wavelength (e.g., about 400-500 nm) and a second portion having a second peak emission wavelength (e.g., about 500-650 nm), which differs from the first peak emission wavelength. In such embodiments, the calibration method may obtain a plurality of optical measurements (e.g., luminous flux, x-chromaticity, and y-chromaticity) from the illumination, which is separately produced by each portion of the phosphor-converted LED spectrum at each of the at least three different drive currents.

For a phosphor-converted LED, the calibration method may also obtain a plurality of electrical measurements (e.g., induced photocurrents and forward voltages) from at least two different photodetectors at each of the at least three different drive currents. Specifically, a first photodetector (e.g., a green photodetector) with a first detection range may be used for measuring the photocurrent induced in the first photodetector by the first portion of the spectrum emitted by the phosphor-converted LED. Likewise, a second photodetector (e.g., a red photodetector) with a second detection range may be used for measuring the photocurrent induced in the second photodetector by the second portion of the spectrum emitted by the phosphor-converted LED. Sometime before, during, or after each induced photocurrent is measured, a non-operative drive current may be supplied to the first and second photodetectors to measure the forward voltages developed there across.

Sometime after the optical and electrical measurement values are obtained at the first ambient temperature, the results of the obtaining steps may be stored within the illumination device to calibrate the first LED at the first temperature. In one embodiment, the optical and electrical measurement values obtained at the first ambient temperature may be stored within a table of calibration values.

In some cases, the calibration method may continue by subjecting the illumination device to a second ambient temperature, which is different from the first ambient temperature, and repeating the steps of successively applying at least three different drive currents to the first LED, obtaining a plurality of optical measurements from the illumination produced by the first LED at each of the at least three different drive currents, obtaining a plurality of electrical measurements from the photodetector, and storing the results of the obtaining steps within the illumination device to calibrate the first LED at the second ambient temperature. In one embodiment, the optical and electrical measurement values obtained at the second ambient temperature may also be stored within the table of calibration values.

In one embodiment, the second ambient temperature may be substantially less than the first ambient temperature. For example, the second ambient temperature may be approximately equal to room temperature (e.g., roughly 25° C.), and the first ambient temperature may be substantially greater than room temperature. In one example, the first ambient temperature may be closer to an elevated temperature (e.g., roughly 70° C.) or a maximum temperature (e.g., roughly 85° C.) at which the device is expected to operate. In an alternative embodiment, the second ambient temperature may be substantially greater than the first ambient temperature.

It is worth noting that the exact values, number, and order in which the temperatures are applied to calibrate the first LED is somewhat unimportant. However, it is generally desired to obtain the luminous flux and chromaticity measurements from the first LED and the photocurrent measurements from the photodetectors at a sufficient number of different drive current levels, so that the non-linear relationship between these measurements and drive current can be accurately characterized across the operating current level range of the first LED. In one exemplary embodiment, the illumination device may be subjected to two (or more) substantially different ambient temperatures selected from across the operating temperature range of the illumination device.

According to another embodiment, an illumination device is provided herein with a plurality of light-emitting diode (LED) chains, which are configured to produce illumination for the illumination device and a photodetector, which is configured for detecting the illumination produced by each of the plurality of LED chains. In general, each LED chain may include one or more LEDs of the same color. In some embodiments, each LED chain may be configured for producing illumination with a different peak emission wavelength. In other embodiments, two or more LED chains may be configured for producing illumination with the same peak emission wavelength.

The illumination device may further comprise an LED driver and receiver circuit, which is configured for successively applying at least three different drive currents to each of the LED chains, one LED chain at a time, to produce illumination at different levels of brightness when the LED chains are subjected to a first ambient temperature, measuring photocurrents induced on the photodetector by the illumination produced by each of the LED chains at each of the at least three different drive currents, and measuring forward voltages that develop across the photodetector

before, during, or after each induced photocurrent is measured. In addition, the LED driver and receiver circuit is also configured for repeating the steps of successively applying at least three different drive currents to each of the LED chains, measuring photocurrents induced on the photodetector, and measuring forward voltages that develop across the photodetector when the LED chains are subjected to a second ambient temperature, which is different from the first ambient temperature.

The illumination device may further comprise an interface that is configured for receiving luminous flux and chromaticity measurements, which are measured by an external calibration tool upon receiving the illumination produced by each of the LED chains at each of the at least three different drive currents when the LED chains are subjected to the first and second ambient temperatures. In one embodiment, the interface may be a wireless interface configured to communicate using radio frequency (RF), infrared (IR) light, or visible light. For example, the wireless interface may be configured to operate according to at least one of the ZigBee, Wi-Fi, or Bluetooth communication protocols. In another embodiment, the interface may be a wired interface, which is configured to communicate over an AC mains, a dedicated conductor, or a set of conductors.

The illumination device may further comprise a storage medium, which is configured for storing the photocurrent, forward voltage, luminous flux, and chromaticity measurements obtained for each of the LED chains within a table of calibration values. In some embodiments, the photocurrent, forward voltage, luminous flux, and chromaticity measurements stored within the storage medium may comprise measured values, which are obtained from the LED driver and receiver circuit and the external calibration tool. In other embodiments, the measured values may be used to calculate calibration coefficients, which are stored within the storage medium in addition to, or in lieu of, the received calibration values.

According to one embodiment, the table of calibration values may include, for each LED chain, a first plurality of photocurrent values obtained from the photodetector upon receiving illumination from the LED chain when the LED chain is subjected to the first ambient temperature and the at least three different drive currents are successively applied to the LED chain, a second plurality of photocurrent values obtained from the photodetector upon receiving illumination from the LED chain when the LED chain is subjected to the second ambient temperature and the at least three different drive currents are successively applied to the LED chain, and a plurality of forward voltage values measured across the photodetector before, during, or after each photocurrent value is measured.

In some embodiments, the LED driver and receiver circuit may be further configured for applying a non-operative drive current to each LED chain and for measuring a forward voltage that develops across each LED chain before, during, or after each induced photocurrent is measured. In such embodiments, the table of calibration values may include, for each LED chain, a first plurality of forward voltage values measured across the LED chain upon applying the non-operative drive current to the LED chain when the LED chain is subjected to the first ambient temperature, and a second plurality of forward voltage values measured across the LED chain upon applying the non-operative drive current to the LED chain when the LED chain is subjected to the second ambient temperature.

In some embodiments, the table of calibration values may further include, for each LED chain, a first plurality of luminous flux values detected from the LED chain upon applying the at least three different drive currents to the LED chain when the LED chain is subjected to the first ambient temperature, and a second plurality of luminous flux values detected from the LED chain upon applying the at least three different drive currents to the LED chain when the LED chain is subjected to the second ambient temperature.

In some embodiments, the table of calibration values may further include, for each LED chain, a first plurality of x-chromaticity values and a first plurality of y-chromaticity values, which are detected from the LED chain upon applying the at least three different drive currents to the LED chain when the LED chain is subjected to the first ambient temperature, and a second plurality of x-chromaticity values and a second plurality of y-chromaticity values, which are detected from the LED chain upon applying the at least three different drive currents to the LED chain when the LED chain is subjected to the second ambient temperature. However, the table of calibration values is not limited to storing chromaticity vales from only the CIE 1931 XYZ color space and may additionally or alternatively store chromaticity values from any color space that can be used to describe the gamut of an LED illumination device comprising combinations of emission LEDs as described herein.

In some embodiments, the illumination device may comprise an additional photodetector, which is configured for detecting the illumination produced by a subset of the plurality of LED chains. In one example, the subset of LED chains may comprise a chain of phosphor-converted LEDs (e.g., a chain of phosphor-converted white LEDs) and a chain of reference LEDs (e.g., a chain of blue LEDs).

In such embodiments, the LED driver and receiver circuit may be further configured for measuring photocurrents induced on the additional photodetector by the illumination individually produced by the chain of phosphor-converted LEDs and the chain of reference LEDs at each of the at least three different drive currents when the LED chains are subjected to the first and second ambient temperatures, and measuring forward voltages that develop across the additional photodetector before, during, or after each induced photocurrent is measured.

In such embodiments, the table of calibration values may further include, for each LED chain within the subset, a first plurality of photocurrent values obtained from the additional photodetector upon receiving illumination from the LED chain when the LED chain is subjected to the first ambient temperature and the at least three different drive currents are successively applied to the LED chain, a second plurality of photocurrent values obtained from the additional photodetector upon receiving illumination from the LED chain when the LED chain is subjected to the second ambient temperature and the at least three different drive currents are successively applied to the LED chain, and a plurality of forward voltage values measured across the additional photodetector before, during, or after each photocurrent value is measured. The induced photocurrents and forward voltages measured across the additional photodetector may be stored within the storage medium as measured values or calculated coefficients.

BRIEF DESCRIPTION OF THE DRAWINGS

Other objects and advantages of the invention will become apparent upon reading the following detailed description and upon reference to the accompanying drawings.

FIG. **1** is a graph of the 1931 CIE chromaticity diagram illustrating the gamut of human color perception and the gamut achievable by an illumination device comprising a plurality of multiple color LEDs (e.g., red, green, and blue);

FIG. **2** is a graph illustrating the non-linear relationship between relative luminous flux and junction temperature for white, blue, and green LEDs;

FIG. **3** is a graph illustrating the substantially more non-linear relationship between relative luminous flux and junction temperature for red, red-orange, and yellow (amber) LEDs;

FIG. **4** is a graph illustrating the non-linear relationship between relative luminous flux and drive current for red and red-orange LEDs;

FIG. **5** is a graph illustrating the substantially more non-linear relationship between relative luminous flux and drive current for white, blue, and green LEDs;

FIG. **6** is a flow chart diagram of an improved method for calibrating an illumination device comprising a plurality of LEDs and one or more photodetectors, in accordance with one embodiment of the invention;

FIG. **7** is a chart illustrating an exemplary table of calibration values that may be obtained in accordance with the calibration method of FIG. **6** and stored within the illumination device;

FIG. **8** is a flowchart diagram of an improved compensation method, in accordance with one embodiment of the invention;

FIG. **9** is an exemplary timing diagram for an illumination device comprising four emission LEDs, illustrating the periodic intervals during which measurements (e.g., emitter forward voltage) are obtained from each emission LED, one LED at a time;

FIG. **10** is a graphical representation depicting how one or more interpolation techniques may be used in the compensation method of FIG. **8** to determine the drive current needed to produce a desired luminous flux for a given LED using the calibration values obtained during the calibration method of FIG. **6** and stored within the illumination device;

FIG. **11** is a graphical representation depicting how one or more interpolation techniques may be used in the compensation method of FIG. **8** to determine the expected x-chromaticity value for a given LED using the present forward voltage, the present drive current and the calibration values obtained during the calibration method of FIG. **6** and stored within the illumination device;

FIG. **12** is a graphical representation depicting how one or more interpolation techniques may be used in the compensation method of FIG. **8** to determine the expected y-chromaticity value for a given LED using the present forward voltage, the present drive current, and the calibration values obtained during the calibration method of FIG. **6** and stored within the illumination device;

FIG. **13** is a flowchart diagram of an improved compensation method, in accordance with another embodiment of the invention;

FIG. **14** is an exemplary timing diagram for an illumination device comprising four emission LEDs, illustrating the periodic intervals during which measurements are obtained from the one or more photodetectors (e.g., induced photocurrent and detector forward voltage) and from each emission LED, one LED at a time (e.g., emitter forward voltage);

FIG. **15** is a graphical representation depicting how one or more interpolation techniques may be used in the compensation method of FIG. **13** to determine the expected photocurrent value for a given LED using the present forward voltage, the present drive current, and the calibration values obtained during the calibration method of FIG. **6** and stored within the illumination device;

FIG. **16**A is a perspective view of an exemplary illumination device;

FIG. **16**B is a top view of an exemplary emitter module that may be included within the exemplary illumination device of FIG. **16**A;

FIG. **17**A is a perspective view of another exemplary illumination device;

FIG. **17**B is a top view of an exemplary emitter module that may be included within the exemplary illumination device of FIG. **17**A;

FIG. **18**A is a side view of an improved emitter module, according to one embodiment of the invention;

FIG. **18**B is a side view of an improved emitter module, according to another embodiment of the invention;

FIG. **19** is an exemplary block diagram of circuit components that may be included within an illumination device, according to one embodiment of the invention;

FIG. **20** is an exemplary block diagram of an LED driver and receiver circuit that may be included within the illumination device of FIG. **19**, according to one embodiment of the invention; and

FIG. **21** is an exemplary graph depicting how the spectrum of a phosphor-converted LED may be divided into two portions and showing how the phosphor efficiency decreases as the phosphor ages.

While the invention is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description thereto are not intended to limit the invention to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present invention as defined by the appended claims.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

An LED generally comprises a chip of semiconducting material doped with impurities to create a p-n junction. As in other diodes, current flows easily from the p-side, or anode, to the n-side, or cathode, but not in the reverse direction. Charge-carriers—electrons and holes—flow into the junction from electrodes with different voltages. When an electron meets a hole, it falls into a lower energy level, and releases energy in the form of a photon (i.e., light). The wavelength of the light emitted by the LED, and thus its color, depends on the band gap energy of the materials forming the p-n junction of the LED.

Red and yellow LEDs are commonly composed of materials (e.g., AlInGaP) having a relatively low band gap energy, and thus produce longer wavelengths of light. For example, most red and yellow LEDs have a peak wavelength in the range of approximately 610-650 nm and approximately 580-600 nm, respectively. On the other hand, green and blue LEDs are commonly composed of materials (e.g., GaN or InGaN) having a larger band gap energy, and thus, produce shorter wavelengths of light. For example, most green and blue LEDs have a peak wavelength in the range of approximately 515-550 nm and approximately 450-490 nm, respectively.

In some cases, a "white" LED may be formed by covering or coating, e.g., a blue LED having a peak emission wavelength of about 450-490 nm with a phosphor (e.g., YAG),

which down-converts the photons emitted by the blue LED to a lower energy level, or a longer peak emission wavelength, such as about 525 nm to about 600 nm. In some cases, such an LED may be configured to produce substantially white light having a correlated color temperature (CCT) of about 3000 K. However, a skilled artisan would understand how different colors of LEDs and/or different phosphors may be used to produce a "white" LED with a potentially different CCT.

When two or more differently-colored LEDs are combined within a single package, the spectral contents of the individual LEDs are combined to produce blended light. In some cases, differently-colored LEDs may be combined to produce white or near-white light within a wide gamut of color points or CCTs ranging from "warm white" (e.g., roughly 2600 K-3000 K), to "neutral white" (e.g., 3000 K-4000 K) to "cool white" (e.g., 4000 K-8300 K). Examples of white light illumination devices include, but are not limited to, those that combine red, green, and blue (RGB) LEDs, red, green, blue, and yellow (RGBY) LEDs, white and red (WR) LEDs, and RGBW LEDs.

The present invention is generally directed to illumination devices having a plurality of light-emitting diodes (LEDs) and one or more photodetectors. For the sake of simplicity, the term "LED" will be used throughout this disclosure to refer to a single LED, or a chain of serially-connected LEDs supplied with the same drive current. According to one embodiment, the present invention provides improved methods for calibrating and compensating individual LEDs within an LED illumination device, so as to accurately maintain a desired luminous flux and a desired chromaticity for the illumination device over changes in drive current, temperature, and/or time.

Although not limited to such, the present invention is particularly well suited to illumination devices (i.e., multi-colored illumination devices) in which two or more different colors of LEDs are combined to produce blended white or near-white light, since the output characteristics of differently-colored LEDs vary differently over drive current, temperature, and time. The present invention is also particularly well suited to illumination devices (i.e., tunable illumination devices) that enable the target dimming level and/or the target chromaticity setting to be changed by adjusting the drive currents supplied to one or more of the LEDs, since changes in drive current inherently affect the lumen output, color, and temperature of the illumination device.

FIGS. **2-3** illustrate how the relative luminous flux of an individual LED changes over junction temperature for different colors of LEDs. As shown in FIGS. **2-3**, the luminous flux output from all LEDs generally decreases with increasing temperature. For some colors (e.g., white, blue, and green), the relationship between luminous flux and junction temperature is relatively linear (see FIG. **2**), while for other colors (e.g., red, orange, and especially yellow) the relationship is significantly non-linear (see, FIG. **3**). The chromaticity of an LED also changes with temperature, due to shifts in the dominant wavelength (for both phosphor-converted and non-phosphor-converted LEDs) and changes in the phosphor efficiency (for phosphor-converted LEDs). In general, the peak emission wavelength of green LEDs tends to decrease with increasing temperature, while the peak emission wavelength of red and blue LEDs tends to increase with increasing temperature. While the change in chromaticity is relatively linear with temperature for most colors, red and yellow LEDs tend to exhibit a more significant non-linear change.

As LEDs age, the luminous flux output from both phosphor-converted and non-phosphor-converted LEDs, and the chromaticity of phosphor-converted LEDs, also changes over time. Early on in life, the luminous flux can either increase (get brighter) or decrease (get dimmer), while late in life, the luminous flux generally decreases. As expected, the lumen output decreases faster over time when the LEDs are subjected to higher drive currents and higher temperatures. As a phosphor-converted LED ages, the phosphor becomes less efficient and the amount of blue light that passes through the phosphor increases. This decrease in phosphor efficiency causes the overall color produced by the phosphor-converted LED to appear "cooler" over time. Although the dominant wavelength and chromaticity of a non-phosphor-converted LED does not change over time, the luminous flux decreases as the LED ages, which in effect causes the chromaticity of a multi-colored LED illumination device to change over time.

When differently-colored LEDs are combined within a multi-colored illumination device, the color point of the resulting device often changes significantly with variations in temperature and over time. For example, when red, green, and blue LEDs are combined within a white light illumination device, the color point of the device may appear increasingly "cooler" as the temperature rises. This is because the luminous flux produced by the red LEDs decreases significantly as temperatures increase, while the luminous flux produced by the green and blue LEDs remains relatively stable (see, FIGS. **2-3**).

To account for temperature and aging effects, some prior art illumination devices attempt to maintain a consistent lumen output and/or a consistent chromaticity over temperature and time by measuring characteristics of the emission LEDs and increasing the drive current supplied to one or more of the emission LEDs. For example, some prior art illumination devices measure the temperature of the illumination device (either directly through an ambient temperature sensor or heat sink measurement, or indirectly through a forward voltage measurement), and adjust the drive currents supplied to one or more of the emission LEDs to account for temperature-related changes in lumen output. Other prior art illumination devices measure the lumen output from individual emission LEDs, and if the measured value differs from a target value, the drive currents supplied to the emission LED are increased to account for changes in luminous flux that occur over time.

However, changing the drive currents supplied to the emission LEDs inherently affects the luminous flux and the chromaticity produced by the LED illumination device. FIGS. **4** and **5** illustrate the relationship between luminous flux and drive current for different colors of LEDs (e.g., red, red-orange, white, blue, and green LEDs). In general, the luminous flux increases with larger drive currents, and decreases with smaller drive currents. However, the change in luminous flux with drive current is non-linear for all colors of LEDs, and this non-linear relationship is substantially more pronounced for certain colors of LEDs (e.g., blue and green LEDs) than others. The chromaticity of the illumination also changes when drive currents are increased to combat temperature and/or aging effects, since larger drive currents inherently result in higher LED junction temperatures (see, FIGS. **2-3**). While the change in chromaticity with drive current/temperature is relatively linear for all colors of LEDs, the rate of change is different for different LED colors and even from part to part.

Although some prior art illumination devices may adjust the drive currents supplied to the emission LEDs, these

devices fail to provide accurate temperature and age compensation by failing to account for the non-linear relationship that exists between luminous flux and junction temperature for certain colors of LEDs (FIGS. **2-3**), the non-linear relationship that exists between luminous flux and drive current for all colors of LEDs (FIGS. **4-5**), and the fact that these relationships differ for different colors of LEDs. These devices also fail to account for the fact that the rate of change in chromaticity with drive current/temperature is different for different colors of LEDs. Without accounting for these behaviors, prior art illumination devices cannot provide accurate temperature and age compensation for all LEDs included within a multi-colored LED illumination device.

A need remains for improved illumination devices and methods for calibrating and compensating individual LEDs included within an illumination device, so as to maintain a desired luminous flux and a desired chromaticity over variations in drive current and temperature and over time, as the LEDs age. This need is particularly relevant to multi-colored LED illumination devices, since different LED colors respond differently over temperature and time, and to illumination devices that provide dimming and/or color tuning capabilities, since changes in drive current inherently affect the lumen output, color, and temperature of the illumination device.

To meet these needs, improved illumination devices and methods are provided herein to individually calibrate and compensate each LED used in the LED illumination device. The improved calibration and compensation methods described herein overcome the disadvantages of conventional methods, which fail to provide accurate temperature and age compensation for all LEDs included within an LED illumination device.

Exemplary Embodiments of Improved Methods for Calibrating an Illumination Device

FIG. **6** illustrates one embodiment of an improved method for calibrating an illumination device comprising a plurality of LEDs and at least one dedicated photodetector. In some embodiments, the calibration method shown in FIG. **6** may be used to calibrate an illumination device having LEDs all of the same color. However, the calibration method described herein is particularly well-suited for calibrating an illumination device comprising two or more differently-colored LEDs (i.e., a multi-colored LED illumination device), since output characteristics of differently-colored LEDs vary differently over drive current, temperature, and time. The calibration method described herein is also particularly well-suited for calibrating an illumination device that provides dimming and/or color tuning capabilities (i.e., a tunable LED illumination device), since changes in drive current inherently affect the lumen output, color, and temperature of the illumination device.

Exemplary embodiments of an improved illumination device will be described below with reference to FIGS. **16-20**, which show different types of LED illumination devices, each having one or more emitter modules. As described below, each emitter module may generally include a plurality of emission LEDs arranged in an array, and at least one dedicated photodetector spaced about a periphery of the array. In one exemplary embodiment, the array of emission LEDs may include red, green, blue, and white (or yellow) LEDs, and the at least one dedicated photodetector may include one or more red, orange, yellow, and/or green LEDs. However, the present invention is not limited to any particular color, number, combination, or arrangement of emission LEDs or photodetectors. A skilled artisan would understand how the method steps described herein may be applied to other LED illumination devices having substantially different emitter modules.

As shown in FIG. **6**, the improved calibration method may generally begin by subjecting the illumination device to a first ambient temperature (in step **10**). Once subjected to this temperature, a plurality of different drive current levels may be applied to the emission LEDs (in step **12**) and a plurality of measurements may be obtained from both the emission LEDs and the dedicated photodetector LED(s) at each of the different drive current levels (in steps **14** and **16**). Specifically, two or more different drive current levels may be successively applied to each emission LED, one LED at a time, for the purpose of obtaining measurements from the illumination device. These measurements may generally include optical measurements and electrical measurements.

For example, a plurality of optical measurements may be obtained from the illumination produced by each emission LED at each of the different drive current levels (in step **14**). According to one embodiment, the optical measurements may include a plurality of luminous flux, x-chromaticity and y-chromaticity measurements, which are obtained for each emission LED at two or more different drive current levels. However, the optical measurements described herein are not limited to luminous flux, a-chromaticity and y-chromaticity, and may include additional or alternative optical measurements in other embodiments of the invention.

In general, the chromaticity calibration values described herein may correspond to the CIE 1931 XYZ color space, the CIE 1931 RGB color space, the CIE 1976 LUV color space, and various other RGB color spaces (e.g., sRGB, Adobe RGB, etc.). Although the calibration and compensation methods described herein acquire and utilize only x- and y-chromaticity calibration values, one skilled in the art would understand how chromaticity values from other color spaces could be alternatively acquired and used in the methods described herein. As such, the calibration and compensation methods described herein and recited in the claims are considered to encompass chromaticity calibration values from any color space that can be used to describe the gamut of an LED illumination device comprising substantially any combination of emission LEDs as described herein.

In one preferred embodiment, three luminous flux (Luma) measurements, three x-chromaticity (x chrom) measurements, and three y-chromaticity (y chrom) measurements are measured from each emission LED at roughly a maximum drive current level (typically about 500 mA, depending on LED part number and manufacturer), roughly 30% of the maximum drive current, and roughly 10% of the maximum drive current, as shown in FIG. **7** and discussed below. In some embodiments, the luminous flux and x-, y-chromaticity measurements may be obtained from the emission LEDs using an external calibration tool, such as a spectrophotometer. In some embodiments, the measurement values obtained from the external calibration tool may be transmitted wirelessly to the illumination device, as described in more detail below with respect to FIG. **19**.

In addition, a plurality of electrical measurements may be obtained from each of the emission LEDs and each of the dedicated photodetectors at each of the different drive current levels (in step **16**). These electrical measurements may include, but are not limited to, photocurrents induced on the dedicated photodetectors and forward voltages measured across the dedicated photodetectors and/or the emission

LEDs. Unlike the optical measurements described above, the electrical measurements may be obtained from the dedicated photodetectors and the emission LEDs using the LED driver and receiver circuit included within the illumination device. An exemplary embodiment of such a circuit is shown in FIGS. **19-20** and described in more detail below.

At each of the different drive currents levels, the LED driver and receiver circuit measures the photocurrents that are induced on the dedicated photodetector by the illumination individually produced by each emission LED. In one preferred embodiment, three photocurrent (Iph_d1) measurements may be obtained from the dedicated photodetector for each emission LED when the emission LEDs are successively driven to produce illumination at three different drive current levels (e.g., 100%, 30% and % of a max drive level). In some embodiments, the LED driver and receiver circuit may obtain the photocurrent (Iph_d1) measurements at substantially the same time the external calibration tool is measuring the luminous flux and x- and y-chromaticity of the illumination produced by the emission LEDs at each of the different drive current levels.

In general, the drive currents applied to the emission LEDs to measure luminous flux, chromaticity, and induced photocurrent may be operative drive current levels (e.g., about 20 mA to about 500 mA). In some cases, increasingly greater drive current levels may be successively applied to each of the emission LEDs to obtain the measurements described herein. In other cases, the measurements may be obtained upon successively applying decreasing levels of drive current to the emission LEDs. The order in which the drive current levels are applied is largely unimportant, only that the drive currents be different from one another.

Although examples are provided herein, the present invention is not limited to any particular value or any particular number of drive current levels and may apply substantially any value and any number of drive current levels to an emission LED within the operating current level range of that LED. However, it is generally desired to obtain the luminous flux and chromaticity measurements from the emission LEDs and the photocurrent measurements from the photodetector at a sufficient number of different drive current levels, so that the non-linear relationship between these measurements and drive current can be accurately characterized across the operating current level range of the LED.

While increasing the number of measurements does improve the accuracy with which the non-linear relationships are characterized, it also increases the calibration time and costs. While the increase in calibration time and cost may not be warranted in all cases, it may be beneficial in some. For example, additional luminous flux measurements may be beneficial when attempting to characterize the luminous flux vs. drive current relationship for certain colors of LEDs (e.g., blue and green LEDs), which tend to exhibit a significantly more non-linear relationship than other colors of LEDs (see, FIGS. **4-5**). Thus, a balance should be struck between accuracy and calibration time/costs when selecting a desired number of drive current levels with which to obtain measurements for a particular color of LED.

Since increasing drive currents affect the junction temperature of the emission LEDs, a forward voltage may be measured across each emission LED and each photodetector immediately after each operative drive current level is supplied to the emission LEDs (in step **16**). For each operative drive current level, the forward voltages can be measured across each emission LED and each photodetector before or after the photocurrent measurements for that operative drive current level are obtained. Unlike the optical measurements, however, relatively small drive currents are applied to the emission LEDs and the dedicated photodetectors to measure the forward voltages developed there across.

In one preferred embodiment, three forward voltage (Vfe) measurements may be obtained from each emission LED and three forward voltage (Vfd1) measurements may be obtained from each dedicated photodetector (in step **16**) immediately after each of the different drive current levels (e.g., 100%, 30% and 10% of a max drive level) is applied to the emission LEDs to measure the luminous flux, x-chromaticity and y-chromaticity. The forward voltage (Vfe and Vfd1) measurements can be obtained before or after the induced photocurrents (Iph_d1) are measured at each of the different drive current levels. By measuring the forward voltage (Vfe) across each emission LED and the forward voltage (Vfd1) across each dedicated photodetector immediately after each operative drive current level is applied to the emission LEDs, the Vfe and Vfd1 measurements may be used to provide a good indication of how the junction temperature of the emission LEDs and the dedicated photodetector change with changes in drive current.

When taking forward voltage measurements, a relatively small drive current is supplied to each of the emission LEDs and each of the dedicated photodetector LEDs, one LED at a time, so that a forward voltage (Vfe or Vfd1) developed across the anode and cathode of the individual LEDs can be measured (in step **16**). When taking these measurements, all other emission LEDs in the illumination device are preferably turned "off" to avoid inaccurate forward voltage measurements (since light from other emission LEDs would induce additional photocurrents in the LED being measured).

As used herein, a "relatively small drive current" may be broadly defined as a non-operative drive current, or a drive current level which is insufficient to produce significant illumination from the LED. Most LED device manufacturers, which use forward voltage measurements to compensate for temperature variations, supply a relatively large drive current to the LEDs (e.g., an operative drive current level sufficient to produce illumination from the LEDs) when taking forward voltage measurements. Unfortunately, forward voltages measured at operative drive current levels tend to vary significantly over the lifetime of an LED. As an LED ages, the parasitic resistance within the junction increases, which in turn causes the forward voltage measured at operating current levels to increase over time, regardless of temperature. For this reason, a relatively small (i.e., non-operative) drive current is used herein when obtaining forward voltage measurements to limit the resistive portion of the forward voltage drop.

For some common types of emission LEDs with one square millimeter of junction area, the optimum drive current used herein to obtain forward voltage measurements from the emission LEDs may be roughly 0.1-10 mA, and more preferably may be about 0.3-3 mA. In one embodiment, the optimum drive current level may be about 1 mA for obtaining forward voltage measurements from the emission LEDs. However, smaller/larger LEDs may use proportionally less/more current to keep the current density roughly the same. In the embodiments that use a significantly smaller LED as the dedicated photodetector, the optimum drive current level for obtaining forward voltage measurements from a single photodetector may range between about 100 μA to about 300 μA. In one embodiment, the optimum drive current level used for obtaining forward voltage measurements from a plurality of dedicated photo-

detectors connected in parallel may be about 1 mA. The relatively small, non-operative drive currents used to obtain forward voltage measurements from the emission LEDs (e.g., about 0.3 mA to about 3 mA) and the relatively small, non-operative drive currents used to obtain forward voltage measurements from a dedicated photodetector (e.g., about 100 μA to about 300 μA) are substantially smaller than the operative drive current levels (e.g., about 20 mA to about 500 mA) used in steps **14** and **16** to measure luminous flux, chromaticity, and induced photocurrent.

After the measurements described in steps **14-16** are obtained at the first temperature, the illumination device is subjected to a second ambient temperature, which is substantially different from the first ambient temperature (in step **18**). Once subjected to this second temperature, steps **12-16** are repeated (in step **20**) to obtain an additional plurality of optical measurements from each of the emission LEDs (in step **14**), and an additional plurality of electrical measurements from the emission LEDs and the dedicated photodetector (in step **16**). The additional measurements may be obtained at the second ambient temperature in the same manner described above for the first ambient temperature.

In one embodiment, the second ambient temperature may be substantially less than the first ambient temperature. For example, the second ambient temperature may be approximately equal to room temperature (e.g., roughly 25° C.), and the first ambient temperature may be substantially greater than room temperature. In one example, the first ambient temperature may be closer to an elevated temperature (e.g., roughly 70° C.) or a maximum temperature (e.g., roughly 85° C.) at which the device is expected to operate. In an alternative embodiment, the second ambient temperature may be substantially greater than the first ambient temperature.

It is worth noting that the exact values, number, and order in which the temperatures are applied to calibrate the individual LEDs is somewhat unimportant. However, it is generally desired to obtain the luminous flux, x- and y-chromaticity, and photocurrent calibration values at a number of different temperatures, so that the non-linear relationships between these measurements and drive current can be accurately characterized across the operating temperature range of each LED. In one preferred embodiment, the illumination device may be subjected to two substantially different ambient temperatures, which are selected from across the operating temperature range of the illumination device. While it is possible to obtain the measurements described herein at three (or more) temperatures, doing so may add significant expense, complexity and/or time to the calibration process. For this reason, it is generally preferred that the emission LEDs and the dedicated photodetector(s) be calibrated at only two different temperatures (e.g., about 25° C. and about 70° C.).

In some embodiments, the illumination device may be subjected to the first and second ambient temperatures by artificially generating the temperatures during the calibration process. However, it is generally preferred that the first and second ambient temperatures are ones which occur naturally during production of the illumination device, as this simplifies the calibration process and significantly decreases the costs associated therewith. In one embodiment, the measurements obtained at the elevated temperature may be taken after burn-in of the LEDs when the illumination device is relatively hot (e.g., roughly 50° C. to 85° C.), and sometime thereafter (e.g., at the end of the manufacturing line), a room temperature calibration may be performed to obtain measurements when the illumination device is relatively cool (e.g., roughly 20° C. to 30° C.).

Once the calibration measurements are obtained, the calibration values may be stored within the illumination device (in step **22**), so that the stored values can be later used to compensate the illumination device for changes in luminous flux and/or chromaticity that may occur over variations in drive current, temperature, and time. In one embodiment, the calibration values may be stored within a table of calibration values as shown, for example, in FIG. **7**. The table of calibration values may be stored within a storage medium of the illumination device, as discussed below with reference to FIG. **19**.

FIG. **7** illustrates one embodiment of a calibration table that may be generated in accordance with the calibration method shown in FIG. **6**. In the illustrated embodiment, the calibration table includes six luminous flux measurements (Luma), six x-chromaticity measurements (x chrom), and six y-chromaticity measurements (y chrom), which were obtained from each emission LED (e.g., white, blue, green and red emission LEDs) at the three different drive currents (e.g., 10%, 30% and 100% of a max drive current) and the two different temperatures (T0, T1) in steps **10**, **12**, **14**, **18**, **20** and **22** of the calibration method. The calibration table shown in FIG. **7** also includes six photocurrent measurements (Iph_d1) that were induced on the photodetector by the illumination produced by each of the emission LEDs at the three different drive currents levels and the two different temperatures in steps **10**, **12**, **16**, **20** and **22** of the calibration method.

For each emission LED (e.g., each white, blue, green, and red emission LED) and each ambient temperature (T0, T1), the calibration table shown in FIG. **7** also includes the forward voltage (Vfe) that was measured across the emission LED and the forward voltage (Vfd1) that was measured across the dedicated photodetector immediately after each of the three different drive currents levels is supplied to the emission LEDs. In this example embodiment, steps **10**, **12**, **16**, **18**, **20** and **22** of the calibration method result in six Vfe measurements and six Vfd1 measurements being stored for each emission LED, as shown in FIG. **7**.

The calibration table shown in FIG. **7** represents only one example of the calibration values that may be stored within an LED illumination device, in accordance with the calibration method described herein. In some embodiments, the calibration method shown in FIG. **6** may be used to store substantially different calibration values, or substantially different numbers of calibration values, within the calibration table of the LED illumination device.

As noted above, the present invention is not limited to the exemplary number of drive current levels and values of drive current shown in FIGS. **6** and **7**. It is certainly possible to obtain a greater/lesser number of optical and electrical measurements from the emission LEDs and the at least one dedicated photodetector by applying a greater/lesser number of drive current levels to the emission LEDs. It is also possible to use substantially different values of drive current, other than the 10%, 30% and 100% of the max drive current illustrated in FIG. **7**.

It is also possible to obtain and store a different number of forward voltage (Vfe) measurements from the emission LEDs, or a different number of forward voltage (Vfd1) measurements from the at least one dedicated photodetector. For example, the calibration table shown in the embodiment of FIG. **7** stores six forward voltage (Vfe) measurements from each emission LED and six*n forward voltage (Vfd1) measurements from each dedicated photodetector, where 'n'

is the number of emission LEDs included within the illumination device. As noted above, the six Vfe measurements and six*n Vfd measurements are preferably obtained at two different ambient temperatures (T0, T1) immediately after each operative drive current level (e.g., 10%, 30% and 100% of a max drive current) is applied to each emission LED. Such an embodiment is generally preferred, as it provides a good indication of how the emitter and detector junction temperatures change with changes in ambient temperature and changes in drive current. In addition, such an embodiment enables the compensation method shown in FIG. **13** (and described below) to compensate for emitter aging when only detector forward voltages (Vfd1) are measured during operation of the device.

As shown in FIGS. **6-7** and described above, the calibration method may obtain only one Vfe and only one Vfd1 measurement for each emission LED at a given temperature (e.g., TO) and a given drive current (e.g., 10% of the max drive current). In one alternative embodiment, the calibration method of FIG. **6** may obtain a plurality of Vfe and a plurality of Vfd1 measurements for each emission LED at a given temperature (e.g., TO) and a given drive current (e.g., 10% of the max drive current). The plurality of Vfe and Vfd1 measurements may be obtained over a short period of time (e.g., 100 msec), and the plurality of Vfe measurements and the plurality of Vfd1 measurements obtained during each time period may be averaged and filtered before they are stored within the calibration table of FIG. **7**.

In another alternative embodiment, the calibration method of FIG. **6** may obtain only two forward voltage (Vfe) measurements from each emission LED, one for each of the two different temperatures (TO, T1), as described in commonly assigned U.S. patent application Ser. Nos. 13/970,944, 13/970,964 and 13/970,990. Likewise, only two*n forward voltage (Vfd1) measurements may be obtained from the dedicated photodetector, where 'n' is the number of emission LEDs included within the illumination device. In this embodiment, however, the forward voltage (Vfe and Vfd1) measurements stored in the calibration table would only provide an indication of how the emitter and detector junction temperatures change with changes in ambient temperature, not with drive current induced temperature changes.

In another alternative embodiment of the invention, the calibration method shown in FIG. **6** may omit the emitter forward voltage (Vfe) measurements altogether and rely solely on the photodetector forward voltage (Vfd1) measurements to provide an indication of temperature. However, the Vfe measurements may only be omitted if the temperature difference between the emission LEDs and the dedicated photodetector(s) remains relatively the same over the operating temperature range. To maintain a consistent temperature difference between the emission LEDs and the photodetector(s), an improved emitter module is provided herein and described below with reference to FIG. **18**A.

In yet another alternative embodiment of the invention, the calibration method shown in FIG. **6** may be used to obtain additional measurements, which may be later used to compensate for phosphor aging, and thereby, control the chromaticity of a phosphor-converted white LED over time.

As noted above, some embodiments of the invention may include a phosphor-converted white emission LED within the emitter module. These LEDs may be formed by coating or covering, e.g., a blue LED, having a peak emission wavelength of about 400-500 nm, with a phosphor material (e.g., YAG) to produce substantially white light with a CCT of about 3000 K. Other combinations of LEDs and phosphors may be used to form a phosphor-converted LED, which is capable of producing white or near-white light with a CCT in the range of about 2700 K to about 10,000 K.

In phosphor-converted LEDs, the spectral content of the LED combines with the spectral content of the phosphor to produce white or near-white light. As shown in FIG. **21**, the combined spectrum may include a first portion having a first peak emission wavelength (e.g., about 400-500 nm), and a second portion having a second peak emission wavelength (e.g., about 500-650 nm), which is substantially different from the first peak emission wavelength. In this example, the first portion of the spectrum is generated by the light emitted by the blue LED, and the second portion is generated by the light that passes through the phosphor (e.g., YAG).

As the phosphor-converted LED ages, the efficiency of the phosphor decreases, which causes the chromaticity of the phosphor-converted LED to appear "cooler" over time. In order to account for age-related chromaticity shifts in a phosphor-converted LED, it may be desirable in some embodiments of the calibration method shown in FIG. **6** to measure the photocurrents induced by the LED portion and the photocurrents induced by the phosphor portion of the phosphor-converted LED separately. Thus, some embodiments of the invention may use two different colors of photodetectors to measure photocurrents, which are separately induced by different portions of the phosphor-converted LED spectrum. In particular, an emitter module of the illumination device may include a first photodetector whose detection range is configured for detecting only the first portion of the spectrum emitted by the phosphor-converted LED, and a second photodetector whose detection range is configured for detecting only the second portion of the spectrum emitted by the phosphor-converted LED.

In general, the detection range of the first and second photodetectors may be selected based on the spectrum of the phosphor-converted LED being measured. In the exemplary embodiment described above, in which a phosphor-converted white emission LED is included within the emitter module and implemented as described above, the detection range of the first photodetector may range between about 400 nm and about 500 nm for measuring the photocurrents induced by light emitted by the blue LED portion, and the detection range of the second photodetector may range between about 500 nm and about 650 nm for measuring the photocurrents induced by light that passes through the phosphor portion of the phosphor-converted white LED. The first and second photodetectors may include dedicated photodetectors and/or emission LEDs, which are sometimes configured for detecting incident light.

As noted above, the emitter module of the illumination device preferably includes at least one dedicated photodetector. In one embodiment, the emitter module may include two different colors of dedicated photodetectors, such as one or more dedicated green photodetectors and one or more dedicated red photodetectors (see, e.g., FIG. **17**B). In another embodiment, the emitter module may include only one dedicated photodetector, such as a single red, orange, or yellow photodetector (see, e.g., FIG. **16**B). In such an embodiment, one of the emission LEDs (e.g., the green emission LED) may be configured, at times, as a photodetector for measuring a portion of the phosphor-converted LED spectrum.

In the calibration method described above and shown in FIG. **6**, the at least one dedicated photodetector may be used in step **16** to measure the photocurrents (Iph_d1), which are induced in the dedicated photodetector by the illumination produced by each of the emission LEDs when the emission

LEDs are successively driven to produce illumination at the plurality of different drive current levels (e.g., 100%, 30% and 10% of a max drive level) and the plurality of different temperatures (e.g., TO and T1). Sometime before or after each of the photocurrent measurements (Iph_d1) is obtained from the dedicated photodetector, a forward voltage (Vfd1) is measured across the dedicated photodetector to provide an indication of the detector junction temperature at each of the calibrated drive current levels.

In some embodiments of the calibration method shown in FIG. **6**, the dedicated photodetector used to obtain the photocurrent (Iph_d1) and forward voltage (Vfd1) measurements may be, e.g., a red LED. When calibrating a phosphor-converted white LED, the dedicated red photodetector may be used to measure the photocurrent (Iph_d1) induced by the light that passes through the phosphor (i.e., the "second portion" of the spectrum shown in FIG. **21**). In some embodiments, another dedicated photodetector (or one of the emission LEDs) may be used to measure the photocurrent (Iph_d2), which is induced by the light emitted by the LED portion (i.e., the "first portion" of the spectrum shown in FIG. **21**) of the phosphor-converted white LED. This photodetector may be, for example, a dedicated green photodetector or one of the green emission LEDs.

As shown in FIG. **7**, the additional photodetector may be used in step **16** of the calibration method shown in FIG. **6** to measure the photocurrents (Iph_d2), which are induced in the additional photodetector by the illumination produced by the LED portion of the phosphor-converted white LED when that LED is successively driven to produce illumination at a plurality of different drive current levels (e.g., 100%, 30% and 10% of a max drive level) and a plurality of different temperatures (e.g., TO and T1). In addition to measuring the photocurrents induced by the LED portion of the phosphor-converted white LED, the photocurrents (Iph_d2) induced by the illumination produced by the blue emission LED may also be obtained from the additional photodetector in step **16** and used as a reference in the compensation method of FIG. **13**. Before or after each of the photocurrent measurements (Iph_d2) is obtained from the additional photodetector, a forward voltage (Vfd2) is measured across the additional photodetector to provide an indication of the detector junction temperature at each of the calibrated drive current levels.

In addition to storing separate photocurrent measurements (Iph_d2 and Iph_d1) for the phosphor-converted white LED, the calibration table may also store separate luminous flux (Luma), x-chromaticity (x chrom) and y-chromaticity (y chrom) measurements for the LED portion and the phosphor portion of the phosphor-converted white LED spectrum at each of the calibrated drive currents and temperatures. While this is not explicitly shown in FIG. **7**, measuring the luminous flux (Luma), x-chromaticity (x chrom) and y-chromaticity (y chrom) attributed to each portion of the phosphor-converted white LED spectrum, and storing these values within the calibration table, the stored calibration values may be later used during one or more of the compensation methods described herein to control the luminous flux and chromaticity of the LED portion and the phosphor portion of the phosphor-converted white LED, separately, as if the LED were two different LEDs.

Exemplary methods for calibrating an illumination device comprising a plurality of emission LEDs and one or more photodetectors has now been described with reference to FIGS. **6-7**. Although the method steps shown in FIG. **6** are described as occurring in a particular order, one or more of the steps of the illustrated method may be performed in a substantially different order. In one alternative embodiment, for example, the plurality of electrical measurements (e.g., Iph, Vfd, and Vfe) may be obtained from the one or more photodetector(s) and emission LEDs in step **16** before the plurality of optical measurements (e.g., Luma, x chrom, y chrom) are obtained from the emission LEDs in step **14**. In another alternative embodiment, the external calibration tool may obtain the optical measurements (in step **14**) at substantially the same time as the LED driver and receiver circuit is obtaining the electrical measurements (in step **16**). While the calibration method shown in FIG. **6** stores the calibration values within the illumination device at the end of the calibration method (e.g., in step **22**), a skilled artisan would recognize that these values may be stored at substantially any time during the calibration process without departing from the scope of the invention. The calibration method described herein is considered to encompass all such variations and alternative embodiments.

The calibration method provided herein improves upon conventional calibration methods in a number of ways. First, the method described herein calibrates each emission LED (or chain of LEDs) individually, while turning off all other emission LEDs not currently under test. This not only improves the accuracy of the stored calibration values, but also enables the stored calibration values to account for process variations between individual LEDs, as well as differences in output characteristics that inherently occur between different colors of LEDs.

Accuracy is further improved herein by supplying a relatively small (i.e., non-operative) drive current to the emission LEDs and the photodetector(s) when obtaining forward voltage measurements, as opposed to the operative drive current levels typically used in conventional calibration methods. By using non-operative drive currents to obtain the forward voltage measurements, the present invention avoids inaccurate compensation by ensuring that the forward voltage measurements for a given temperature and fixed drive current do not change significantly over time (due to parasitic resistances in the junction when operative drive currents are used to obtain forward voltage measurements).

As another advantage, the calibration method described herein obtains a plurality of optical measurements from each emission LED and a plurality of electrical measurements from each photodetector at a plurality of different drive current levels and a plurality of different temperatures. This further improves calibration accuracy by enabling the non-linear relationship between luminous flux and drive current and the non-linear relationship between photocurrent and drive current to be precisely characterized for each individual LED. Furthermore, obtaining the calibration values at a number of different ambient temperatures improves compensation accuracy by enabling the compensation methods (described below) to interpolate between the stored calibration values, so that accurate compensation values may be determined for current operating temperatures.

As yet another advantage, the calibration method described herein may use different colors of photodetectors to measure photocurrents, which are induced by different portions (e.g., an LED portion and a phosphor portion) of a phosphor-converted LED spectrum. The different colors of photodetectors may also be used to measure the photocurrent, which is induced by a reference emission LED, whose peak emission wavelength falls within the LED portion of the spectrum emitted by the phosphor-converted LED. By storing these calibration values within the illumination

device, the calibration values may be later used to detect and account for chromaticity shifts that may occur in a phosphor-converted LED over time.

As described in more detail below, the calibration values stored within the calibration table can be used in one or more compensation methods described herein to adjust the individual drive currents supplied to the emission LEDs, so as to obtain a desired luminous flux and a desired chromaticity over changes in drive current, changes in temperature and over time, as the LEDs age. For example, the luminous flux (Luma) measurements may be used in some embodiments to maintain a consistent lumen output and chromaticity over changes in temperature. In other embodiments, the luminous flux (Luma) measurements may be used along with the chromaticity (e.g., x chrom, y chrom) measurements to obtain a new target lumen output or a new target chromaticity when the dimming level or color point setting for the illumination device is changed. Regardless of the particular luminous flux or chromaticity setting used, the photocurrent (Iph) measurements may be used to adjust the individual drive currents supplied to the emission LEDs to account for LED aging effects. While the most accurate results may be obtained by utilizing all such measurements when compensating an LED illumination device, one skilled in the art would understand how one or more of the calibration values described herein may be used to improve upon the compensation methods performed by prior art illumination devices.

Exemplary Embodiments of Improved Methods for Controlling an Illumination Device

FIGS. **8-15** illustrate exemplary embodiments of improved methods for controlling an illumination device that generally includes a plurality of emission LEDs and at least one dedicated photodetector. More specifically, FIGS. **8-15** illustrate exemplary embodiments of improved compensation methods that may be used to adjust the drive currents supplied to individual LEDs of an LED illumination device, so as to obtain a desired luminous flux and a desired chromaticity over changes in drive current, changes in temperature and over time, as the LEDs age.

In some embodiments, the compensation methods shown in FIGS. **8-15** may be used to control an illumination device having LEDs all of the same color. However, the compensation methods described herein are particularly well-suited for controlling an illumination device comprising two or more differently colored LEDs (i.e., a multi-colored LED illumination device), since output characteristics of differently colored LEDs vary differently over drive current, temperature and time. The compensation methods described herein are also particularly well-suited for controlling an illumination device that provides dimming and/or color tuning capabilities (i.e., a tunable LED illumination device), since changes in drive current inherently affect the lumen output, color, and temperature of the illumination device.

Exemplary embodiments of an illumination device will be described below with reference to FIGS. **16-20**, which show different types of LED illumination devices, each having one or more emitter modules. As shown in FIGS. **16**B, **17**B, **18**A and **18**B, each emitter module may generally include a plurality of emission LEDs arranged in an array, and one or more photodetectors spaced about a periphery of the array. In one exemplary embodiment, the array of emission LEDs may include red, green, blue, and white (or yellow) LEDs, and the one or more photodetectors may include one or more red, orange, yellow, and/or green LEDs. However, the present invention is not limited to any particular color, number, combination, or arrangement of emission LEDs and photodetectors. A skilled artisan would understand how the method steps described herein may be applied to other LED illumination devices having substantially different emitter modules.

According to one embodiment, FIG. **8** illustrates an exemplary compensation method that may be used to adjust the drive currents supplied to individual LEDs of an LED illumination device, so as to maintain a desired luminous flux and a desired chromaticity over changes in temperature. The compensation method steps shown in solid outline in FIG. **8** are similar to the compensation method steps described in FIG. 5 of commonly assigned U.S. patent application Ser. Nos. 13/970,944, 13/970,964, and 13/970,990, which provided improved illumination devices and temperature compensation methods.

However, the previously-filed applications failed to disclose methods for controlling an illumination device, so as to maintain a desired luminous flux and a desired chromaticity over time, or to obtain a new luminous flux or a new chromaticity when the dimming level or the color point setting is changed for the illumination device. The optional method steps shown in dashed outline in FIG. **8** improve upon the compensation method described in the previously filed applications by enabling the luminous flux and chromaticity to be precisely controlled over a range of dimming levels and a range of color point settings. The compensation method shown in FIG. **13**, and discussed below, further improves upon the compensation method described in the previously filed applications by providing a method for maintaining a desired luminous flux and a desired chromaticity over time, as the LEDs age. The compensation methods shown in FIGS. **8** and **13** may be used alone, or together to provide accurate compensation for all LEDs used in the illumination device over changes in drive current, temperature, and time.

As shown in FIG. **8**, the improved compensation method may generally begin by driving the plurality of emission LEDs substantially continuously to produce illumination (in step **30**). When the illumination device is first turned "on," the ambient temperature surrounding the LEDs steadily increases over time, until the temperature stabilizes. In some embodiments, the compensation method may be performed soon after the illumination device is turned "on," and may be repeated a number of times until the temperature surrounding the LEDs stabilizes. However, since changes in drive current and other factors affect the ambient temperature surrounding the LEDs, the compensation method shown in FIG. **8** may be performed at other times during operation of the illumination device.

In one preferred embodiment, the compensation method may be performed only if a significant change in ambient temperature is detected (in step **32**). In some embodiments, the ambient temperature may be monitored by a dedicated temperature sensor or an additional LED, which is used as both a temperature sensor and an optical sensor, as discussed below with reference to FIG. **20**. A "significant change" may be any incremental increase or decrease in ambient temperature (e.g., 1° C.) and is generally set within the control circuitry of the illumination device as a specified amount. If no significant temperature change is detected ("No" branch in step **32**), the compensation method may continue to drive the plurality of emission LEDs substantially continuously to produce illumination (in step **30**). Otherwise, the compensation method may proceed to step **36**.

In some embodiments, the compensation method may also be performed if the dimming level and/or the color point

setting is changed in a tunable illumination device. To accommodate such changes, the compensation method may monitor the target luminance (Ym) and the target chromaticity setting (xm, ym) stored within the control circuitry of the illumination device. In some embodiments, the target luminance (Ym) may be expressed as a finite set of dimming levels, and the target chromaticity setting (xm, ym) may be expressed as a range of x,y coordinates within the color gamut of the illumination device. In some embodiments, the target luminance (Ym) and the target chromaticity setting (xm, ym) may be stored, e.g., as 16-bit integer values within a hardware or software register of the control circuitry and may be changed by a user or a building controller by storing alternative values within the illumination device. If no changes to the stored Ym, xm and ym values are detected by the control circuitry ("No" branch in step **34**), the compensation method may continue to drive the plurality of emission LEDs substantially continuously to produce illumination (in step **30**). Otherwise, the compensation method may proceed to step **36**.

As used herein, the term "substantially continuously" means that an operative drive current is supplied to the plurality of emission LEDs almost continuously, with the exception of periodic intervals during which the plurality of emission LEDs are momentarily turned "off" for short durations of time (in step **36**). In the exemplary embodiment of FIG. **8**, the periodic intervals are utilized for obtaining forward voltage (Vfe) measurements from each of the emission LEDs, one LED at a time (in step **38**). These periodic intervals may also be used for other purposes, as shown in FIGS. **13-14** and discussed in more detail below.

FIG. **9** is an exemplary timing diagram illustrating steps **30**, **36** and **38** of the compensation method shown in FIG. **8**, according to one embodiment of the invention. As shown in FIGS. **8** and **9**, the plurality of emission LEDS are driven substantially continuously with operative drive current levels (denoted generically as I1 in FIG. **9**) to produce illumination (in step **30** of FIG. **8**). At periodic intervals, the plurality of emission LEDs are turned off for short durations of time (in step **36** of FIG. **8**) by removing the drive currents, or at least reducing the drive currents to non-operative levels (denoted generically as I0 in FIG. **9**). Between the periodic intervals, the illumination device produces continuous illumination with DC current supplied to the LEDs.

During each periodic interval, one emission LED is driven with a relatively small, non-operative drive current (e.g., approximately 0.1-10 mA, not shown in FIG. **9**) and the forward voltage developed across that LED is measured (e.g., Vfe1). In this manner, forward voltages (e.g., Vfe1, Vfe2, Vfe3, and Vfe4) are measured across each emission LED, one LED at a time, as shown in FIG. **9**. In some cases, the forward voltages associated with each individual emission LED may be averaged over a period of time, filtered to eliminate erroneous data, and stored for example in a register of the illumination device.

FIG. **9** provides an exemplary timing diagram for measuring the present forward voltages (i.e., Vfe_present, or the forward voltages that develop across the emission LEDs during operation of the device), which provide an indication of the current junction temperatures of each emission LED included in the illumination device. Although FIG. **9** provides an exemplary timing diagram for an illumination device comprising four emission LEDs, such as RGBY or RGBW, the timing diagram and method described herein can be easily modified to accommodate a fewer or greater number of emission LEDs. As described in more detail below, the timing diagram described herein can also be easily modified to accommodate additional measurements, as shown for example in FIG. **14**.

In one alternative embodiment, the compensation method of FIG. **8** may measure the detector forward voltage (Vfd_present) that develops across the one or more photodetectors in step **38**, instead of the emitter forward voltages (Vfe_present) shown in FIGS. **8-9**. However, the detector forward voltage may only be used in the presently described compensation method if a temperature difference between the emission LEDs and the photodetector(s) remains relatively the same over the operating temperature range. To maintain a substantially consistent temperature difference between the emission LEDs and the photodetector(s), an improved emitter module is provided herein and described below with reference to FIG. **18**A.

In some embodiments of the invention, the compensation method shown in FIG. **8** may continue to step **46** if a temperature change was detected in step **32**, but the target luminance (Ym) and the target chromaticity setting (xm, ym) were not changed in step **34**. As shown in FIGS. **8** and **10**, the compensation method may determine the drive current (Ix) needed to achieve a desired luminous flux (Lx) from each emission LED using the present forward voltages (Vfe_present), the table of stored calibration values generated during the calibration method of FIG. **6** and one or more interpolation techniques (in step **46** of FIG. **8**). For example, the compensation method may interpolate between the stored luminous flux calibration values to calculate a luminous flux value that would be expected for the present forward voltage at each of the previously calibrated (i.e., known) drive current levels. Once a luminous flux value is calculated at each of the calibrated drive current levels, another interpolation technique may be used to determine an unknown drive current (Ix) needed to produce a desired luminous flux (Lx), should the desired luminous flux differ from one of the calculated luminous flux values.

FIG. **10** is a graphical illustration depicting how one or more interpolation technique(s) may be used to determine the drive current (Ix) needed to produce a desired luminous flux (Lx) from the stored calibration values. In FIG. **10**, the six solid dots ("•"s) represent the luminous flux calibration values, which were obtained during calibration at three different drive currents (e.g., 10%, 30% and 100% of the maximum drive current) and two different temperatures (e.g., TO and T1). As noted above, the luminous flux calibration values ("•"s) were previously stored within a table of calibration values (see, e.g., FIG. **7**) for each emission LED included in the illumination device.

To determine the drive current (Ix) needed to produce a desired luminous flux (Lx) for a given LED, the compensation method of FIG. **8** interpolates between each of the stored calibration values ("•"s) to calculate the luminous flux values ("Δ"s), which should be produced at the present operating temperature (Vfe_present) when using the same three drive currents (e.g., 10%, 30% and 100% of the maximum drive current). If the desired luminous flux (Lx) differs from one of the calculated luminous flux values ("Δ"s), the compensation method of FIG. **8** may apply another interpolation technique to the calculated luminous flux values ("Δ"s) to generate a relationship there between (denoted by a solid line in FIG. **10**). From this relationship, the drive current (Ix) needed to produce a desired luminous flux (Lx) may be determined.

In some embodiments, the interpolation technique(s) used to calculate the luminous flux values ("Δ"s) and to generate the relationship between the calculated values may depend on the color of the LED being compensated. For example,

the luminous flux vs. junction temperature (or forward voltage) relationship for blue, green, and white LEDs is substantially linear over the operating temperature range (see, FIG. **2**). Because of this linear relationship, the compensation method is able to calculate luminous flux values (Δ's) at the present forward voltage for blue, green and white LEDs by linearly interpolating between the calibration values stored at each of the calibrated drive current levels. However, red, red-orange, and yellow LEDs exhibit a substantially more non-linear relationship between luminous flux vs. junction temperature (see, FIG. **3**). For these LEDs, a higher-order interpolation technique may be used to calculate the luminous flux values (Δ's) at the present forward voltage for each of the calibrated drive current levels.

In one embodiment, the higher-order interpolation technique may be in the form of a quadratic interpolation, which follows the general equation:

$$ax^2+bx+c=y \quad \text{(EQ. 1)}$$

where 'x' is Vf (or temperature), 'y' is luminous flux, and 'a,' 'b,' and 'c' are coefficients. If forward voltage and luminous flux values were previously obtained during the calibration phase at three different temperatures, the 'a,' 'b' and 'c' coefficient values may be precisely determined by inserting the stored calibration values into EQ. 1 and separately solving the equation for 'a,' 'b,' and 'c'. If, on the other hand, the LED was calibrated at only two different temperatures, as described above, the 'a' coefficient may be obtained from data sheets provided by the LED manufacturer, while the 'b' and 'c' coefficients are determined from the calibration values. While the latter method (sometimes referred to as a "poor man's quadratic interpolation") may sacrifice a small amount of accuracy, it may represent an acceptable trade-off between accuracy and calibration costs. In some embodiments, the calibration table may store the actual (i.e., measured) luminous flux calibration values, or just the coefficients needed for the higher-order interpolation. In other embodiments, a combination of measured values and coefficients may be stored in the calibration table.

In some embodiments, the relationship (solid line in FIG. **10**) between the calculated luminous flux values (Δ's) may be determined through another interpolation technique, if a desired luminous flux (Lx) differs from one of the calculated values. However, since the relationship between luminous flux and drive current is non-linear for all LED colors (see, FIGS. **4-5**), the relationship is preferably derived through a higher-order interpolation of the calculated luminous flux values. Alternatively, a piece-wise linear interpolation could be used to characterize the non-linear relationship between the calculated luminous flux values, or a typical curvature could be assumed from data sheets provided by the LED manufacturer.

In some embodiments, an appropriate interpolation technique may be selected based on trade-offs between memory and processing requirements, and/or based upon the particular color of LED being compensated. As noted above, some LED colors, such as blue and green, exhibit a comparatively more non-linear luminous flux vs. drive current relationship than other LED colors, such as red and red-orange (see, FIGS. **4-5**). LED colors exhibiting substantially greater non-linear behaviors (such as blue and green) may be more accurately compensated by obtaining more luminous flux calibration values and using a piece-wise linear interpolation technique, or by obtaining fewer calibration values and using a higher-order interpolation technique or an assumed curvature to generate the non-linear relationship between luminous flux and drive current.

Once the relationship between luminous flux and drive current is derived for a given LED, the drive current (Ix) needed to produce a desired luminous flux (Lx) may be selected from the generated relationship, as shown in the example of FIG. **10**. The selected drive current may then be used to drive the LED to produce illumination with the desired luminous flux (in step **48** of FIG. **8**). This process is performed for each emission LED, until each LED is configured for producing a desired luminous flux (Lx) at the present operating temperature (Vfe_present). The drive currents supplied to the LEDs may be adjusted to meet the selected drive currents either by adjusting the drive current level (i.e., current dimming), or by changing the duty cycle of the drive current through Pulse Width Modulation (PWM) dimming.

In some embodiments, the compensation method shown in FIG. **8** may perform additional steps (e.g., steps **40**, **42**, and **44**), if the target luminance (Ym) and/or the target chromaticity setting (xm, ym) was changed in step **34**. As noted above, the target luminance (Ym) or target chromaticity setting (xm, ym) may be changed by a user or a building controller to adjust the dimming level or the color point setting of a tunable LED illumination device. If a change in either target luminance or chromaticity is detected in step **34**, the compensation method may determine the drive currents (Idrv), which are presently supplied to each of the emission LEDs (in step **40**). The present drive currents (Idrv) may be obtained from the LED driver circuitry. Although illustrated in FIG. **8** as occurring after the present forward voltages (Vfe_present) are measured in step **38**, the present drive currents (Idrv) may be determined before the forward voltages are measured, in alternative embodiments of the invention.

Once the present forward voltages and the present drive currents are determined, the compensation method may determine the chromaticity values (x_i, y_i) that are expected for each emission LED (in step **42**) using the present forward voltage (Vfe_present) measured across the LED, the present drive current (Idrv) supplied to the LED, the table of stored calibration values generated during the calibration method of FIG. **6**, and one or more interpolation techniques. FIGS. **11-12** depict how one or more interpolation technique(s) may be used to determine the expected x- and y-chromaticity values (x_i, y_i) for a given LED at the present operating temperature (Vfe_present) and the present drive current (Idrv) from the table of stored calibration values.

In FIGS. **11-12**, the solid dots (•'s) represent the x- and y-chromaticity calibration values, which were obtained during calibration at three different drive currents (e.g., 10%, 30% and 100% of the maximum drive current) and two different temperatures (e.g., TO and T1). As noted above, these x- and y-chromaticity calibration values (•'s) were stored within a table of calibration values (see, e.g., FIG. **7**) for each emission LED included within the illumination device. To determine the expected x- and y-chromaticity values (x_i, y_i) for a given LED, the compensation method of FIG. **8** interpolates between the stored calibration values (•'s) to calculate the x- and y-chromaticity values (Δ's), which should be produced at the present operating temperature (Vfe_present) when using the same three drive currents (e.g., 10%, 30%, and 100% of the maximum drive current). In most cases, a linear interpolation technique may be used to calculate the x- and y-chromaticity values (Δ's) at the present operating temperature (Vfe_present). However, if the x- and y-chromaticity values were calibrated at more than two temperatures, a non-linear interpolation technique

may be used to calculate the x- and y-chromaticity values (Δ's) at the present operating temperature (Vfe_present).

If the drive current (Idrv) presently supplied to the emission LED differs from one of the calibrated drive current levels, the compensation method of FIG. **8** may apply another interpolation technique to the calculated x- and y-chromaticity values (Δ's) to generate a relationship there between (denoted by a solid line in FIGS. **11-12**). From this relationship, the expected x- and y-chromaticity values (x_i, y_i) may be determined for the present drive current (Idrv).

The interpolation technique used to generate the relationship between the calculated x- and y-chromaticity values (Δ's) may generally depend on the color of LED being compensated. As noted above, the change in chromaticity with drive current and temperature is significantly different for different colors of LEDs. For a red LED, the peak emission wavelength increases relatively linearly with drive current yet increases significantly with temperature. The peak emission wavelength of a blue LED decreases somewhat non-linearly with drive current and increases slightly with temperature. Although the peak emission wavelength of a green LED varies very little with temperature, it decreases very non-linearly with drive current. In some embodiments, a piecewise linear interpolation technique may be used to generate the relationship between the calculated x- and y-chromaticity values (Δ's), or a typical curvature may be assumed from data sheets provided by an LED manufacturer. In other embodiments, however, a higher-order interpolation technique may be used to increase accuracy when compensating certain colors of LEDs (e.g., green LEDs), which exhibit more significant non-linear changes in chromaticity with drive current.

The x- and y-chromaticity values expected for each emission LED may be expressed as a color point in the form of (x_i, y_i). In an illumination device comprising four emission LEDs, step **42** of the compensation method may result in the generation of four expected color points: (x_1, y_1), (x_2, y_2), (x_3, y_3), and (x_4, y_4). Once the expected color points are determined in step **42**, the compensation method may calculate the relative lumens needed from each of the emission LEDs to achieve the target luminance (Ym) and the target chromaticity setting (xm, ym) in step **44**. For example, when light from four emission LEDs is combined, the target luminance (Ym) of the combined light may be expressed as:

$$Ym=Y_1+Y_2+Y_3+Y_4 \quad \text{(EQ. 2)}$$

where Y_1, Y_2, Y_3, and Y_4 represent the relative lumens of the four emission LEDs. The relative lumen values (Y_1, Y_2, Y_3, and Y_4) may be calculated using well-known color mixing equations, the target luminance (Ym) and target chromaticity (xm, ym) values set within the illumination device, and the expected color points (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4) determined in step **42** of the compensation method. As these equations are well-known and readily understood by a skilled artisan, further description of such equations will be omitted herein.

Once the relative lumens (e.g., Y_1, Y_2, Y_3, and Y_4) are calculated for each emission LED (in step **44**), the drive currents that should be supplied to each emission LED to achieve the target luminance (Ym) are determined in step **46** similar to the manner described above. However, instead of determining a drive current (Ix) needed to achieve a "desired luminous flux" (Lx) from each emission LED, the current embodiment determines the drive current (Ix) needed to achieve the relative lumen value (Y_1, Y_2, Y_3, or Y_4), which was calculated for each emission LED in step **44**. Once the individual drive currents (Ix) are determined in step **46**, the emission LEDs are driven with the determined drive currents to produce illumination having a desired luminous flux and a desired chromaticity in step **48**.

The compensation method shown in FIG. **8** may be used to adjust the drive currents supplied to the emission LEDs whenever a significant change in the temperature is detected (in step **32**) and/or the target luminance or target chromaticity setting is changed (in step **34**). However, since changes in drive current inherently affect the LED junction temperature, the method steps shown in FIG. **8** may generally be repeated a number of times until the temperature stabilizes and the target luminance and chromaticity values remain unchanged.

One embodiment of a compensation method for controlling an LED illumination device over changes in drive current and temperature has now been described with reference to FIGS. **8-12**. Although the compensation method described herein references target chromaticity settings and chromaticity calibration values from the CIE 1931 XYZ color space, one skilled in the art would readily understand how the compensation method could be modified to use target chromaticity settings and chromaticity calibration values from other color spaces, such as the CIE 1931 RGB color space, the CIE 1976 LUV color space, and various other RGB color spaces (e.g., sRGB, Adobe RGB, etc.). For this reason, the compensation method described herein and recited in the claims is considered to encompass any color space that can be used to describe the gamut of an LED illumination device comprising substantially any combination of emission LEDs as described herein.

The compensation method shown in FIG. **8** provides many advantages over conventional compensation methods. As noted above, conventional methods typically measure forward voltages by applying operative drive current levels to the emission LEDs. Unfortunately, forward voltages measured at operative drive current levels vary significantly over the lifetime of an LED. As an LED ages, the parasitic resistance within the junction increases, which in turn, causes the forward voltage measured at operating current levels to increase over time, regardless of temperature. For this reason, the present compensation method uses a relatively small drive current (e.g., about 0.1 mA to about 10 mA) to obtain forward voltage measurements from each LED individually, while turning off all emission LEDs not currently under test. This improves the accuracy of the operating forward voltage values and enables each emission LED to be individually compensated for temperature and process.

Conventional methods often rely on typical values or linear relationships between luminous flux and drive current when performing temperature compensation. In contrast, the compensation method described herein interpolates between a plurality of stored luminous flux calibration values taken at different drive currents and different temperatures and derives a non-linear relationship between luminous flux and drive current for each LED at the present operating temperature (Vfe_present). This enables the present compensation method to accurately and individually characterize the luminous flux vs. drive current relationship for each LED included within the illumination device, and to provide accurate temperature compensation, regardless of process. As a consequence, the compensation method described herein provides more accurate control of the luminous flux over temperature changes.

The compensation method shown in FIG. **8** further improves upon conventional compensation methods by providing a more accurate method of controlling the luminance

and chromaticity of the illumination device when the dimming level or color point setting is changed by a user or building controller. When a change in target luminance or target chromaticity is detected by the illumination device, the drive currents supplied to the emission LEDs must be adjusted to achieve the new target values. The compensation method described herein increases the accuracy with which new drive currents are determined by interpolating between a plurality of stored chromaticity calibration values taken at different temperatures and deriving a relationship between chromaticity and drive current for each emission LED at the present operating temperature (Vfe_present). This enables the present compensation method to accurately determine the expected chromaticity values that should be produced for each emission LED at the present operating temperature and the present drive current. As a consequence, the compensation method described herein provides more accurate control of the chromaticity as drive current changes.

While the compensation method shown in FIG. **8** provides an accurate method for controlling the luminous flux and chromaticity of an LED illumination device over changes in drive current and temperature, it does not account for LED aging effects. In order to mitigate such effects, FIG. **13** provides an improved method for maintaining a desired luminous flux and a desired chromaticity over time, as the LEDs age. As noted above, the compensation methods shown in FIGS. **8** and **13** may be used alone, or together to provide accurate compensation for all LEDs used in the illumination device over changes in drive current, temperature, and time.

In general, the compensation method shown in FIG. **13** may be performed repeatedly throughout the lifetime of the illumination device to account for LED aging effects. The method shown in FIG. **13** may be performed at substantially any time, such as when the illumination device is first turned "on," or at periodic or random intervals throughout the lifetime of the device. In some embodiments, the compensation method shown in FIG. **13** may be performed after a change in temperature, dimming level, or color point setting is detected to fine tune the drive current values determined in the compensation method of FIG. **8**.

As shown in FIG. **13**, the age compensation method may generally begin by driving the plurality of emission LEDs substantially continuously to produce illumination, e.g., by applying operative drive currents (Idrv) to each of the plurality of emission LEDs (in step **50**). As noted above, the term "substantially continuously" means that an operative drive current is applied to the plurality of emission LEDs almost continuously, with the exception of periodic intervals during which the plurality of emission LEDs are momentarily turned off for short durations of time (in step **52**). In the method shown in FIG. **13**, the periodic intervals may be used for obtaining various measurements from a dedicated photodetector (e.g., a red photodetector) included within the illumination device or an emission LED (e.g., a red emission LED) configured to detect incident light. In step **54**, for example, the periodic intervals are used for measuring a photocurrent (Iph), which is induced on the photodetector in response to illumination that is produced by each emission LED, one LED at a time, and received by the photodetector. In step **56**, the periodic intervals are used for obtaining a forward voltage (Vfd), which develops across the photodetector upon applying a relatively small (i.e., non-operative) drive current thereto.

FIG. **14** is an exemplary timing diagram illustrating steps **50**, **52**, **54** and **56** of the compensation method shown in FIG. **13**, according to one embodiment of the invention. As shown in FIGS. **13** and **14**, the plurality of emission LEDS are driven substantially continuously with operative drive current levels (denoted generically as I1 in FIG. **14**) to produce illumination (in step **50** of FIG. **13**). At periodic intervals, the plurality of emission LEDs are turned "off" for short durations of time (in step **52** of FIG. **13**) by removing the drive currents, or at least reducing the drive currents to non-operative levels (denoted generically as I0 in FIG. **14**). Between the periodic intervals, the illumination device produces continuous illumination with DC current supplied to the emission LEDs.

During some of the periodic intervals, one emission LED is driven with an operative drive current level (I1) to produce illumination, while the remaining LEDs remain "off," and the photocurrent (e.g., Iph1) induced in the photodetector by the illumination from the driven LED is measured. The photocurrents (e.g., Iph1, Iph2, Iph3, and Iph4) induced in the photodetector by the illumination produced by each of the emission LEDs are measured, one LED at a time, as shown in FIG. **14** and step **54** of FIG. **13**. Although FIG. **14** provides an exemplary timing diagram for an illumination device comprising four emission LEDs, such as RGBY or RGBW, the timing diagram described herein can be easily modified to accommodate a fewer or greater number of emission LEDs.

Sometime before or after the photocurrent (Iph) measurements are obtained, a forward voltage (Vfd) is measured across the photodetector by applying a relatively small, non-operative drive current (e.g., approximately 0.1-0.3 mA) to the photodetector (in step **56** of FIG. **13**). This forward voltage measurement (also referred to herein as Vfd_present) provides an indication of the current junction temperature of the photodetector. Although the timing diagram of FIG. **14** shows only one forward voltage (Vfd) measurement obtained from a single photodetector, the timing diagram can be easily modified to accommodate a greater number of photodetectors.

In one exemplary embodiment, the presently described compensation method may be utilized within an illumination device comprising a plurality of photodetectors implemented with differently colored LEDs. In particular, each emitter module of the illumination device may include one or more red LEDs and one or more green LEDs as photodetectors. In such an embodiment, a forward voltage measurement (Vfd) may be obtained from each photodetector by applying a small drive current thereto (in step **56**). In some cases, the photocurrents associated with each emission LED (e.g., Iph1, Iph2, Iph3, Iph4) and the forward voltage(s) associated with each photodetector (Vfd) may be independently averaged over a period of time, filtered to eliminate erroneous data, and stored, for example, in a register of the illumination device.

In addition to the photocurrents and detector forward voltages, the periodic intervals shown in FIG. **14** may be used to obtain other measurements, such as the emitter forward voltages (Vfe1, Vfe2, etc.) described above with respect to FIGS. **8** and **9**. The periodic intervals may also be used for other purposes not specifically illustrated herein. For example, some of the periodic intervals may be used by the photodetector to detect light originating from outside of the illumination device, such as ambient light or light from other illumination devices. In some cases, ambient light measurements may be used to turn the illumination device on when the ambient light level drops below a threshold (i.e., when it gets dark), and turn the illumination device off when the ambient light level exceeds another threshold (i.e., when it gets light). In other cases, the ambient light mea-

surements may be used to adjust the lumen output of the illumination device over changes in ambient light level, for example, to maintain a consistent level of brightness in a room. If the periodic intervals are used to detect light from other illumination devices, the detected light may be used to avoid interference from the other illumination devices when obtaining the photocurrent and detector forward voltage measurements in the compensation method of FIG. **13**.

In other embodiments, the periodic intervals may be used to measure different portions of a particular LED's spectrum using two or more different colors of photodetectors. For example, the spectrum of a phosphor-converted white LED may be divided into two portions, and each portion may be measured separately during two different periodic intervals using two different photodetectors. Specifically, a first periodic interval may be used to detect the photocurrent, which is induced on a first photodetector (e.g., a green photodetector) by a first spectral portion (e.g., about 400 nm to about 500 nm) of the phosphor-converted white LED. A second periodic interval may then be used to detect the photocurrent, which is induced on a second photodetector (e.g., a red photodetector) by a second spectral portion (e.g., about 500 nm to about 650 nm) of the phosphor-converted white LED. As described in more detail below, dividing the spectrum of the phosphor-converted white LED into two portions, and measuring each portion separately with two different colors of photodetectors may enable the compensation method of FIG. **13** to detect and account for changes in chromaticity that occur as the phosphor-converted LED ages.

Once the photocurrents and detector forward voltage(s) are measured, the compensation method shown in FIG. **13** may determine the photocurrents (Iph_exp) that are expected for each emission LED (in step **58**) using the forward voltage (Vfd_present) presently measured across the photodetector, the drive current (Idrv) presently applied to the LED, the table of stored calibration values generated during the calibration method of FIG. **6**, and one or more interpolation techniques. FIG. **15** depicts how one or more interpolation techniques may be used to determine the expected photocurrent (Iph_exp) for a given LED at the present operating temperature (Vfd_present) and the present drive current (Idrv) from the table of stored calibration values.

In FIG. **15**, the solid dots (•'s) represent the photocurrent calibration values, which were obtained during calibration at three different drive currents (e.g., 10%, 30% and 100% of the maximum drive current) and two different temperatures (e.g., TO and T1). As noted above, the photocurrent calibration values (•'s) were stored within a table of calibration values for each emission LED included within the illumination device (see, e.g., FIG. **7**). To determine the expected photocurrent value (Iph_exp) for a given LED, the compensation method of FIG. **13** interpolates between the stored calibration values (•'s) to calculate the photocurrent values (Δ's), which should be produced at the present operating temperature (Vfd_present) when using the same three drive currents (e.g., 10%, 30% and 100% of the maximum drive current).

The change in photocurrent over temperature is non-linear for all LED colors, since both the emitted power and the responsivity of the photodetector decrease linearly with temperature. For this reason, a non-linear interpolation technique may be used to calculate the photocurrent values (Δ's) at the present operating temperature for all LED colors. In some embodiments, the non-linear interpolation technique may be a higher-order interpolation, a "poor man's" quadratic interpolation, or an assumed curvature. In other embodiments, the results of two different linear interpolations (e.g., a linear interpolation between emitted power and temperature, and a linear interpolation between detector responsivity and temperature) may be multiplied together to calculate the photocurrent values (Δ's) at the present operating temperature.

If the drive current (Idrv) presently supplied to the emission LED differs from one of the calibrated drive current levels, the compensation method of FIG. **13** may apply another interpolation technique to the calculated photocurrent values (Δ's) to generate a relationship there between (denoted by a solid line in FIG. **15**). In some cases, a higher-order interpolation of the calculated photocurrent values (Δ's) may be used to generate a non-linear relationship between photocurrent and drive current. In other cases, a piece-wise linear interpolation could be used to characterize the non-linear relationship between the calculated photocurrent values, or a typical curvature could be assumed from data sheets provided by the LED manufacturer. From the generated relationship, the expected photocurrent value (Iph_exp) may be determined for the present drive current (Idrv).

Once expected photocurrents (Iph_exp) are determined for each emission LED (in step **58**), the compensation method shown in FIG. **13** calculates a scale factor for each emission LED (in step **60**) by dividing the photocurrent (Iph_exp) expected for each LED by the photocurrent (e.g., Ipd1) measured for each LED. Next, the compensation method applies each scale factor to a desired luminous flux value for each emission LED to obtain an adjusted luminous flux value for each emission LED (in step **62**). In some embodiments, the desired luminous flux value may be one of the relative lumen values (Y_1, Y_2, Y_3, or Y_4) calculated, e.g., in step **44** of the compensation method shown in FIG. **8** to account for any changes in the target luminance (Ym) and/or target chromaticity (xm, ym) settings stored within the illumination device. Finally, the drive currents presently applied to the emission LEDs are adjusted (in step **64**) to achieve the adjusted luminous flux values if a difference exists between the expected and measured photocurrents for any of the emission LEDs.

The compensation method described above and illustrated in FIG. **13** provides an accurate method for adjusting the individual drive currents applied to the emission LEDs, so as to compensate for the degradation in lumen output that occurs over time as the LEDs age. By accurately controlling the luminous flux produced by each emission LED, the compensation method accurately controls the color of an LED illumination device comprising a plurality of multi-colored, non-phosphor-converted emission LEDs. However, and as noted above, some embodiments of the invention may include a phosphor-converted emission LED (e.g., a white or yellow LED) within the emitter module. In such embodiments, additional steps may be taken to control the luminous flux and chromaticity of the phosphor-converted LED over time.

Like non-phosphor-converted LEDs, the luminous flux produced by a phosphor-converted LED generally decreases over time. Unlike non-phosphor-converted LEDs, however, phosphor-converted LEDs are also susceptible to changes in chromaticity over time since the efficiency of the phosphor degrades as the phosphor ages. When a phosphor-converted LED is included in the emitter module, the compensation method shown in FIG. **13** may be used to detect and account for chromaticity shifts that occur in the phosphor-converted LED by separately measuring and compensating for the chromaticity shifts caused by phosphor aging. In order to do

so, two separate photodetectors may be included within the emitter module for measuring the photocurrents, which are induced by two different portions of the phosphor-converted LED spectrum. In some embodiments, the emitter module may include two different colors of dedicated photodetectors (e.g., red and green photodetectors as shown in FIG. **17**B) for measuring the photocurrents induced by the two different portions of the phosphor-converted LED spectrum (as shown, e.g., in FIG. **21**). In other embodiments, only one dedicated photodetector may be included within the emitter module (e.g., a red photodetector, as shown in FIG. **16**B), and one of the emission LEDs (e.g., a green emission LED) may be used, at times, as an additional photodetector.

In order to detect and account for chromaticity shifts that occur over time in a phosphor-converted white LED, the different portions of the phosphor-converted LED spectrum must first be calibrated. For example, the dedicated red photodetector and the dedicated green photodetector (or the green emission LED) may both be used in step **16** of the calibration method of FIG. **6** to obtain photocurrent measurements from the phosphor-converted white LED. The white LED-on-red detector photocurrent measurements are stored in the calibration table of FIG. **7** as "Iph_d1". The white LED-on-green detector photocurrent measurements are stored in the calibration table of FIG. **7** as "Iph_d2". The Iph_d2 calibration values are italicized in FIG. **7** to show that they are optional values, which may only be obtained when a phosphor-converted LED is included within the emitter module.

As described above and shown in FIG. **7**, the Iph_d1 and Iph_d2 calibration values may be obtained for a phosphor-converted white LED at each of the three different drive currents (e.g., 10%, 30% and 100% of the max drive current) and at each of the two different temperatures (e.g., TO and T1). The white LED-on-green detector photocurrent (Iph_d2) measurements indicate the photocurrents that were induced in the green photodetector (or the green emission LED) by light emitted by the blue LED portion of the phosphor-converted white LED (i.e., the first portion shown in FIG. **21**). The white LED-on-red detector photocurrent (Iph_d1) measurements indicate the photocurrents that were induced in the red photodetector by the light that passed through the phosphor portion of the phosphor-converted white LED (i.e., the second portion shown in FIG. **21**).

Sometime before or after each Iph_d1 and Iph_d2 measurement is obtained, a forward voltage (Vfd1 or Vfd2) is measured across the dedicated red photodetector and the dedicated green photodetector (or the green emission LED) to provide an indication of the red detector junction temperature and the green detector junction temperature at each of the calibrated drive current levels and ambient temperatures. The Vfd1 and Vfd2 measurements may be stored within the calibration table, as shown in FIG. **7**.

During the compensation method of FIG. **13**, the white LED-on-red detector photocurrent (Iph_d1) and the white LED-on-green detector photocurrent (Iph_d2) can be measured as described in step **54**, and the expected photocurrents (Iph_exp) for the white LED-on-red detector and the white LED-on-green detector can be determined as described in steps **56-58**. Next, the expected photocurrents (Iph_exp) determined in step **58** can be respectively divided by the photocurrents measured in step **54** to produce a white LED-on-red detector scale factor and a white LED-on-green detector scale factor, as described in step **60**. The white LED-on-green detector scale factor provides an indication of how the relative lumens of the LED portion of the phosphor-converted LED has changed over time. The white LED-on-red detector scale factor provides an indication of how the relative lumens of the phosphor portion of the phosphor-converted LED has changed over time. In some embodiments, the white LED-on-green detector scale factor and the white LED-on-red detector scale factor can be used to control the luminous flux and chromaticity of the phosphor-converted white LED, as if it were two separate LEDs.

For example, the white LED-on-red detector scale factor can be applied in step **62** to an overall desired luminous flux value for the phosphor-converted LED to account for lumen changes caused by LED and phosphor aging. In some cases, the overall desired luminous flux value for the phosphor-converted LED may be determined by adding together the relative lumen values, which were separately calculated for the LED portion and the phosphor portion of the phosphor-converted LED in steps **40-44** of the compensation method of FIG. **8**. In order to calculate the relative lumen values for the LED portion and the phosphor portion separately, the compensation method of FIG. **8** may use the luminous flux, x-chromaticity and y-chromaticity calibration values, which were separately stored in the calibration table for each portion of the phosphor-converted white LED spectrum, as described above with reference to FIG. **6**.

To account for chromaticity shifts in the phosphor-converted white LED over time, the white LED-on-green detector scale factor can be compared to the white LED-on-red detector scale factor. Based on such comparison, the actual chromaticity of the phosphor-converted LED can be determined using well known color mixing equations, and the overall chromaticity of the illumination device can be maintained by adjusting the drive currents applied to all emission LEDs.

While the method described above provides an acceptable solution for controlling the luminous flux and chromaticity of a phosphor-converted LED over time, the chromaticity of the phosphor-converted white LED may be more accurately determined, in some embodiments, by accounting for detector aging. For example, the photocurrents induced on the red and green photodetectors (or the green emission LED) by a given amount of incident light generally decreases over time. Although such decrease is typically not large, it may cause the white LED-on-green detector scale factor and the white LED-on-red detector scale factor to likewise decrease over time, which in turn, decreases the accuracy with which the chromaticity of the phosphor-converted white LED is adjusted. To eliminate this source of measurement error, the scale factors generated in step **60** for the phosphor-converted LED may be adjusted, in some embodiments, to account for detector aging. This may be achieved, for example, by using the blue emission LED as a reference for the phosphor-converted white LED.

In some embodiments, the dedicated red photodetector and the dedicated green photodetector (or the green emission LED) may both be used in step **54** of the compensation method of FIG. **13** to obtain Iph_d1 and Iph_d2 photocurrent measurements from the blue emission LED. After measuring the forward voltages measured across the red and green detectors in step **56**, steps **58** and **60** of the compensation method of FIG. **13** may be repeated to produce a blue LED-on-red detector scale factor and a blue LED-on-green detector scale factor for the blue emission/reference LED. In some embodiments, the blue LED-on-red detector scale factor may be applied in step **62** to a desired luminous flux value for the blue emission/reference LED to adjust the luminous flux produced by the blue emission/reference LED. In some embodiments, the desired luminous flux value for the blue emission/reference LED may be a relative

lumen value, which is calculated as described in steps **40-44** of the compensation method of FIG. **8**.

In order to account for detector aging in the phosphor-converted LED scale factors, the white LED-on-green detector scale factor may be divided by the blue LED-on-green detector scale factor to produce a white-over-blue-on-green scale factor ratio (WoBoG). Likewise, the white LED-on-red detector scale factor may be divided by the blue LED-on-red detector scale factor to produce a white-over-blue-on-red scale factor ratio (WoBoR). Next, the WoBoR scale factor ratio may be further divided by the WoBoG scale factor ratio to produce a White Phosphor over White Blue pump (WPoWB) ratio. The White Phosphor over White Blue pump (WPoWB) ratio provides an indication of how the spectrum of the phosphor-converted white LED changes over time and accounts for any degradation of the detector responsivity that may over time. In some embodiments, the WPoWB ratio may be used to adjust the overall chromaticity of the phosphor-converted LED. Once the overall chromaticity of the phosphor-converted LED is set, the white-on-red detector scale factor may be applied in step **62** to the overall desired luminous flux value for the phosphor-converted white LED to account for age-related lumen changes.

The compensation method shown in FIG. **13** and described above provides many advantages over conventional compensation methods. As with the other methods described herein, the compensation method of FIG. **13** improves the accuracy with which the detector forward voltages are measured by applying a relatively small drive current (e.g., about 0.1 mA to about 0.3 mA) to the photodetector(s). In addition, the compensation method of FIG. **13** interpolates between a plurality of stored photocurrent values taken at different drive currents and different temperatures and derives a non-linear relationship between photocurrent and drive current for each emission LED at the present operating temperature (Vf_present). This enables the present compensation method to accurately and individually characterize the photocurrent vs. drive current relationship for each individual LED, thereby providing accurate age compensation for all emission LEDs included within the illumination device. Furthermore, the compensation method described herein accounts for both emitter and detector aging and provides compensation for chromaticity shifts that occur when phosphor-converted white emission LEDs age. As a consequence, the age compensation method described herein provides more accurate control of the luminous flux and chromaticity of the individual emission LEDs over time.

Exemplary Embodiments of Improved Illumination Devices

The improved methods described herein for calibrating and controlling an illumination device may be used within substantially any LED illumination device having a plurality of emission LEDs and one or more photodetectors. As described in more detail below, the improved methods described herein may be implemented within an LED illumination device in the form of hardware, software, or a combination of both.

Illumination devices, which benefit from the improved methods described herein, may have substantially any form factor including, but not limited to, parabolic lamps (e.g., PAR **20**, **30**, or **38**), linear lamps, flood lights and mini-reflectors. In some cases, the illumination devices may be installed in a ceiling or wall of a building and may be connected to an AC mains or some other AC power source. However, a skilled artisan would understand how the improved methods described herein may be used within other types of illumination devices powered by other power sources (e.g., batteries or solar energy).

Exemplary embodiments of an improved illumination device will now be described with reference to FIGS. **16-20**, which show different types of LED illumination devices, each having one or more emitter modules. Although examples are provided herein, the present invention is not limited to any particular type of LED illumination device or emitter module design. A skilled artisan would understand how the method steps described herein may be applied to other types of LED illumination devices having substantially different emitter module designs.

FIG. **16**A is a photograph of a linear lamp **70** comprising a plurality of emitter modules (not shown in FIG. **16**A), which are spaced apart from one another and arranged generally in a line. Each emitter module included within linear lamp **70** includes a plurality of emission LEDs and at least one dedicated photodetector, all of which are mounted onto a common substrate and encapsulated within a primary optics structure. The primary optics structure may be formed from a variety of different materials and may have substantially any shape and/or dimensions necessary to shape the light emitted by the emission LEDs in a desirable manner. Although the primary optics structure is described below as a dome, one skilled in the art would understand how the primary optics structure may have substantially any other shape or configuration, which encapsulates the emission LEDs and the at least one photodetector.

An exemplary emitter module **72** that may be included within the linear lamp **70** of FIG. **16**A is shown in FIG. **16**B. In the illustrated embodiment, emitter module **72** includes four differently colored emission LEDs **74**, which are arranged in a square array and placed as close as possible together in the center of a primary optics structure (e.g., a dome) **76**, so as to approximate a centrally located point source. In some embodiments, the emission LEDs **74** may each be configured for producing illumination at a different peak emission wavelength. For example, the emission LEDs **74** may include RGBW LEDs or RGBY LEDs. In addition to the emission LEDs **74**, a dedicated photodetector **78** is included within the dome **76** and arranged somewhere around the periphery of the array. The dedicated photodetector **78** may be any device (such as a silicon photodiode or an LED) that produces current indicative of incident light.

In at least one embodiment, photodetector **78** is an LED with a peak emission wavelength in the range of approximately 550 nm to 700 nm. A photodetector with such a peak emission wavelength will not produce photocurrent in response to infrared light, which reduces interference from ambient light sources. In at least one preferred embodiment, photodetector **78** may comprise a small red, orange, or yellow LED. In some embodiments, the dedicated photodetector **78** may be arranged to capture a maximum amount light, which is reflected from a surface of the dome **76** from the emission LEDs having the shortest wavelengths (e.g., the blue and green emission LEDs).

In some embodiments, the emitter module **72** may include a phosphor-converted white (W) emission LED **74**, and the dedicated photodetector **78** may be used to measure the photocurrents (Iph_d1) that are induced by the light that passes through the phosphor portion of the phosphor-converted LED. In order to measure the photocurrents (Iph_d2) induced by the blue LED portion of the phosphor-converted LED, the green emission LED **74** may be configured, at times, as an additional photodetector. These photocurrent measurements may be stored within the calibration table of

FIG. **7** and used within the compensation method of FIG. **13** to compensate for chromaticity shifts caused by phosphor aging.

FIGS. **17**A and **17**B illustrate a substantially different type of illumination device and emitter module design. Specifically, FIG. **17**A depicts an illumination device **80** having a parabolic form factor (e.g., a PAR **38**) and only one emitter module (not shown in FIG. **17**A). As these illumination devices have only one emitter module, the emitter modules included in such devices typically include a plurality of differently-colored chains of LEDs, where each chain includes two or more LEDs of the same color. FIG. **17**B illustrates an exemplary emitter module **82** that may be included within the PAR lamp **80** shown in FIG. **17**A.

In the illustrated embodiment, emitter module **82** includes an array of emission LEDs **84** and a plurality of dedicated photodetectors **88**, all of which are mounted on a common substrate and encapsulated within a primary optics structure (e.g., a dome) **86**. In some embodiments, the array of emission LEDs **84** may include a number of differently-colored chains of LEDS, wherein each chain is configured for producing illumination at a different peak emission wavelength. According to one embodiment, the array of emission LEDs **84** may include a chain of four red LEDs, a chain of four green LEDs, a chain of four blue LEDs, and a chain of four white or yellow LEDs. Each chain of LEDs is coupled in series and driven with the same drive current. In some embodiments, the individual LEDs in each chain may be scattered about the array and arranged so that no color appears twice in any row, column, or diagonal, to improve color mixing within the emitter module **82**.

In the exemplary embodiment of FIG. **17**B, four dedicated photodetectors **88** are included within the dome **86** and arranged around the periphery of the array. In some embodiments, the dedicated photodetectors **88** may be placed close to, and in the middle of, each edge of the array and may be connected in parallel to a receiver of the illumination device. By connecting the dedicated photodetectors **88** in parallel with the receiver, the photocurrents induced on each photodetector may be summed to minimize the spatial variation between the similarly colored LEDs, which may be scattered about the array. The dedicated photodetectors **88** may be any devices that produce current indicative of incident light (such as a silicon photodiode or an LED). In one embodiment, however, the dedicated photodetectors **88** are preferably LEDs with peak emission wavelengths in the range of 500 nm to 700 nm. Photodetectors with such peak emission wavelengths will not produce photocurrent in response to infrared light, which reduces interference from ambient light.

In at least one preferred embodiment, emitter module **82** includes a phosphor-converted white (W) emission LED **84** and two different colors of photodetectors for measuring different portions of the phosphor-converted LED spectrum. In one example, the dedicated photodetectors **88** may include one or more small red LEDs and one or more small green LEDs. In such an example, the dedicated red photodetector(s) **88** may be used to measure the photocurrents (Iph_d1) that are induced by the light that passes through the phosphor portion of the phosphor-converted LED, while the dedicated green photodetector(s) **88** are used to measure the photocurrents (Iph_d2) induced by the light emitted by blue LED portion of the phosphor-converted LED. These photocurrent measurements (Iph_d1 and Iph_d2) may be stored within the calibration table of FIG. **7** and used within the compensation method of FIG. **13** to compensate for chromaticity shifts caused by phosphor aging.

The illumination devices shown in FIGS. **16**A and **17**A and the emitter modules shown in FIGS. **16**B and **17**B are provided merely as examples of illumination devices in which the improved calibration and compensation methods may be used. Further description of these illumination devices and emitter modules may be found in related U.S. patent application Ser. No. 14/097,339 and related U.S. Provisional Patent Application No. 61/886,471, which are commonly assigned and incorporated herein by reference in their entirety. However, the inventive concepts described herein are not limited to any particular type of LED illumination device, any particular number of emitter modules that may be included within an LED illumination device, or any particular number, color, or arrangement of emission LEDs and photodetectors that may be included within an emitter module. Instead, the present invention may only require an LED illumination device to include at least one emitter module comprising a plurality of emission LEDs and at least one dedicated photodetector. In some embodiments, a dedicated photodetector may not be required, if one or more of the emission LEDs is configured, at times, to provide such functionality.

FIG. **18**A is a side view of one embodiment of an improved emitter module **90** comprising a plurality of emission LEDs **92** and one or more dedicated photodetectors **94**, all of which are mounted on a common substrate **96** and encapsulated within a primary optics structure (e.g., a dome) **98**. A heat sink **100** is coupled to a bottom surface of the substrate **96** for drawing heat away from the heat generating components of the emitter module **90**. The heat sink **100** may comprise substantially any material with relatively high thermal and electrical conductivity. In some embodiments, the heat sink **100** is formed from a material having a thermal conductivity that ranges between about 200 W/(mK) and about 400 W/(mK). In one embodiment, the heat sink is formed from a copper or copper-alloy material, or an aluminum or aluminum alloy material. The heat sink **100** may be a relatively thick layer ranging between about 1 mm and about 10 mm, and in one embodiment, may be about 3 mm thick.

Emitter module **90** may include substantially any number and color of emission LEDs **92** and substantially any number and color of dedicated photodetectors **94**. In one exemplary embodiment, the emission LEDs **92** include one or more red LEDs, one or more blue LEDs, one or more green LEDs and one or more white or yellow LEDs, as shown in FIGS. **16**B and **17**B. The emission LEDs **92** may generally be arranged in an array near the center of the dome **98**, and the dedicated photodetectors **94** may generally be arranged about a periphery of the array. In one exemplary embodiment, the dedicated photodetectors **94** may include one or more red, orange, yellow and/or green LEDs. The LEDs used to implement the dedicated photodetectors **94** are generally smaller than the emission LEDs **92** and are generally arranged to capture a maximum amount of light that is emitted from the emission LEDs **92** and reflected from the dome **98**. In some embodiments, dedicated photodetectors **114** may be omitted if one or more of the emission LEDs **112** are configured, at times, for detecting incident light.

The primary optics structure **98** may be formed from a variety of different materials and may have substantially any shape and/or dimensions necessary to shape the light emitted by the emission LEDs in a desirable manner. Although the primary optics structure **98** is described herein as a dome, one skilled in the art would understand how the primary optics structure may have substantially any other shape or configuration, which encapsulates the emission LEDs **92**

and the at least one photodetector **94**. In some embodiments, the shape, size, and material of the dome **98** may be generally designed to improve optical efficiency and color mixing within the emitter module **90**.

In one embodiment, substrate **96** may comprise a laminate material such as a printed circuit board (PCB) FR4 material, or a metal clad PCB material. However, substrate **96** may be formed from a ceramic material (or some other optically reflective material), in at least one preferred embodiment of the invention, so that the substrate may generally function to improve output efficiency by reflecting light back out of the emitter module **90**. In addition, substrate **96** may be configured to provide a relatively high thermal impedance, or low thermal conductivity, in the lateral direction (i.e., the direction in the plane of the substrate). In one embodiment, substrate **96** may be formed from a material (e.g., aluminum nitride, AlN) having a thermal conductivity less than or equal to about 150 W/(mK). In another embodiment, substrate **96** may be formed from a material (e.g., an aluminum oxide Al_2O_3 material) having a thermal conductivity less than about 30 W/(mK), or material (e.g., a PCB laminate material) having a thermal conductivity less than about 1 W/(mK). The high thermal impedance, or low thermal conductivity, provided by substrate **96** in the lateral direction advantageously isolates the junction temperatures of the emission LEDs **92** and the photodetectors **94**, and avoids inaccurate Vfe and Vfd measurements.

In some embodiments, substrate **96** may be further configured to provide a relatively low thermal impedance, or high thermal conductivity, in the vertical direction (i.e., the direction perpendicular to the plane of the substrate). In particular, a relatively low thermal impedance path **102** may be provided between each emission LED **92** and each photodetector **94** to the heat sink **100**. In addition to improving heat dissipation, the low thermal impedance paths **102** enable the substrate **96** to maintain a consistent temperature difference between the emitter and detector junction temperatures over operating conditions.

As noted above, alternative embodiments of the temperature compensation method shown in FIG. **8** may use the forward voltage (Vfd) measured across the photodetector(s), instead of the forward voltages (Vfe) measured across the emission LEDs, to provide an indication of the current operating temperature. However, these alternative embodiments note that the detector forward voltage (Vfd) may only be used if the temperature difference between the emission LEDs and the photodetector(s) remains substantially the same over the operating temperature range.

The improved emitter module **90** shown in FIG. **18**A maintains a relatively fixed temperature difference between the emission LEDs **92** and the dedicated photodetectors **94** by providing each of the emission LEDs **92** and each of the photodetectors **94** with a low thermal impedance path **102** to the heat sink **100**. This may be achieved in a number of different ways. In the particular embodiment shown in FIG. **18**A, a low thermal impedance path **102** is provided by minimizing the thickness of the substrate **96** (which minimizes the vertical distance between the emission LEDs **92**, the photodetectors **94** and the heat sink **100**), and by connecting each of the emission LEDs **92** and each of the photodetectors **94** to the heat sink **100** with a plurality of thermally conductive lines **102**. In one example, the thickness (T) of substrate **96** may range between about 300 μm and about 500 μm.

In general, the plurality of thermally conductive lines **102** may comprise substantially any thermally conductive material. In some embodiments, the thermally conductive lines **102** are formed from a material having a thermal conductivity that ranges between about 200 W/(mK) and about 400 W/(mK). The material used for the thermally conductive lines **102** may be the same material used for the heat sink **100** or may be different. In one embodiment, the thermally conductive lines **102** are formed from an aluminum, aluminum-alloy, copper, or copper-alloy material. The plurality of thermally conductive lines **102** may be formed by drilling vertical holes through the substrate (using any mechanical or optical means), and filling or plating the holes (or vias) with a metal material using any appropriate method. In some embodiments, each thermally conductive line **102** may comprise a plurality (e.g., about 10-20) of densely-packed vias, with each via being on the order of a couple of hundred microns wide.

If the emitter module **90** shown in FIG. **18**A is utilized within an illumination device, the detector forward voltage (Vfd) may be used within the temperature compensation method of FIG. **8** to provide a good indication of the junction temperatures of the emission LEDs **92**. Since the temperature difference between the emission LEDs **92** and the photodetectors **94** is fixed, the detector forward voltage (Vfd) may be measured in step **38** of FIG. **8** (instead of the emitter forward voltages), and the emitter forward voltages (Vfe) may be calculated using the detector forward voltage (Vfd) measurement and the emitter forward voltages (Vfe) stored in the calibration table of FIG. **7**. This would advantageously reduce the number of forward voltage measurements that need to be obtained during the temperature compensation method of FIG. **8**.

While the emitter module shown in FIG. **18**A provides desirable thermal characteristics, it may not provide sufficient electrical isolation between the emission LEDs, photodetectors and heat sink. For both the emission LEDs and the photodetectors, the electrical contacts to either the anode, cathode, or both (in a flip-chip LED design) are generally provided on the backside of the LEDs. These contacts cannot be electrically coupled to the heat sink by directly connecting the contacts to the heat sink with the metal lines shown in FIG. **18**A. In order to provide electrical isolation between the LEDs and the heat sink, routing layers are provided in FIG. **18**B for connecting a chain of LEDs together and for connecting the LED anodes/cathodes to external contacts outside of the dome.

FIG. **18**B is a side view of another embodiment of an improved emitter module **1810** comprising a plurality of emission LEDs **92** and one or more dedicated photodetectors **94**, all of which are mounted on a common substrate **1812** and encapsulated within a primary optics structure **98**. Many of the components shown in FIG. **18**B are similar to those shown in FIG. **18**A. Like components are denoted with like reference numerals.

Emitter module **1810** may include substantially any number, color, and arrangement of emission LEDs **92** and substantially any number, color, and arrangement of dedicated photodetectors **94**. The emission LEDs **92** and the dedicated photodetectors **94** may be similar to those described above but are not limited to such. In some embodiments, one or more of the dedicated photodetectors **94** may be omitted if one or more of the emission LEDs **92** are configured, at times, for detecting incident light.

The primary optics structure **98** may be formed from a variety of different materials and may have substantially any shape and/or dimensions necessary to shape the light emitted by the emission LEDs **92** in a desirable manner. Although the primary optics structure **98** is described herein as a dome, one skilled in the art would understand how the

primary optics structure may have substantially any other shape or configuration, which encapsulates the emission LEDs **92** and the at least one photodetector **94**. In some embodiments, the shape, size, and material of the dome **98** may be generally designed to improve optical efficiency and color mixing within the emitter module **110**.

Heat sink **100** is coupled to a bottom surface of the substrate **1812** for drawing heat away from the heat-generating components of the emitter module **1810**. The heat sink **100** may comprise substantially any material with relatively high thermal and electrical conductivity. In some embodiments, heat sink **100** is formed from a material having a thermal conductivity that ranges between about 200 W/(mK) and about 400 W/(mK). In one embodiment, the heat sink is formed from a copper or copper-alloy material, or an aluminum or aluminum-alloy material. In some embodiments, the heat sink **100** may be a relatively thick layer ranging between about 1 mm and about 10 mm, and in one embodiment, may be about 3 mm thick.

Emitter module **1810** differs from emitter module **90**, in at least one aspect, by electrically isolating the electrical contacts of the emission LEDs **92** and the photodetectors **94** from the heat sink **100**. This is achieved in the embodiment of FIG. **18**B by utilizing a substrate **1812** having multiple layers. While an overall thickness (e.g., about 300 μm to about 500 μm) of the multiple-layer substrate **1812** may be similar to the single-layer substrate **96** shown in FIG. **18**A, in some embodiments, the multiple-layer substrate **1812** shown in FIG. **18**B is generally formed to include multiple routing and dielectric layers, which not only provide electrical isolation between the electrical contacts of the LEDs and the heat sink, but also improve routing flexibility.

According to one embodiment, multiple-layer substrate **1812** may include a first routing layer **1814**, a first dielectric layer **1816**, a second routing layer **1818**, and a second dielectric layer **1820**. The first routing layer **1814** is coupled to the electrical contacts of the emission LEDs **92** and the one or more photodetectors **94** and may be formed on the first dielectric layer **1816**. The first routing layer **1814** may have a thickness that ranges between about 10 μm to about 20 μm, may be formed of a material (e.g., a copper or aluminum material, or an alloy thereof) having a thermal conductivity that ranges between 200 W/(mK) and about 400 W/(mK), and may be formed by any well-known process on the upper surface of the first dielectric layer **1816**. For example, the first routing layer **1814** may be formed by printing or depositing metal lines on the upper surface of the first dielectric layer **1816**.

The first dielectric layer **1816** is sandwiched between the first routing layer **1814** and the second routing layer **1818** for electrically isolating the electrical contacts of the LEDs from the heat sink **100**. In some embodiments, the first dielectric layer **1816** may be a relatively thin layer having a thickness between about 10 μm and about 100 μm and may be formed from a dielectric material having a relative permittivity that ranges between about 3 and 12. In one example, the first dielectric layer **1816** may be formed from an aluminum nitride material or an aluminum oxide material but is not limited to such materials.

In addition to providing electrical isolation, first dielectric layer **1816** provides a relatively high thermal impedance in the lateral direction by using a material with a relatively low thermal conductivity, which is less than about 150 W/(mK), and keeping the thickness of the layer small relative to the spacing between the emission LEDs and the photodetectors. In one exemplary embodiment, the first dielectric layer **1816** may have a thickness of about 30 μm, and the emission LEDs **92** and photodetectors **94** may be spaced at least 200-300 μm apart on an upper surface of the substrate **1812**. Such an embodiment would provide at least 10 times higher thermal conductivity in the vertical direction than in the lateral direction.

The second routing layer **1818** is coupled between the first dielectric layer **1816** and the second dielectric layer **1820** and is generally configured for routing signals between the first routing layer **1814** and external electrical contacts (not shown) arranged outside of the primary optics structure. Like the first routing layer **1814**, the second routing layer **1818** may have a thickness that ranges between about 10 μm to about 20 μm, may be formed of a material (e.g., a copper or aluminum material, or an alloy thereof) having a thermal conductivity that ranges between 200 W/(mK) and about 400 W/(mK), and may be formed by any well-known process on the upper surface of the second dielectric layer **1820**. For example, the second routing layer **1818** may be formed by printing or depositing metal lines on the upper surface of the second dielectric layer **1820**. In order to route signals between the first routing layer **1814** and the second routing layer **1818**, vias **1824** may be formed within the first dielectric layer **1816**. These vias may be formed in accordance with any known process.

In some embodiments, the second dielectric layer **1820** may be coupled between the second routing layer **1818** and the heat sink **100** and may be generally configured for providing a relatively high thermal impedance in the lateral direction, and a relatively low thermal impedance in the vertical direction. In some embodiments, the second dielectric layer **1820** may be a relatively thick layer having a thickness between about 100 μm and about 1000 μm, which imparts rigidity to the emitter module **1810**. In some embodiments, the second dielectric layer **1820** may be formed from a dielectric material having a relative permittivity that ranges between about 3 and 12 and a thermal conductivity that is less than about 150 W/(mK). In one example, the second dielectric layer **1820** may be formed from an aluminum nitride material or an aluminum oxide material but is not limited to such materials.

The second dielectric layer **1820** is similar to the substrate **96** shown in FIG. **18**A, in that it provides a relatively high thermal impedance in the lateral direction by implementing the second dielectric layer **1820** with a material having a thermal conductivity less than about 150 W/(mK), and a relatively low thermal impedance in the vertical direction by including a plurality of thermally conductive lines **1822**, which extend vertically through the second dielectric layer **1820** between the second routing layer **1818** and the heat sink **100**. As noted above, the plurality of thermally conductive lines **1822** may be formed from a material having a thermal conductivity that ranges between about 200 W/(mK) and about 400 W/(mK), such as a copper or aluminum material, or an alloy thereof. The plurality of thermally conductive lines **1822** may be formed by drilling vertical holes through the second dielectric layer (using any mechanical or optical means), and filling or plating the holes (or vias) with an appropriate metal material using any appropriate method. In some embodiments, each thermally conductive line **1822** may comprise a plurality (e.g., about of densely-packed vias, with each via being a couple of hundred microns wide.

In some embodiments, a third routing layer **1826** may be coupled between the second dielectric layer **1820** and the heat sink **100**. Unlike the first and second routing layers, which comprise metal lines printed on the first and second dielectric layers, the third routing layer **1826** may extend

substantially continuously across an upper surface of the heat sink **100** for improving the thermal contact between the plurality of thermally conductive lines **1822** and the heat sink **100** and improving heat spreading there across.

Like the substrate **96** shown in FIG. **18**A, the multiple-layer substrate **1812** shown in FIG. **18**B provides good thermal isolation between the emission LEDs **92**, and between the emission LEDs and the photodetectors, while providing rigidity and maintaining good overall thermal conductivity to the heat sink **100**. This allows the detector forward voltage (Vfd) measurements to be used in lieu of emitter forward voltage (Vfe) measurements in the compensation method of FIG. **8**, which reduces the number of forward voltage measurements that need to be obtained during the compensation method.

The multiple-layer substrate **1812** shown in FIG. **18**B also provides other advantages, which the single-layer substrate **96** cannot provide. For example, substrate **1812** includes multiple routing and dielectric layers, which improves routing flexibility for connecting chains of the emission LEDs **92** together and provides electrical isolation between the electrical contacts of the emission LEDs and photodetectors and the heat sink **100**.

The multiple-layer substrate **1812** can also be implemented somewhat differently in alternative embodiments of the invention. For example, instead of using dielectric or ceramic materials for layers **1816** and **1820**, these layers could be using a laminate material such as a printed circuit board (PCB) FR4 material, or a metal-clad PCB material. However, since the thermal conductivity of laminate materials (e.g., less than about 1 W/(mK)) is much less than that of ceramic materials, using a laminate material in lieu of a ceramic material would reduce the thermal conductivity of layer **1820**. Regardless of the material used for layer **1820**, thermal conductivity may be increased in some embodiments of the invention by increasing the number of thermally conductive lines **1822** included under the LED array. While this approach would provide better overall thermal conductivity from the LED array to the heat sink, it would provide worse thermal separation between the emission LEDs.

FIG. **19** is one example of a block diagram of an illumination device **110** configured to accurately maintain a desired luminous flux and a desired chromaticity over variations in drive current, temperature, and time. The illumination device illustrated in FIG. **19** provides one example of the hardware and/or software that may be used to implement the calibration method shown in FIG. **6** and the compensation methods shown in FIGS. **8** and **13**.

In the illustrated embodiment, illumination device **110** comprises a plurality of emission LEDs **126** and one or more dedicated photodetectors **128**. The emission LEDs **126**, in this example, comprise four chains of any number of LEDs. In typical embodiments, each chain may have 2 to 4 LEDs of the same color, which are coupled in series and configured to receive the same drive current. In one example, the emission LEDs **126** may include a chain of red LEDs, a chain of green LEDs, a chain of blue LEDs, and a chain of white or yellow LEDs. However, the present invention is not limited to any particular number of LED chains, any particular number of LEDs within the chains, or any particular color or combination of LED colors.

Although the one or more dedicated photodetectors **128** are also illustrated in FIG. **19** as including a chain of LEDs, the present invention is not limited to any particular type, number, color, combination, or arrangement of photodetectors. In one embodiment, the one or more dedicated photodetectors **128** may include a small red, orange, or yellow LED, as discussed above with respect to FIG. **16**B. In another embodiment, the one or more dedicated photodetectors **128** may include one or more small red LEDs and one or more small green LEDs, as discussed above with respect to FIG. **17**B. In some embodiments, one or more of the dedicated photodetectors **128** shown in FIG. **19** may be omitted if one or more of the emission LEDs **126** are configured, at times, to function as a photodetector. The plurality of emission LEDs **126** and the (optional) dedicated photodetectors **128** may be included within an emitter module, as discussed above. In some embodiments, an illumination device may include more than one emitter module, as discussed above.

In addition to including one or more emitter modules, illumination device **110** includes various hardware and software components, which are configured for powering the illumination device and controlling the light output from the emitter module(s). In one embodiment, the illumination device is connected to an AC mains **112**, and includes an AC/DC converter **114** for converting AC mains power (e.g., 120 V or 240 V) to a DC voltage (V_{DC}). As shown in FIG. **19**, this DC voltage (e.g., 15 V) is supplied to the LED driver and receiver circuit **124** for producing the operative drive currents applied to the emission LEDs **126** for producing illumination. In addition to the AC/DC converter, a DC/DC converter **116** is included for converting the DC voltage V_{DC} (e.g., 15 V) to a lower voltage V_L (e.g., 3.3 V), which is used to power the low-voltage circuitry included within the illumination device, such as PLL **118**, wireless interface **120**, and control circuit **122**.

In the illustrated embodiment, PLL **118** locks to the AC mains frequency (e.g., 50 or 60 HZ) and produces a high-speed clock (CLK) signal and a synchronization signal (SYNC). The CLK signal provides the timing for control circuit **122** and LED driver and receiver circuit **124**. In one example, the CLK signal frequency is in the tens of MHz range (e.g., 23 MHz), and is precisely synchronized to the AC Mains frequency and phase. The SYNC signal is used by the control circuit **122** to create the timing used to obtain the various optical and electrical measurements described above. In one example, the SYNC signal frequency is equal to the AC Mains frequency (e.g., 50 or 60 HZ) and also has a precise phase alignment with the AC Mains.

In some embodiments, a wireless interface **120** may be included and used to calibrate the illumination device **110** during manufacturing. As noted above, for example, an external calibration tool (not shown in FIG. **19**) may communicate luminous flux and chromaticity calibration values to an illumination device under test via the wireless interface **120**. The calibration values received via the wireless interface **120** may be stored in the table of calibration values within a storage medium **121** of the control circuit **122**, for example. In some embodiments, the control circuit **122** may use the calibration values to generate calibration coefficients, which are stored within the storage medium **121** in addition to, or in lieu of, the received calibration values.

Wireless interface **120** is not limited to receiving only calibration data and may be used for communicating information and commands for many other purposes. For example, wireless interface **120** could be used during normal operation to communicate commands, which may be used to control the illumination device **110**, or to obtain information about the illumination device **110**. For instance, commands may be communicated to the illumination device **110** via the wireless interface **120** to turn the illumination device on/off, to control the dimming level and/or color set point of the

illumination device, to initiate the calibration procedure, or to store calibration results in memory. In other examples, wireless interface **120** may be used to obtain status information or fault condition codes associated with illumination device **110**.

In some embodiments, wireless interface **120** could operate according to ZigBee, Wi-Fi, Bluetooth, or any other proprietary or standard wireless data communication protocol. In other embodiments, wireless interface **120** could communicate using radio frequency (RF), infrared (IR) light or visible light. In alternative embodiments, a wired interface could be used, in place of the wireless interface **120** shown, to communicate information, data and/or commands over the AC mains or a dedicated conductor or set of conductors.

Using the timing signals received from PLL **118**, the control circuit **122** calculates and produces values indicating the desired drive current to be used for each LED chain **126**. This information may be communicated from the control circuit **122** to the LED driver and receiver circuit **124** over a serial bus conforming to a standard, such as SPI or I^2C, for example. In addition, the control circuit **122** may provide a latching signal that instructs the LED driver and receiver circuit **124** to simultaneously change the drive currents supplied to each of the LEDs **126** to prevent brightness and color artifacts.

In general, the control circuit **122** may be configured for determining the respective drive currents needed to achieve a desired luminous flux and/or a desired chromaticity for the illumination device in accordance with one or more of the compensation methods shown in FIGS. **8** and **13** and described above. In some embodiments, the control circuit **122** may determine the respective drive currents by executing program instructions stored within the storage medium **121**. In one embodiment, the storage medium may be a non-volatile memory, and may be configured for storing the program instructions along with a table of calibration values, such as the table described above with respect to FIG. **7**. Alternatively, the control circuit **122** may include combinatorial logic for determining the desired drive currents, and the storage medium **121** may only be used for storing the table of calibration values.

In general, the LED driver and receiver circuit **124** may include a number (N) of driver blocks **130** equal to the number of emission LED chains **126** included within the illumination device. In the exemplary embodiment discussed herein, LED driver and receiver circuit **124** comprises four driver blocks **130**, each configured to produce illumination from a different one of the emission LED chains **126**. The LED driver and receiver circuit **124** also comprises the circuitry needed to measure ambient temperature (optional), the detector and/or emitter forward voltages, and the detector photocurrents, and to adjust the LED drive currents accordingly. Each driver block **130** receives data indicating a desired drive current from the control circuit **122**, along with a latching signal indicating when the driver block **130** should change the drive current.

FIG. **20** is an exemplary block diagram of an LED driver and receiver circuit **124**, according to one embodiment of the invention. As shown in FIG. **20**, the LED driver and receiver circuit **124** includes four driver blocks **130**, each block including a buck converter **132**, a current source **134**, and an LC filter **138** for generating the drive currents that are supplied to a connected chain of emission LEDs **126** to produce illumination and obtain forward voltage (Vfe) measurements. In some embodiments, buck converter **132** may produce a pulse width modulated (PWM) voltage output (Vdr) when the controller **154** drives the "Out_En" signal high. This voltage signal (Vdr) is filtered by the LC filter **138** to produce a forward voltage on the anode of the connected LED chain **126**. The cathode of the LED chain is connected to the current source **134**, which forces a fixed drive current equal to the value provided by the "Emitter Current" signal through the LED chain **126** when the "Led_On" signal is high. The "Vc" signal from the current source **134** provides feedback to the buck converter **132** to output the proper duty cycle and minimize the voltage drop across the current source **134**.

As shown in FIG. **20**, each driver block **130** includes a difference amplifier **137** for measuring the forward voltage drop (Vfe) across the chain of emission LEDs **126**. When measuring Vfe, the buck converter **132** is turned off and the current source **134** is configured for drawing a relatively small drive current (e.g., about 1 mA) through the connected chain of emission LEDs **126**. The voltage drop (Vfe) produced across the LED chain **126** by that current is measured by the difference amplifier **137**. The difference amplifier **137** produces a signal that is equal to the forward voltage (Vfe) drop across the emission LED chain **126** during forward voltage measurements.

As noted above, some embodiments of the invention may use one of the emission LEDs (e.g., a green emission LED), at times, as a photodetector. In such embodiments, the driver blocks **130** may include additional circuitry for measuring the photocurrents (Iph_d2), which are induced across an emission LED, when the emission LED is configured for detecting incident light. For example, each driver block **130** may include a transimpedance amplifier **135**, which generally functions to convert an input current to an output voltage proportional to a feedback resistance. As shown in FIG. **20**, the positive terminal of transimpedance amplifier **135** is connected to the Vdr output of the buck converter **132**, while the negative terminal is connected to the cathode of the last LED in the LED chain **126**. Transimpedance amplifier **135** is enabled when the "LED_On" signal is low. When the "LED_On" signal is high, the output of transimpedance amplifier **135** is tri-stated.

When measuring the photocurrents (Iph_d2) induced by an emission LED, the buck converters **132** connected to all other emission LEDs should be turned off to avoid visual artifacts produced by LED current transients. In addition, the buck converter **132** coupled to the emission LED under test should also be turned off to prevent switching noise within the buck converter from interfering with the photocurrent measurements. Although turned off, the Vdr output of the buck converter **132** coupled to the emission LED under test is held to a particular value (e.g., about 2-3.5 volts times the number of emission LEDs in the chain) by the capacitor within LC filter **138**. When this voltage (Vdr) is supplied to the anode of emission LED under test and the positive terminal of the transimpedance amplifier **135**, the transimpedance amplifier produces an output voltage (relative to Vdr) that is supplied to the positive terminal of difference amplifier **136**. Difference amplifier **136** compares the output voltage of transimpedance amplifier **135** to Vdr and generates a difference signal, which corresponds to the photocurrent (Iph_d2) induced across the LED chain **126**.

In addition to including a plurality of driver blocks **130**, the LED driver and receiver circuit **124** may include one or more receiver blocks **140** for measuring the forward voltages (Vfd) and photocurrents (Iph_d1 or Iph_d2) induced across the one or more dedicated photodetectors **128**. Although only one receiver block **140** is shown in FIG. **20**, the LED driver and receiver circuit **124** may generally

include a number of receiver blocks **140** equal to the number of dedicated photodetectors included within the emitter module.

In the illustrated embodiment, receiver block **140** comprises a voltage source **142**, which is coupled for supplying a DC voltage (Vdr) to the anode of the dedicated photodetector **128** coupled to the receiver block, while the cathode of the photodetector **128** is connected to current source **144**. When photodetector **128** is configured for obtaining forward voltage (Vfd), the controller **154** supplies a "Detector_On" signal to the current source **144**, which forces a fixed drive current (Idrv) equal to the value provided by the "Detector Current" signal through photodetector **128**.

When obtaining detector forward voltage (Vfd) measurements, current source **144** is configured for drawing a relatively small amount of drive current (Idrv) through photodetector **128**. The voltage drop (Vfd) produced across photodetector **128** by that current is measured by difference amplifier **147**, which produces a signal equal to the forward voltage (Vfd) drop across photodetector **128**. As noted above, the drive current (Idrv) forced through photodetector **128** by the current source **144** is generally a relatively small, non-operative drive current. In the embodiment in which four dedicated photodetectors **128** are coupled in parallel, the non-operative drive current may be roughly 1 mA. However, smaller/larger drive currents may be used in embodiments that include fewer/greater numbers of photodetectors, or embodiments that do not connect the photodetectors in parallel.

Similar to driver block **130**, receiver block **140** also includes circuitry for measuring the photocurrents (Iph_d1 or Iph_d2) induced on photodetector **128** by light emitted by the emission LEDs. As shown in FIG. **20**, the positive terminal of transimpedance amplifier **145** is coupled to the Vdr output of voltage source **142**, while the negative terminal is connected to the cathode of photodetector **128**. When connected in this manner, the transimpedance amplifier **145** produces an output voltage relative to Vdr (e.g., about 0-1 V), which is supplied to the positive terminal of difference amplifier **146**. Difference amplifier **146** compares the output voltage to Vdr and generates a difference signal, which corresponds to the photocurrent (Iph_d1 or Iph_d2) induced across photodetector **128**. Transimpedance amplifier **145** is enabled when the "Detector_On" signal is low. When the "Detector_On" signal is high, the output of transimpedance amplifier **145** is tri-stated.

As noted above, some embodiments of the invention may scatter the individual LEDs within each chain of LEDs **126** about the array of LEDs, so that no two LEDs of the same color exist in any row, column, or diagonal (see, e.g., FIG. **17**B). By connecting a plurality of dedicated photodetectors **128** in parallel with the receiver block **140**, the photocurrents (Iph_d1 or Iph_d2) induced on each photodetector **128** by the LEDs of a given color may be summed to minimize the spatial variation between the similarly-colored LEDs, which are scattered about the array.

As shown in FIG. **20**, the LED driver and receiver circuit **124** may also include a multiplexor (Mux) **150**, an analog-to-digital converter (ADC) **152**, a controller **154**, and an optional temperature sensor **156**. In some embodiments, multiplexor **150** may be coupled for receiving the emitter forward voltage (Vfe) and the (optional) photocurrent (Iph_d2) measurements from the driver blocks **130**, and the detector forward voltage (Vfd) and detector photocurrent (Iph_d1 and/or Iph_d2) measurements from the receiver block **140**. The ADC **152** digitizes the emitter forward voltage (Vfe) and the optional photocurrent (Iph_d2) measurements output from the driver blocks **130**, and the detector forward voltage (Vfd) and detector photocurrent (Iph_d1 and/or Iph_d2) measurements output from the receiver block **140**, and provides the results to the controller **154**. The controller **154** determines when to take forward voltage and photocurrent measurements and produces the Out_En, Emitter Current, and Led_On signals, which are supplied to the driver blocks **130**, and the Detector Current and Detector_On signals, which are supplied to the receiver block **140**, as shown in FIG. **20**.

In some embodiments, the LED driver and receiver circuit **124** may include an optional temperature sensor **156** for taking ambient temperature (Ta) measurements. In such embodiments, multiplexor **150** may also be coupled for multiplexing the ambient temperature (Ta) with the forward voltage and photocurrent measurements sent to the ADC **152**. In some embodiments, the temperature sensor **156** may be a thermistor, and may be included on the driver circuit chip for measuring the ambient temperature surrounding the LEDs, or a temperature from the heat sink of the emitter module. In other embodiments, the temperature sensor **156** may be an LED, which is used as both a temperature sensor and an optical sensor to measure ambient light conditions or output characteristics of the LED emission chains **126**. If the optional temperature sensor **156** is included, the output of the temperature sensor may be used in some embodiments to determine if a significant change in temperature is detected in step **32** of FIG. **8**.

One implementation of an improved illumination device **110** has now been described in reference to FIGS. **19-20**. Further description of such an illumination device may be found in commonly assigned U.S. application Ser. Nos. 13/970,944, 13/970,964, and 13/970,990. A skilled artisan would understand how the illumination device could be alternatively implemented within the scope of the present invention.

It will be appreciated to those skilled in the art having the benefit of this disclosure that this invention is believed to provide an improved illumination device and improved methods for calibrating and compensating individual LEDs in the illumination device, so as to maintain a desired luminous flux and a desired chromaticity over variations in drive current, temperature and time. In addition, emitter modules having improved thermal and electrical characteristics are also provided herein. Further modifications and alternative embodiments of various aspects of the invention will be apparent to those skilled in the art in view of this description. It is intended, therefore, that the following claims be interpreted to embrace all such modifications and changes and, accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense.

The invention claimed is:

1. A light-emitting diode (LED) lighting controller disposed in an LED lighting fixture, the LED lighting controller comprising:

LED lighting control circuitry to, at each of a plurality of ambient temperatures:

cause a sequential application of each of a plurality of drive currents to one of a plurality of LED emitters while maintaining the remaining plurality of LED emitters in a non-illuminated condition;

receive data representative of one or more optical measurements at each of the plurality of drive currents;

receive data representative of one or more electrical measurements from one or more operatively coupled photodetectors at each of the plurality of drive currents;

determine at least one LED emitter operating temperature dependent relationship between the optical output parameters of the respective LED emitter and the drive current applied to the respective LED emitter using the received data representative of the one or more optical measurements and the received data representative of the one or more electrical measurements; and

cause a storage of data representative of the determined at least one LED emitter operating temperature dependent relationship in operatively coupled memory circuitry.

2. The LED lighting controller of claim **1**, the control circuitry to further:

responsive to receipt of an instruction to provide illumination having a defined spectral content, determine for each of the plurality of LEDs included in the LED lighting fixture, a drive current to provide to each of the plurality of LEDs using the stored data representative of the optical output parameters of the respective LED emitter and the drive current applied to the respective LED emitter.

3. The LED lighting controller of claim **1**, wherein to cause a sequential application of each of the plurality of drive currents to each of the plurality of LED emitters, the control circuitry to further:

determine a maximum drive current for the respective LED emitter; and

cause a sequential application of:

a first drive current equal to 100% of the maximum drive current for the respective LED emitter;

a second drive current equal to 30% of the maximum drive current for the respective LED emitter; and

a third drive current equal to 10% of the maximum drive current for the respective LED emitter.

4. The LED lighting controller of claim **1**, wherein to receive the data representative of the one or more optical measurements at each of the plurality of drive currents, the control circuitry to further:

receive data representative of one or more of the following at each of the plurality of drive currents:

a luminous flux of the respective LED emitter;

an x-chromaticity value of the respective LED emitter; or

a y-chromaticity value of the respective LED emitter.

5. The LED lighting controller of claim **1**, wherein to receive the data representative of the one or more electrical measurements from the one or more operatively coupled photodetectors at each of the plurality of drive currents, the control circuitry to further:

receive, from the one or more photodetectors, data representative of one or more of the following at each of the plurality of drive currents:

an induced photocurrent in the one or more photodetectors; or

a forward voltage across the one or more photodetectors.

6. The LED lighting controller of claim **1**, wherein to receive the data representative of the one or more optical measurements at each of the plurality of drive currents, the control circuitry to further:

receive, from a remote device via wireless communication interface circuitry, the data representative of the one or more optical measurements at each of the plurality of drive currents.

7. A light-emitting diode (LED) lighting control method, comprising:

at each of a plurality of ambient temperatures:

causing, by LED lighting controller control circuitry, a sequential application of each of a plurality of drive currents to one of a plurality of LED emitters while maintaining the remaining plurality of LED emitters in a non-illuminated condition;

receiving, by the LED lighting controller control circuitry, data representative of one or more optical measurements at each of the plurality of drive currents;

receiving, by the LED lighting controller control circuitry, data representative of one or more electrical measurements from one or more operatively coupled photodetectors at each of the plurality of drive currents;

determining, by the LED lighting controller control circuitry, at least one LED emitter operating temperature dependent relationship between the optical output parameters of the respective LED emitter and the drive current applied to the respective LED emitter using the received data representative of the one or more optical measurements and the received data representative of the one or more electrical measurements; and

causing, by the LED lighting controller control circuitry, a storage of data representative of the determined at least one LED emitter operating temperature dependent relationship in operatively coupled memory circuitry.

8. The LED lighting control method of claim **7**, further comprising:

determining, by the LED lighting controller control circuitry, for each of the plurality of LEDs included in the LED lighting fixture, a drive current to provide to each of the plurality of LEDs using the stored data representative of the optical output parameters of the respective LED emitter and the drive current applied to the respective LED emitter responsive to receipt of an instruction to provide illumination having a defined spectral content.

9. The LED lighting control method of claim **7**, wherein causing the sequential application of each of the plurality of drive currents to each of the plurality of LED emitters further comprises:

determining, by the LED lighting controller control circuitry, a maximum drive current for the respective LED emitter; and

causing, by the LED lighting controller control circuitry, a sequential application of:

a first drive current equal to 100% of the maximum drive current for the respective LED emitter;

a second drive current equal to 30% of the maximum drive current for the respective LED emitter; and

a third drive current equal to 10% of the maximum drive current for the respective LED emitter.

10. The LED lighting control method of claim **7**, wherein receiving the data representative of the one or more optical measurements at each of the plurality of drive currents further comprises:

receiving, by the LED lighting controller control circuitry, data representative of one or more of the following at each of the plurality of drive currents:

a luminous flux of the respective LED emitter;

an x-chromaticity value of the respective LED emitter; or

a y-chromaticity value of the respective LED emitter.

11. The LED lighting control method of claim **10**, wherein receiving the data representative of the one or more electrical measurements from the one or more operatively coupled photodetectors at each of the plurality of drive currents, further comprises:

receiving, by the LED lighting controller control circuitry from the one or more photodetectors, data representative of one or more of the following at each of the plurality of drive currents:

an induced photocurrent in the one or more photodetectors; or

a forward voltage across the one or more photodetectors.

12. The LED lighting control method of claim **7**, wherein receiving the data representative of the one or more optical measurements at each of the plurality of drive currents, further comprises:

receiving, by the LED lighting controller control circuitry from a remote device via wireless communication interface circuitry, the data representative of the one or more optical measurements at each of the plurality of drive currents.

13. A non-transitory, machine-readable, storage device that includes instructions that, when executed by light-emitting diode (LED) lighting controller control circuitry, causes the control circuitry to:

at each of a plurality of ambient temperatures:

cause a sequential application of each of a plurality of drive currents to one of a plurality of LED emitters while maintaining the remaining plurality of LED emitters in a non-illuminated condition;

receive data representative of one or more optical measurements at each of the plurality of drive currents;

receive data representative of one or more electrical measurements from one or more operatively coupled photodetectors at each of the plurality of drive currents;

determine at least one LED emitter operating temperature dependent relationship between the optical output parameters of the respective LED emitter and the drive current applied to the respective LED emitter using the received data representative of the one or more optical measurements and the received data representative of the one or more electrical measurements; and

cause a storage of data representative of the determined at least one LED emitter operating temperature dependent relationship in operatively coupled memory circuitry.

14. The non-transitory, machine-readable, storage device of claim **13**, wherein the instructions, when executed by the LED lighting controller control circuitry, further cause the control circuitry to:

determine for each of the plurality of LEDs included in the LED lighting fixture, a drive current to provide to each of the plurality of LEDs using the stored data representative of the optical output parameters of the respective LED emitter and the drive current applied to the respective LED emitter responsive to receipt of an instruction to provide illumination having a defined spectral content.

15. The non-transitory, machine-readable, storage device of claim **13**, wherein the instructions that cause the LED lighting controller control circuitry to cause the sequential application of each of the plurality of drive currents to each of the plurality of LED emitters further cause the control circuitry to:

determine a maximum drive current for the respective LED emitter; and

cause a sequential application of:

a first drive current equal to 100% of the maximum drive current for the respective LED emitter;

a second drive current equal to 30% of the maximum drive current for the respective LED emitter; and

a third drive current equal to 10% of the maximum drive current for the respective LED emitter.

16. The non-transitory, machine-readable, storage device of claim **13**, wherein the instructions that cause the LED lighting controller control circuitry to receive the data representative of the one or more optical measurements at each of the plurality of drive currents further cause the control circuitry to:

receive data representative of one or more of the following at each of the plurality of drive currents:

a luminous flux of the respective LED emitter;

an x-chromaticity value of the respective LED emitter; or

a y-chromaticity value of the respective LED emitter.

17. The non-transitory, machine-readable, storage device of claim **16**, wherein the instructions that cause the LED lighting controller control circuitry to receive the data representative of the one or more electrical measurements from the one or more operatively coupled photodetectors at each of the plurality of drive currents, further cause the control circuitry to:

receive, from the one or more photodetectors, data representative of one or more of the following at each of the plurality of drive currents:

an induced photocurrent in the one or more photodetectors; or

a forward voltage across the one or more photodetectors.

18. The non-transitory, machine-readable, storage device of claim **13**, wherein the instructions that cause the LED lighting controller control circuitry to receive the data representative of the one or more optical measurements at each of the plurality of drive currents, further cause the control circuitry to:

receive, from a remote device via wireless communication interface circuitry, the data representative of the one or more optical measurements at each of the plurality of drive currents.

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